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Necromancing Diels: computerising the phonological analysis of early Slavonic texts using existing treebank data and a Late Common Slavonic computerised inflectional morphology

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ABSTRACT

This article demonstrates a new way of massively speeding up traditional historical phonological investigations into early Slavonic philological material by furnishing each word of a manuscript-text with its direct Late Common Slavonic (LCS) reconstructed ancestor. Such reconstructions act as a type of "phonological annotation", or index, since they make the reflexes of any LCS phoneme-sequence reliably and predictably searchable, regardless of a manuscript's surface-spellings. To automate the production of this LCS reconstructed layer, a comprehensive computerised LCS inflectional morphology has been developed, which is capable of using existing lemmatisation and morphology-tagging annotation-data from texts in the Tromsø Old Russian and Old Church Slavonic (TOROT) corpus in order to inflect LCS lemmas according to each text-word's morphological shape. A web-interface, ocstexts.co.uk, is provided to showcase some of the benefits of the resulting "phonological annotations", including searchable autoreconstructed versions of OCS texts from the TOROT corpus, with virtually 100% coverage of the Codex Marianus, Codex Zographensis, and Psalterium Sinaiticum.

0. Introduction

Much progress has been made in the last twenty years in early Slavonic corpus linguistics as a result of the Old Church Slavonic part of the PROIEL project (Haug & Jøhndal 2008) and its subsequent expansion as the TOROT treebank (Eckhoff & Berdičevskis 2015), such that currently just over 240,000 words of canonical OCS have been manually lemmatised, part-of-speech and morphologically-tagged, and syntactically parsed. The focus of these projects, however, has been exclusively on the higher-level linguistic domains of syntax, semantics, and pragmatics: surface-morphology has been of only incidental concern, for example in investigations into differential-object marking (Eckhoff 2015, 2022), but the sort of inflection-class-marking which would enable the retrieval of, say, masculine *o-stems vs *i-stems is lacking. The needs of historical phonologists especially are ill-served, since some of the texts contain quite severe typographical inconsistencies and errors that make them impossible to search reliably and dangerous to use without reference to the manuscripts.

That being said, enough information is included in TOROT's lemmatisation and morphology-tagging that, with a few exceptions (e.g. certain comparatives), the morphological shape of the inflected text-forms can be predicted from just the tag-information, provided that inflection-class annotations are added to the lemmas. This means that the immediate Late Common Slavonic ancestors of surface-text forms can be generated by reconstructing and inflection-class-marking the LCS lemmas, and then using a computerised LCS inflectional-morphology to inflect each text-word's lemma according to its morphology-tag[1]. Such LCS reconstructions are an extremely useful form of 'phonological annotation', because theoretically all the information required to give rise to an attested form must by definition be present in any correct reconstructed proto-form, and the complete regularity of the idealised LCS forms makes texts predictably searchable regardless of orthographic variability, abbreviations, or other irregularities in the surface-texts. When applied to whole texts, they make the exhaustive investigation of almost any phonological or orthographic question trivially easy compared to manually reading and extracting relevant forms, or using TOROT's existing lemmatisation and morphology-tagging to try to gather morphological categories which might contain the sound-groups one is interested in.

[1] Morphological innovations and variations are detected by inspecting the text-forms and then applying 'alternative' endings as specified in the inflectional-endings database; see Section 2.3.

The goal of this article is to describe just such a computerised LCS inflectional-morphology, and to show how it can be used to "autoreconstruct" different OCS texts from the TOROT corpus, while explaining how difficulties caused by things like morphological innovations, badly-integrated foreign loanwords, or insufficiently-precise tagging-data can be overcome. The resulting ‘phonologically annotated’ texts should allow investigators of the lower-level linguistic domains (phonology, orthography, and morphology) to benefit from the huge amount of manual annotation work that has gone into the OCS part of TOROT, and the computerised LCS inflectional-morphology by itself is valuable in that it allows linguistically-rigorous and comprehensive inflection and conjugation-tables to be generated[2].

A web-interface for autoreconstructed TOROT OCS texts is being developed at https://ocstexts.co.uk, which aims to demonstrate the benefits of this extra layer of annotation by offering things like LCS phoneme-search of texts, conversion of each autoreconstructed word into canonical "normalised" OCS, for easy comparison with the manuscript-form, and computer-generated inflection and conjugation-tables for every reconstructed lemma. It also includes TOROT's existing morphology, lemmatisation, and (for three of the texts) Greek source-alignment data, plus integration of the recently digitised OCS dictionaries hosted at gorazd.org.

The usefulness of computational methods for diachronic phonology has of course not escaped the notice of previous scholars, and Sims-Williams (2018) offers a useful survey of a field he calls "Mechanising Historical Phonology". Of the three main strands of work considered there, (“automatic cognate-recognition”, “computerised forwards-reconstruction”, and “computerised backwards-reconstruction”), only forwards-reconstruction is of any relevance to the task of holistically investigating the sound-systems of manuscripts, in that one could apply sound-changes and orthographic rules to the LCS reconstructed layer of individual texts to try to “reverse engineer” the manuscript spellings, and in so doing test hypotheses about the phonological features of the underlying dialect[s] that gave rise to them. The forwards-reconstruction work mentioned by Sims-Williams, though, is invariably aimed at producing sets of isolated, regularised forms, either from modern standard languages, or theoretical idealised forms from old languages, e.g. Borin’s (1988) OCSGen system that tried to produce “normalised” OCS from PIE. This type of work is useful for identifying problems with traditional diachronic accounts of ordered sound-rules[3], but it is of a fundamentally different nature to what I'm doing here, which is about elucidating the concrete philological facts of the manuscripts in order to better understand the synchronic linguistic systems responsible for them.
Most early Slavic texts are from dialect-areas (East Slavic and Bulgaro-Macedonian) whose modern phonological systems show evidence of sharp differentiation along deep structural lines, most importantly in the phonologisation of secondary consonant-palatalisation and the tolerance for bimoraic syllables (i.e. distinctive vowel-length). The seeds of these differences must have been sown around the time of the radical restructuring occasioned by loss of the reduced-vowels *ъ,ь (the so-called "Jer Shift"), i.e. the very time at or shortly after which our manuscripts were written, but the limitations of the original writing-system (which was devised for a language which had *not*

[2] The lemma and inflection-class data included as a supplement to Polivanova’s (2013) comprehensive OCS grammar, https://integral.github.io/osd/, would in principle allow similar tables to be generated, but as far as I know Polivanova has not digitised the actual endings, nor written any scripts to generate paradigms. Additionally, because she bases her forms on a normalised ***OCS***, rather than ***LCS***, phonemic system, a massive amount of information has been needlessly destroyed. To give just one obvious example, the syllabic-liquids *ŕ̥ *ĺ̥ *r̥ *l̥ are represented by their OCS reflexes /рь ль ръ лъ/, meaning there’s no distinction between the syllable-rimes of roots like /-срьд-/ <*sŕ̥d and /-крьст-/ <*krьst. An inflectional-morphology based on her data would therefore only be good for generating the (in reality unattested and fictional) “normalised” form of OCS, whereas my LCS forms are trivially convertible into *any* dialect of OCS, as well as pre-Jer Shift Old East Slavic (see pg. 17 and Figure 6), because they represent the language at a stage before any such dialect-specific phonemic mergers.

[3] For a recent example see Marr & Mortensen (2020), who found many problems with the traditional account of how Latin became French, and were able to improve upon it using their suite of software for performing forwards-reconstruction and "debugging" problematic sets of rules.

yet undergone the Jer Shift) mean that they do not find clear expression in the texts and can only be dimly groped at through indirect means. Much more thorough investigation of *all* aspects of the manuscript-spellings are thus required in order to uncover the nature of the linguistic systems they reflect.

Slavonicists are in the fortunate position of having not only a widely and diversely attested set of daughter-dialects from which to reconstruct the proto-language, but also a historical written record which is so close in time to the Common Slavonic period that the far-southeastern Bulgarian dialect of Thessaloniki was considered a suitable vehicle for proselytising the proto-Czech speakers of Moravia. The hypothetical LCS system which I employ for my reconstructions is therefore mostly uncontroversial and widely agreed-upon (except in trivial matters of notation); nevertheless, a corpus-focused project like mine where the aim is to reconstruct *all* words of a text requires an exhaustiveness and rigour that leaves no room for the dodging of difficult points, so Section 1 of this article is taken up with a comprehensive exposition and justification of my chosen LCS system. Section 2 then looks in more detail at my computerised LCS inflectional-morphology and the structure of the existing TOROT data, and shows that together they can be used to produce the direct phonemic ancestor-forms of text-words, despite various challenges, including the morphological variation and restructuring which is occurring in even the earliest texts. Section 3 highlights some of the benefits gained by having texts indexed with LCS reconstructions and explores some ways this could be used to improve and speed up investigations of manuscripts.

1. My Late Common Slavonic (LCS) phoneme-system

The premise of my chosen form of "phonological annotation" is that the earliest Slavic texts reflect languages which are **structurally** close enough to the broadly-agreed-upon system of Late Common Slavonic that the forms underlying the manuscript-spellings are more or less trivially derivable (by the application of sound-change rules) from their theoretical LCS ancestors.
By 'structurally' I am referring to structure at the phonological level; structural changes at higher levels of analysis (i.e. inflectional morphology, derivational morphology) are of no concern unless they are **made possible only by intervening phonological changes**.
My contention is that before about 1100 not enough of these structural changes are in evidence in any Slavic text, and thus texts can be relatively straightforwardly indexed using a well-chosen LCS system. Before giving examples of structural changes that are problematic for such an indexing-system, it's necessary to first lay out my LCS system in full:

In order to account for as much of the subsequently attested Slavic as possible, a point after the monophthongisation of diphthongs, but before the Second and Third Velar Palatalisations (PV2 and PV3) is chosen as the point of departure, because of the difference between the West Slavic /š/ and South/East /ś/ reflex of these two palatalisations of *x (Cz. loc. pl. *dušich* vs Suprasliensis доусѣхъ <*duxěxъ; Polish *wszak* vs Supr. вьсакъ , Ru. *всяк[ий]* <*vьx-akъ), as well as the probable complete absence of PV2 in northern East Slavic[4] (Old Novgorodian, see Zaliznjak 2004: 42-45 for

[4] The evidence regarding the possible absence of PV3 from Novgorodian is far less convincing: the Birchbark letters abound with examples of the PV3 reflex of *k (e.g. letter №439 from around 1200 has свинеце <*svinьkь and полотенеца <*poltьnьka), and those of *g are not unknown: Zaliznjak (2004: 47) admits that palatalised forms of the Germanic loan кънѧз- <*kъnęg- are the rule, but considers this to be a "supradialectal" word originating outside of the Novgorodian dialect-area; Галинская (2014: 10) is less convinced and adduces the form оусьрѧзи (cf. Russian *серьга*, commonly assumed to be an Oghur, i.e. Bulgar, Turkic loan, cognate with e.g. Kazakh *сырға*) 'earrings' from letter №429 as a word of "вполне бытового характера" which thus supposedly shows a native Novgorodian reflex of PV3 of *g.
More importantly, as Галинская (op. cit.) points out, in all of the well-known Novgorodian forms of the pronoun *vьxъ 'all' which supposedly show a lack of PV3 by retaining both /x/ and back/hard desinences (e.g. fem. gen. sg. въхоѣ <*vьxoję̌ from letter №850), and which come from letters which otherwise correctly convey the jers (by writing <ь,е> for *ь and <о,ъ> for *ъ), the weak-jer is always written with <ъ,о>, unambiguously suggesting a /ъ/ pronunciation. These forms therefore more likely point to a LCS doublet-form *vъxъ which would never contain the conditioning environment for PV3 anyway, and thus you can't use them as evidence of a lack of PV3 in Novgorodian (on the plausibility of such a doublet see Галинская (2014: 14), though cf. Zaliznjak's (2004: 54) less

the evidence), and the blocking of PV2 by an intervening *v in West Slavic (Pol. *gwiazda*, Cz. *květ* <*gvězda, *květъ, etc.).
To be explicit, the native phonemes in my LCS system are given in the tables below:

Table 1: LCS Vowels after the monophthongisation of diphthongs

	Front		Back
High	i		y u
	ŕ̥ ь ĺ̥	y̨	r̥ ъ l̥
Mid	e ę		ǫ o
	ě ę̌		
Low	Ǣ		
		a	

Table 2: LCS consonants before PV2/PV3 (adapted from Winslow 2022: 304)

Labial	Dental	Palatal	Velar
m	n	ń	
b p	t d	ħ ђ	k g
	s z	š ž	x
		č	
	l	ĺ	
	r	ŕ	
v		j	

1.1 Foreign sounds

In addition, the following symbols are used for phonemes of wholly foreign origin in order to represent badly-integrated foreign borrowings, whose level of integration into the native system we deliberately do not take a position on: /ḱ ǵ x́ f ü/, e.g. in respectively к̑итъ <*ḱitъ, □□емонъ <*iǵemonъ, хитонъ <*x́itonъ, иосифъ <*ijosifъ[5], and мѵро <*müro. Almost none of the words containing these symbols would actually have existed in the language during Common Slavonic times, but they need to be included in the indexing-system because they often contain native Slavic elements (f.ex. inflectional endings). Normally they represent specific sounds in the source-language (usually Greek), so including them is useful for investigating the process of these sounds' integration into the native systems. For instance, the extent to which Greek /ü/ is integrated into either native /i/ or /u/ can be seen in variations in the OCS spellings of the word for 'Egypt' (*eǵüpьtъ): □□□□□□ {eǵüpta} vs □□□□□□ {egıpta} vs □□□□□□ {egüptŭ}vs е́гоуптъ vs □□□□□□ {eǵüpta}[6], etc.. One might also ask whether a separate <□> letter for /ǵ/ (and the writing of <к̑>/<□̑> with the palatalisation-diacritic) could be linked to the inadmissibility in the native systems of soft [kʲ, gʲ] sounds, and whether their replacement with regular <□,□> or <г, к> was more likely in systems with some level of native [kʲ, gʲ] (for instance, in Rus' after the so-called Fourth Velar Palatalisation *ky,*gy,*xy > [kʲi, gʲi, xʲi], or in Novgorod due to the retention of native velars before front-vowels because of the non-action of PV2, etc.). In any case, such questions are far easier to investigate if all relevant forms can be reliably retrieved by giving them even a consciously artificial LCS representation.

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convincing explanation of the /ъ/ in these words as an assimilation of original /ь/ to the back-vowels of the following syllable).

5 Of course the sequence /jo/ violates LCS phonotactics as well.

6 Forms are given as they appear in the manuscripts; modern fonts and Unicode symbols mean that the misleading and unhelpful practice of transcribing Glagolitic into Cyrillic is no longer defensible. Latin-based transliterations of Glagolitic forms are given in {curly braces} according to the table in Appendix 1, but it should be emphasised that no transliteration system can adequately convey the internal logic of the Glagolitic, and those who study OCS through transliterations are hamstringing themselves. Where specific forms from Eckhoff's corpus-texts are cited, they are hyperlinked to the corresponding part of the text as it appears on the ocstexts.co.uk website. The Glagolitic texts there are unfortunately still Cyrillicised and should be used with care, particularly ones like Psalterium Sinaiticum where certain ruesome decisions from the editors Severjanov (1922) and Mareš (1997) have been compounded by further information-destroying simplifications (for instance, Severjanov transliterates <□> with <г̑>, but Eckhoff then replaces Severjanov's roundy diacritic with a titlo, leading to nonsense transcriptions like Psalm 8 <анг҃г҃лъ> for ms. <□□□□□>). Cleaned up digitisations, using the Glagolitic originals as the base-text, and a mechanism for displaying manuscript-images, are planned for the near future.

1.2 Vowels

I have deliberately not included accentual information in my reconstruction of vowels, even though such information is in fact required to explain certain differing manuscript-reflexes, e.g. Russkaja Pravda fem. acc. sg. робу < *orb-ǫ vs Uspenskij Sbornik nt. acc. sg. рало < *ordl-o, because for too large a proportion of the vocabulary this information is not sufficiently securely and uncontroversially reconstructed, and anyway the (often post-LCS) derivational processes which are responsible for most of the actual words in the attested texts (and the inevitable accentual levelling processes likely to have occurred in the course of these derivations) complicate things even further.

The two extra nasal-vowels /y̨/ and /ę̌/ are required to account for the split between North (East and West) and South Slavic forms of certain inflectional-endings:

- *y̨ is used for the nom. sg. masc./nt. pres. act. participle of certain verb-classes whose present-stem ends on a hard-consonant, which in South Slavic remains high and backed, e.g. Supr. □ов□ <*zovy̨, Psalterium Sinaiticum □□□□□□□□ {strěgǫï}<*stergy̨-jь, Codex Marianus □□□□ {ědn̨i} <*jǞdy̨-jь (these forms lead Kortlandt (1979:260) to posit that some dialects of early OCS retained some kind of nasal character in this vowel and may even have developed the special "hooked" nasal letter <□> for it), but which in most of North Slavic lowered to /a/: Old Polish (Kazania Świętokrzyskie) has both *reca* and *rec□* (with the special Old Polish letter for the merged reflex of *ę and *ǫ) <*reky̨, and in other texts also *bior□* <*bery̨; Russkaja Pravda река, Uspenskij Sbornik доида <*dojьdy̨, Ru.Ch.Sl. (Vita Methodii) вьсемогаи <*vьxemogy̨-jь. Kortlandt's positing of a CS *y̨ (which he writes as *aN) is far from universally accepted, and others consider these forms the result of various dialect-specific analogical process; see references and discussion in Olander (2015: 88-92). Whatever the truth of the matter, our *y̨ is a convenient placeholder which allows all the relevant evidence to be retrieved.
- *ę̌ is responsible for the NSl. /ě/ vs SSl. /ę/ shapes of jo-stem masc. acc. pl. and the ja-stem nom./acc. pl. and gen. sg. endings, which are reflected in respectively the post- and pre-revolutionary spellings of the Russian nom./acc. pl. long-adjective endings *-ые* < *yjě < *yję̌ vs *-ыя* < *yja < *yję < *yję̌.[7]

The need for the retention of the /Ǣ/ archiphoneme, which represents merged Early Common Slavonic *ē *ā in the position after palatal consonants, up to this point of LCS, is explored in detail in Winslow (2022), but the same archiphoneme (along with its short counterpart /Æ/) was explicitly posited by Kortlandt as far back as 1979 (p.266) as part of his ECS system. In short, a combination of:

1.) the lack of any device in the Glagolitic alphabet to render /ja ńa ŕa ĺa/ sequences (for which Glagolitic texts must use the jat' <□> letter whose base-value is /ě/);

2.) overwhelming spellings of palatal-letter (<□□□□□>) + jat' in the Kiev Folia (the oldest and therefore least distant ms. from the 'original' OCS, as first codified by Cyrill and Methodius and for which the Glagolitic alphabet was devised) for the reflexes of LCS *č,*š,*ž,*ħ + *Ǣ (e.g. □□□□□□□ {oběcělŭ} <*ob-věħǢlъ, □□□□□□ {dušěmı}<*dušǢmi), as well as occasional traces of such spellings in later Glagolitic OCS (e.g. Psal. □□□□ {čěšę} <*čǢšę̌); and

3.) the evidence of certain modern Bulgarian dialects, which have reflexes of LCS *ě in words like *ж'еба* <*žǢba 'toad' (Stojkov 1954: 74–78),

[7] Other troublesome pre-LCS morphological isoglosses reflected in the texts include the masc./nt. instr. sg. *o- and *jo-stem endings *-ъmь/*-ьmь (N.Sl., e.g. KF □□□□□□□□ {obrazŭmĭ}, Uspensk. Sbor. кнѧзьмь) and *-omь/*-emь (S.Sl., e.g. Supr. о҄бра□омъ, кн□□емь), which are most commonly (e.g. Olander 2015: 168) thought to be analogical replacements of the original instr. sg. ending ECS *-ā which is preserved in the adverb *vьčera 'yesterday'; and the *-tь (N.Sl.) vs *-tъ (S.Sl.) verbal endings of 3rd sg. and pl. present (plus its extension to 2nd and 3rd sg. aorists like OCS начѧтъ, OR (Uspensk. Sbor.) б□сть, начать). Here I have no choice but to index them with dummy-symbols in the database: *-Omь/*-Emь for the instr. sg. ending and *-tQ for the verb-endings.

all together point very strongly towards there having occurred a split on the Southeastern periphery of Slavic between LCS dialects which have /ě/ < *Ǣ and the majority of the rest which got /a/, and that original OCS ('Urkirchenslavisch') was an *Ǣ > /ě/ dialect. I posit that *Ǣ remained until the opposition /a/:/ě/ after palatal consonants was reintroduced when PV2 and PV3 brought new soft-consonants /c ś dź/ into the system, which could be followed by both /a/ and /ě/: fem. nom. sg. /stьdźa/ < *stьga (PV3) vs fem. loc. sg. /nodźě/ <*nogě (PV2) (Winslow 2022: 304-305). Thus since my LCS system is based on a point just *before* PV2 and PV3, I must also retain the *Ǣ archiphoneme.[8]

The syllabic liquids /ŕ̥ ĺ̥ r̥ l̥/ are included as unitary vocalic phonemes, following Schenker (1995: 94), rather than as combinations of /ь ъ/ + /ŕ ĺ r l/, because these groups descend from PIE syllabic liquids and many descendant South Slavic dialects which retain syllabic liquids in this position (including most of those underlying canonical OCS) do not show any evidence of an intervening oral-vowel + liquid stage (such a view is shared by Bethin (1998: 71-72); cf. also Bulgarian dialectal evidence in Stojkov (1954: 130-131), where hard consonants precede reflexes of the LCS /ĺ̥ ŕ̥/ even in dialects with secondarily-palatalised consonants before fallen weak LCS /ь/).
The need for both front and back *ŕ̥ *r̥ is unambiguously shown by the East Slavic reflexes /er/ and /or/ (Ru. *смерть*, *морковь*), but *ĺ̥ vs *l̥ is more complicated: PIE *pl̥nos, *wl̥kʷos > Lithuanian *pilnas* 'full', *wilkas* 'wolf' (LCS *pĺ̥nъ, *vĺ̥kъ) vs Lith. *stulpas* (LCS *stl̥pъ 'pillar') suggests that Balto-Slavic had differentiated front/back variants of the PIE syllabic *l̥ (Bethin 1998: 69), but the ancestor to East Slavic backed all vowels preceding tautosyllabic /l/ (Ru. *молоко* < Proto-ESl. *molko < LCS *melko > OCS млѣко), and thus only has /ol/ reflexes here: Ru. *волк*, *столб*, *полный*. It's true that Polish has *wilk* and *milczeć* (<*mĺ̥čǢti), but the Polish reflexes are complicated and likely have more to do with the surrounding consonants: *pĺ̥nъ by contrast gives *pełny* with hardened /l/ and the Polish non-palatalising-/e/ reflex of *ъ, and the differing reflexes in *wierzch* <*vŕ̥xъ, *śmierć* <*sъmŕ̥tь and *martwy* <*mŕ̥tvъjь rule out any explanation based on the nature of the LCS syllabic-liquid alone (for more discussion see Bethin op. cit.: 73-75).
While most OCS shows no sign at all of a front-back distinction in the syllabic-liquids and writes the reflexes of these groups overwhelmingly with <ръ> and <лъ>, the Kiev Folia, which is the only OCS text that reflects a pre-Jer Shift stage and is very nearly flawless in its etymologically correct rendering of the jers, also spells *ŕ̥ *r̥ and *ĺ̥ as one would expect: □□□□□□- {skvrĭn-} □□□□□- {tvrĭd-} □□□□□□- {srĭdĭc-} □□□□- {drĭž-} <*ŕ̥, □□□□□□□□ {skrŭbĭnı} <*r̥, and □□□□□□□□□ {naplĭnenï} <*ĺ̥ (Winslow 2022: 313), and even Zographensis spells all 5 occurences of *vĺ̥k- 'wolf' with □□□□-/□□□□- {vlĭk-/vlĭc-} and all 15 instances of its *-mĺ̥č- root with -□□□□- {-mlĭč-}(e.g. □□□□□□□□ {umlĭčašɴ}). Therefore, taken as a whole the Slavic evidence pretty securely points to front and back variants of both syllabic liquids, and for searching

8 It's possible to argue that the short *Æ counterpart to *Ǣ persisted in East Slavic until after the Fall of the Jers, and that the ESl. so-called e > o shift before hard-consonants / back-vowelled syllables is actually just the resolution of this archiphoneme as /o/ (where palatalisation of the preceding consonant remained, in e.g. Ukr. *бджола* <*bьčÆla, or was newly phonemicised, in e.g. Ru. *вёсла* <*v'Æsla <*vesla), and that there was never a stage when these words had /e/ (based among other things on <o> spellings regardless of stress after palatal-letters in very early texts, and even after the letters for secondarily-soft LCS plain consonants in the Birchbark documents (Le Feuvre 1993, Nakonečnyj 1962), but there isn't space to elaborate on the issue here (see Winslow 2022: 304 fn.16). Unlike the situation with long *Ǣ, OCS shows no sign of anything but an /e/ reflex of short *Æ (and indeed the fact that the East Slavs inherited their writing system ultimately from the Urkirchenslavisch system designed for such a dialect, rather than one which had a clear way of writing /soft consonant/ + /o/, is likely the reason that /o/ reflexes are so rarely detectable in the early texts, since <e> had to be used for both /e/ and /'o/, cf. the spelling євшанъ of the Kipchak word /jovşan/ 'wormwood' in the Hypatian Codex, whose modern cognates (Turkmen *ýowşan* /jowşan/, Kazakh *жусан* /žuwsan/, Azeri *yovşan*) unambiguously point to a Kipchak /o/), and the history of the East Slavic /o/ reflexes remains the subject of much disagreement, so it's simpler for everyone if I continue the traditional practice of writing LCS *e after palatals, even if that strictly speaking is inconsistent with my use of *Ǣ.

purposes it's far preferable to denote them with separate symbols[9] rather than as the sequences /ьr ъr ьl ъl/[10].

1.3 Consonants - Dejotation

Reflexes of the so-called jot-palatalisation are all written either as unitary palatal phonemes, or in the case of jot-palatalised labials as /vĺ mĺ bĺ pĺ/, rather than as sequences of consonant + /j/, hence /ń ĺ ŕ/ for *nj *lj *rj. The 'dejotated' reflexes of *tj (and *kt+front-vowel) and *dj are denoted using the modern Serbian Cyrillic letters /ћ/ and /ђ/ respectively, because the commonly used alternatives, i.e. /ť ď/ (as used in e.g. Olander 2015) or /ḱ ǵ/ (as used in Winslow 2022), or variations thereof, are visually too close to symbols used elsewhere in the system. /ḱ, ǵ/ are anyway already used in my system for foreign /k, g/ before front-vowels, and /ť ď/ look too similar to the common denotations of secondarily-palatalised post-Jer Shift /t' d'/, as used in discussions of systems like Russian or Eastern Bulgarian where they arise.
The compelling hypothesis, first proposed by Durnovo (1929: 55-58) but most recently elaborated by Vermeer (2014: 209-214), and accepted by Mathiesen (2014: 197 fn. 22) and Winslow (2022: 310 fn.25), according to which the Urkirchenslavisch reflexes of *ђ,ћ were close enough to foreign /g k/ before front-vowels that the original Glagolitic system used <□ □> for both sets of sounds (i.e. alongside attested □□□□□□□ {iǵemonŭ}< ἡγεμών would have been **□□□□□□□ {osǫǵeni}<*osǫђeni, and alongside attested □□□□□□ {dŭćerĭ} <*dъђerь would have been **□□□□□□ {ćinŭsŭ} < κῆνσος[11]), does not prevent us from keeping the foreign sounds separate for our LCS stage, since clearly they differed enough in all the dialects underlying actually attested OCS to be written separately.

Pre-dejotation *stj and *zdj are differentiated from the PV1 reflexes of *sk and *zg by writing the former as *šћ and *žђ and the latter as *šč and *žǯ, even though their modern reflexes do not differ from each other anywhere and so must've fallen together in the CS period, because they often alternate with their respective un-palatalised counterparts morphologically and derivationally, e.g. očistiti:očišћenьje vs. jьskati:jьščǫ, jǢzditi:jǢžђǫ vs jьzgъnati:jьžǯenǫ.

There are convincing arguments for PV2/3 having preceded dejotation, at least in more central areas, most recently presented in e.g. Vermeer (2014: 197) and Wandl & Kavitskaya (2023: 244-247), and therefore it could be objected that my system, which contains the dejotation reflexes /ћ ђ ń ĺ ŕ/ but not the PV2/3 reflexes /c ś dź/, is ahistorical. However, it should be reemphasised that the primary goal of my LCS reconstructions is to act as an index which allows reflexes in texts to be found, not to be a historically realistic description of some actually-existing LCS dialect. The absence of PV2 in Novgorodian shows that it can't have preceded dejotation everywhere in Slavic, and in any case the replacement of the sequences /tj dj nj lj rj/ by articulatorily distinct combined

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[9] In the database I will have to use the single Unicode characters <ṛ ṝ ḷ ḹ>, rather than what's shown in Table 1, since the latter cannot actually be rendered without using the letters for /r ŕ l ĺ/ plus the 'combining ring below' U+0325 symbol, which means searches for the consonantal liquids on their own will also return results containing syllabic liquids. A similar problem affects /ę̌ y̌/, which I will have to replace with <ę̃ ỹ>.

[10] To my mind the only evidence in support of a genuine jer + liquid stage comes from the paradigms of verbs like OCS сътрьти < *sъtŕ̥ti, where the syllabic /ŕ̥/ in the stem alternates with /ьr/ depending on the vocality of the following morpheme: the e.g. 3sg. pres. *sъtьretъ (Zogr., Supr. сътьретъ) or (one possibility of the) 3rd sg. aorist *sъtьre (Supr. сътьре) must have /ьre/, while the 3rd pl. aorist *sъtŕ̥šę (Supr. сътрьш□) and the other possibility for the 3rd sg. aorist *sъtŕ̥ (Psal. □□□□□ {sŭtrĭ}, or with a different prefix Mar. □□□□ {otrŭ} <*otŕ̥), being word-final or pre-consonantal, must be syllabic /ŕ̥/. The same alternation occurs in the zero-grade forms of verbs like *umerti, as is clear from the Polish reflexes *umarł* <*umŕ̥lъ vs *umrę* <*umьrǫ. The argument could be that at some stage, before the LCS tendency towards Open Syllables became dominant, the stems in these paradigms were surely unitary /tĭr/, /mĭr/, i.e. 3sg. aor. /sŭ.tĭ.re/ vs 3rd. pl. aor. /sŭ.tĭr.šę/, and that the latter's closed /tĭr/ syllable was only forced to open itself up by changing to /tŕ̥/ because of the Law of Open Syllables. Thus at least one source of the syllabic-liquids could be shown to have developed from a vowel + liquid stage, but that still doesn't prove that they all did, or that the change of /ĭr/ to /ŕ̥/ in these verb-forms was not merely a move to an already-existing syllabic-liquid phoneme.

[11] Interestingly, this aspect of the hypothesised Urksl. orthographic system has rearisen in the modern Macedonian standard due to Turkish loanwords: *ќемер* < Tk. *kemer* 'belt', *ќе* < *[хъ]ћe[тъ]; *ѓон* < Tk. *gön* 'leather', *меѓу* <*meђu.

units, no longer associated by speakers with their /t/ and /j/ phonemes, is structurally completely irrelevant unless and until these new units merge with existing phonemes (or new sequences of dental + /j/ are introduced), as e.g. in the KF dialect where /tj/ merged with /c/ from PV2/3, or in ESl. where it merged with /č/ from PV1. A language which had distinct Czech-like palatal [c, ɟ] reflexes of *tj and *dj, and also no new sequences of [tj, dj], could not convincingly be argued to have undergone dejotation at the phonemic level, as these new units would just be phonetic realisations of /tj, dj/. Analysed like that, the symbols /ħ ђ ń ĺ ŕ/ in my system strictly speaking would really just be cover-symbols for the pre-jotation sequences, but such notation is preferable since it prevents searches for groups containing /j/ alone from returning results polluted by all the dejotation-groups. As explored in Winslow (2022), the status of /j/ as a phoneme in the earliest OCS texts is an intricate problem, so the ability to investigate the reflexes of *j in isolation from the dejotation-reflexes is important.

1.4 Issues involving the glide /j/

1.4.1 Word-initial *jǢ-/*a-

The tendency for ECS *ā- to have taken prothetic /j/ by LCS times (in accordance with the drive towards open syllables) can make it difficult to distinguish this group from *jǢ- in the absence of wider Indo-European evidence. Normally I've followed Derksen (2008), or the ESSJa (*Этимологический словарь славянских языков*, Trubačev 1974-), but for certain lexemes, e.g. *ama 'pit', which in OCS is spelt overwhelmingly with □□- {ěm-} or □м-, the single Greek cognate ἄμη adduced by ESSJa I p.70 in favour of jot-less *am- is not enough to categorically exclude the alternative *jǢma. In particular the 1sg. nom. pronoun *azъ/*jǢzъ is especially problematic: I follow ESSJa I p.100 which ultimately plumps for *azъ, but Derksen doesn't discuss it at all. (A lengthy discussion of the evidence can be found in Teneva's (2012) article on the subject.)

Forms with insecure etymologies can't under any methodology be used as good evidence in phonological investigations, so in difficult cases like the above I simply mark the lemma in the database and provide some short discussion, so that eventually the web-interface can flag such forms in some way and inform users of the specific difficulties.

Like Derksen, I assume that roots going back to PIE jot-less long *ē or dipthongal *oi-, e.g. the root for 'to eat', PIE *h₁ēd-, all took prothetic *j and merged with *jǢ- from other sources, unlike Durnovo (1929: 54), who seems to think that such a development was limited to Bulgarian and Macedonian dialects, including those underlying OCS (where in the Cyrillic mss. we get regular □сти etc.). Isolated nominal forms like Ru. *язва* (which Derksen (2008: 155) derives from a Balto-Slavic *oi- based on Lith. *aiža* and Old Prussian *eyswo*) suggest that *ě reflexes in the modern forms of verbs like Ru. *exать*, Pol. *jeść* are later generalisations from prefixed forms like OR сънѣсти, where no jot-prothesis could take place (cf. Schenker 1995: 88, Winslow 2022: 302 fn.14).

1.4.2 Jers before *j

As explored more fully in Winslow (2022: 313-315), OCS spellings seem to suggest that free-variation between <ь,ъ> and <и,ъi> was a feature of the pre-Jer Shift Urkirchenslavisch orthographic system for conveying the reflexes of the sound-groups *ьj and *ъj (so-called 'tense jers'), regardless of whether they were in strong or weak position. The examples given were: Zogr. □□□□□□□□ {znamenĭě}vs □□□□□□□□ {znameniě} <*znamenьjǢ, □□□□□□ {udarĭī} <*udaŕь-jь vs □□□□□□ {omočiı} <*omočь-jь; Mar. □□□□□□□ □ {osǫdɴtŭ ı} vs □□□□□□□□ □ {osǫdɴtŭı i} <*osǫdętъ ‿jь; KF □□□□□□□□ {milostĭǫ̈} vs □□□□□□□□ {milostiǫ̈} <*milostьjǫ, □□□□□□□□□ {vĭsemogŭı} vs □□□□□□□□□□ {vĭsemogŭiı} <*vьxomogy̆-jь). Of importance here is the fact that the same orthographic system characterises both pre- (i.e. KF) and post-Jer Shift texts; that even in strong position in a text like Zographensis, which shows pretty clear signs of having undergone the Jer Shift, spellings like □□□□□□ {udarĭī}, □□□□□ {boĬĭı}, □□□□□□ {vɴštĭı} for what in the live dialect underlying Zogr. must surely have been /udaŕij/, /boĺij/, /vęštij/, are not infrequent[12]. The fact that the same alternation occurs in the pre-Jer Shift KF (i-stem gen. plurals □□□□□□□□□□ {zapovědi̮ï} <*zapovědьjь vs □□□□□ {lüdi̮ï} <*ĺudьjь) suggests that it is a common inheritance from the Urkirchenslavisch spelling system, and thus that in pre-Jer Shift Slavic the difference between /ь ъ/ and /i y/ was neutralised before /j/, and we should perhaps posit archiphonemes (which I call /Î/ and /Ŷ/) in this position. These archiphonemes are, in slightly different terms, effectively posited by Trubetzkoy (1954: 70) in his analysis of the Urkirchenslavisch phoneme-system[13].

However, for simplicity and accessibility's sake it's better to avoid overburdening the indexing-system with unfamiliar and controversial archiphoneme-symbols, so I keep *ьj/*ъj as the denotations for these groups.

Difficulties arise though when deciding how to denote foreign sources of /ij/[14] which may or may not have been integrated into the native system as reflexes of /Îj/: words like мари□ < Μαρία, стадии < στάδιον, which are well-integrated into the morphological system as a fem. ja-stem and masc. jo-stem respectively, could either be reconstructed as consciously-foreign *marijǢ, *stadijь, or as nativised *marьjǢ, *stadьjь, but there are no occurrences of jer-spellings in these words in the OCS texts in TOROT. Other similarly-Greek words like ди□волъ (< διάβολος), however, do show up in OCS with jer-spellings: Supr. дь□ѓвола, Zogr. Luke 8 and Psal. Psalm 108 □□□□□□□□ {dĭ̮ěvolŭ}, which (alongside the modern Macedonian *ѓавол* with the reflex of *ђ produced by the Macedonian so-called 'new jotation' of /d/ after the fallen jer brought it into contact with /j/) clearly suggest an early adaptation of this foreign /ij/-group to native /Îj/. Old Russian texts even show spellings of мари□ suggestive of full nativisation: Laurentian Primary Chronicle м҃рья, м҃рьею, Zadonshchina маря, марья, as well as First Novgorod Chronicle gen. sg. василь□ (jo-stem василии < Βασίλειος).

Since we can't ever be sure of the precise timing or route by which these late borrowings entered the various Slavic dialects, or of the extent of their adoption by Slavs beyond a tiny and often Greek-knowing scribal-class, the best solution is to set all such foreign /ij/ groups apart from the native vocabulary by using an *ij reconstruction, even where we can be pretty sure that early nativisation to reflexes of *ьj occurred: *dijǢvolъ, *vasilijь, *marijǢ etc.

1.4.3 Word-initial *jь-/*ji-/*i-

With native Slavic word-initial *ji-/*jь-, I follow Derksen's (2008: 16) practice of writing *jь-, even though Derksen himself (2003) has argued for a split between *ji- and *jь- conditioned partly by accentological factors (which, as stated above, I have chosen not to consider). Most of the modern languages reflect these groups as just /i-/, except for Czech and Ukrainian: forms like Cz. *jdou* and Ukr. (after vowels) *йдуть* appear to have dropped the weak-jer in *jьdǫtь just like any other and retained the /j/, and Ukr. *ськати* <*jьskati (with the restricted meaning 'look for nits/fleas in someone's hair' after the base-meaning 'seek' was taken over by the Polonism *шукати*) shows the expected Ukr. softening of the /s/ after fallen weak-jer in *ьsk groups (cf. *польський*).

I make an exception for certain forms of the personal-pronoun *jь, however, and write *jimь, *jima, *jixъ *jimъ and *jimi for the masc/nt. instr. sg. and dat./instr. dual/pl., because Czech here has *jim jich jimi*.

In badly-integrated clearly post-LCS foreign words, such as Biblical names like и□ковъ (borrowed via Gk. *Ἰακώβ*), or □□емонъ (< *ἡγεμών*), I keep a bare initial *i-, though this is rather an arbitrary

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[12] Marianus and Psalterium Sinaiticum, on the other hand, frequently show a Russian-style /ej/ reflex of strong tense *ьj: Psal. □□□□□□ {vrabe̲ı} <*vorbьjь, □□□□□□ {plŭte̲ı} <*plъtьjь, □□□□□ {mĭne̲ı} <*mьńьjь; Mar. □□□□□□□□□ {zapověde̲i} <*zapovědьjь, □□□□□□ {udare̲i} <*udaŕь-jь, □□□□□□□□□□□ {gvozde̲inŭıę} <*gvozdьjьny-ję̌. Psal. (Psalm 21) even has a past act. part. Nsg. masc. def. form □□□□□□□ {strŭgo̲ı} <*jьstrg̥ъ-jь that suggests an /oj/ reflex of *ъjь.

[13] Though Trubetzkoy, like me, believes Urkirchenslavisch to have been based on a /j/-less dialect, so in that particular system the archiphonemes would be conditioned by the position before *vowels*, rather than before /j/.

[14] The sequence /ij/ is not totally banned from native words, since it appears to be preserved across morpheme-boundaries, such as in prefixed-verbs like приѩти <*prijęti or long-form adjectives like masc. nom. pl. дроуѕии <*drugi-ji, but within roots it does seem restricted to these post-LCS loanwords.

choice and done partly as a way of marking such words as non-native[15] (cf. my treatment of foreign initial *e- below). An exception is made for исоусъ < Gk. Ἰησοῦς, which I have as *jisusъ, because of the greater likelihood that Slavs will have heard of Jesus even before the first biblical translations, and because spellings like Zogr. □□□ □□□ {kŭı ı̑sü} suggest that it causes the same /Ŷ/ archiphoneme reflex of *ъ before *j as you get in e.g. native Mar. □□□ □□□□□□□ {vŭı istinǫ} < *vъ ‿ jьstinǫ (see above).
Prefixed forms like *do-jьti 'to come, arrive' for morphological reasons have to be distinguished from the class 4 verb *dojiti/dojiši/dojimъ etc. 'to breastfeed' (and its derived noun *dojidlika), a difference which is reflected in the modern Ukrainian *дійти* (<*dojьti with compensatorily-lengthened /o/ > /i/) vs *доїти.* Thus /i/ can follow /j/ when the former is part of a morpheme which just happens to be stuck onto a /j/-ending stem: I similarly allow words like *šujika (шюица) and *vojinъ 'warrior' (воинъ, as opposed to *vojьnъ, the gen. pl. of *vojьna), or the loc. sg./pl. desinences of any jo-stem noun whose stem ends on /j/, e.g. Psal. □□□□□□□ {žrěbĭı} <*žerbьji.

1.4.4 Word-initial *je-/*e-

No Glagolitic text makes any effort to distinguish /je/ (after vowels or word-initially) from post-consonantal /e/, writing both with <□>, unlike the situation with the reflexes of *ję vs *ę, where in Zogr. and Mar. and partially in Assem. (Velcheva 1981: 168) the full front-nasal digraph <□> is reserved for *ję, while just the second 'nasalising component' <□> is used for post-consonatal *ę, e.g. Mar. 3rd pl. aorist □□□ {ęsɴ} <*jęsę, as opposed to KF □□□□□□ {prıęti} <*prijęti vs □□□□□□ {vŭzęlı} <*vъzęli[16].
Glagolitic evidence alone therefore would suggest that foreign borrowings with word-initial /e-/ were simply adapted to whatever the reflex of native LCS *je was. Suprasliensis, though, which uses the jotated <ѥ> letter, does in fact make an extremely consistent spelling distinction between foreign borrowings and native Slavic words: of the 157 occurrences of the 13 foreign lemmas I have so far reconstructed with word-initial *e/*je- which appear in Supr. (*episkupъ, evanǵelьje, eǵüpьtъ, elisavetь, elinъ, evanǵelistъ, eǵüpьtьskъ, elinьskъ, episkupьstvo, evrejьskъ, elisejь, emъmausъ, etijopьskъ*), the only spellings with <ѥ> are ѥлисеи, ѥп҃пъ, ѥ́лини, and ѥ́лина, i.e. 4/157 or 2.5%. By contrast, of the 3172 native Slavic words in Suprasliensis which I Autoreconstruct as starting with *je- (not all of whose *lemmas* start with *je-, e.g. forms of **byti*), just 88 are written with initial <e>, vs 3070 with <ѥ>[17]. Thus 97.2% of native word-initial *je- in Suprasliensis is spelt with <ѥ>, while 97.5% of the occurrences of the clearly post-LCS Greek-mediated foreign borrowings listed above instead use plain <e>, suggesting that *some* sort of difference was felt, at least by the scribes of Suprasliensis, and that we probably shouldn't index these with the same *je- as used for native forms. I therefore use non-jotated *e- for such foreign borrowings, and the extent to which they take prothetic *j- and fall together with the native vocabulary is left as something for investigators to determine based on the evidence of each manuscript.

1.5 Prefixes and consonant-clusters

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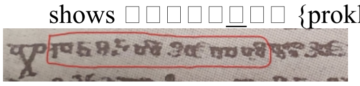
15 Spellings like Zogr. Mark 13:3 "□□□□□. □ □□□□□. □ □□□□□." {petrŭ. ï ěkovŭ. ï oannŭ} "Peter and Jacob and John" would suggest that this initial *i- can get dropped after an /i/ of a preceding word, but whether this points to a dropping of the non-native *i-, simple deletion of a double /i i/ (haplology), or a native-like reflex of a weak-jer /*i *jьjǢkovъ/ > /i jakov/, is not really knowable, so indexing such words with a markedly foreign initial *ij- group is again the best way of allowing such difficult cases to be investigated.

16 Psalterium Sinaiticum contains at least five occurrences of non-digraph <□> for *ę: Eckhoff's digitisation suggests □□□□□ {brěmɴ}, □□□□□□□ {vŭzlačɴ}, □□□□□ {stĭʒɴ}, □□ {tɴ}, □□ {nɴ}, and □□□□□□□□ {proklɴtŭ}, but the last one □□□□□□□□ (which is from Psalm 151 in the part discovered in 1975) comes from a mistake in Mareš's (1997: 50) Cyrillicised edition: the manuscript-photograph in Tarnanidēs (1988: 260) clearly shows □□□□□□□□ {proklętŭ}:



17 The leftover 14 are things like 1st. pres. dual. и́мавѣ which Eckhoff's corpus wrongly lemmatises as имати instead of имѣти, and which thus get reconstructed as *jemĺevě instead of *jьmavě. At the time of writing only 3227/6862 Suprasliensis lemmas have been reconstructed, but those 3227 cover 89713/99194, or 90.4%, of the words.

The last particularity of my LCS indexing-system worth mentioning relates to the handling of consonant-clusters in prefixes: as exhaustively exemplified by Diels (1963: 121-125), Common Slavic permitted only a restricted set of consonant-combinations in the syllable onset, generally either combinations of the continuants *s/*z plus obstruent or sonorant (except *r, see below), or of obstruents plus sonorant (with some curiosities such as the seeming dialectal diversity in the tolerance of *bn but not *pn: OCS гъібнѫти <*gyb-nǫti > Ukr. *гинути,* vs OCS оусънѫти <*usъp-nǫti (cf. 3sg. aor. оусъпе), Ru. *тонуть* <*top-nǫti*,* though see Meillet (1965: 142)). Geminate consonants were banned and either simplified (исѣшти <*jьs-sěkti) or dissimilated (процвисти <*prokvit-ti).

The ban on *sr/*zr is dealt with by insertion of *t and *d respectively, but the commonly-cited examples of *str <*sr (сестра, строу□, остръ) all concern root-internal *sr where insertion of *t is common also to the Germanic and sometimes Baltic cognates. The examples given by Meillet (1965: 136) include: (for строу□) Lith. dial. *srauja* next to Latvian *strauja*, then Germanic *straum- (> Eng. *stream*, Old Norse *straumr* etc.); (for остръ) Lith. *aštrus,* Gk. *ἄκρος* (here the *s is from PIE *ḱ). As Meillet says, "*ce n'est pas un developpement germano-balto-slave; d'une part, le developpement d'un* -t- *dans le groupe* sr *est chose naturelle et se retrouve ailleurs (fr. pop.* castrole *de* casserole*) et, d'autre part, le developpement de* t *en ces conditions n'est pas general en baltique:* str *est regulier en lette, mais* sr *subsiste couramment en lituanien.*", so we can't really be sure when the Slavic change took place or whether it was still active during our LCS stage. The only indication of its activity in OCS is the single Psal. □□□□□□□□ {stramomĭ} <*sorm-omь spelling cited by Diels (1963: 122); otherwise new /sr/ from metathesised *sErC groups is tolerated unchanged.

New occurences of *zr, on the other hand, are regularly generated in the language right up to OCS times, not only in the derivational-morphology because of the verb-prefixes *orz-, *vъz-, *jьz- (e.g. Supr. 3sg. aor. въздроу 'roared', from *vъz-ruti), but also because of the clitic prepositions *jьz and *bez, which form one phonological word with whatever follows them and thus cause OCS spellings like Mar. Luke 1 □□□□□□□ {izdrǫkŭ} <*jь.z ‿rǫ.kъ. Meillet (p.136) also cites the Old Polish adverb *zdręki* <*jьz ‿rǫ.ky, which proves that the phenomenon is not limited to SSl. or OCS. Curiously, though, despite this overwhelming evidence of a synchronic /zr/ > /zdr/ rule in OCS, /zr/ from the metathesised *zork- root is never spelt <здрак> and so seems to be tolerated, even though Diels (p.122) cites prepositional forms like Supr. бе□дра□оума <*bez ‿*orzuma, бездрала[18] <*bez ‿*ordla, which come from metathesised *orT- groups but *do* show inserted /d/. Such inconsistency is hard to explain unless the addition of /d/ has been partly morphologised as a variant of specifically the prepositions before /r/.

With such a sound-change that appears most often at morpheme or straight-up word-boundaries, there is a strong drive to restore the underlying shape of the constituent parts, hence the modern languages have mostly restored /zr/ groups in e.g. Russian *разрешить*, and there are traces of this even in Psalterium Sinaiticum: Psalm 48 □□□□□□□ {izrǫkŭı} (Diels 1963: 122). In Old Russian, the Uspenskij Sbornik is pretty consistent in keeping prefixed verb-forms like раздрушить <*orzrušitь, but by the time of the Laurentian Codex we get forms like възрадуем and неизреченное.

Therefore even though *sr > *str and *zr > *zdr appear to be simply voiced and unvoiced variants of the same sound change, the practical effects are very different because the former is, from the LCS perspective, totally 'opaque', since it only occurs in roots and thus is not analysable by speakers into constituent morphemes without the inserted stop, in the way that /bez ‿drǫky/ can be identified with separate /bez/ and /rǫky/.

For this reason I don't include /zdr/ <*zr at prefix or preposition-boundaries in my LCS system, so that investigators can see for themselves the extent of each text's adherence to the expected phonological development vs restoration of /zr/ under morphological pressure.

Following the same logic I also retain illegal *ss and *sš groups in prefixed-verbs like Psal. □□□□□ {isęče} <*jьs-sęče, Mar. □□□□□ {ïšedŭ} <*jьs-šьdъ, Assemanianus □□□□□□□□̆ {rašırěǫ̈t} <*ors-šiŕǢjǫtъ, because that same drive towards restoration of the underlying shapes of

[18] For some baffling reason this form is missing from Eckhoff's text, so I link instead to the relevant folio of David Birnbaum's online edition.

the prefixes *jьs-/*ors- etc. can be seen in modern Russian spellings *иссякнуть, расширять,* and Laurentian Codex расщедше.[19] This treatment is also more consistent with my handling of verbs like *jьs-kěliti (> OCS ицѣлити/исцѣлити) where simplification *must* have occurred posterior to our pre-PV2 LCS stage (since /sk/ is always totally permissible), and where manuscripts show great diversity, e.g. Zogr. and Assem. consistently have □□□□- {icěl-} while Mar. and Psal. keep □□□□□- {iscěl-} (more discussion of the wider Slavic reflexes of this group, including the OCS <ст> spellings, can be found in Meillet 1965: 133).

1.6 Morphological innovations that scupper LCS reconstruction

The units of the phoneme-system sketched above serve as the building-blocks for all higher-level linguistic systems, most immediately the inflectional-morphology and derivational-morphology systems, whose features are thus constrained by said phoneme-system and the distributional-restrictions of its units (i.e. *phonotactics*). Changes which occur in the phoneme-system between the time of our theoretical LCS and the time of our texts can therefore trigger (or allow) restructuring of these morphological systems, which in turn can produce forms containing phoneme-sequences with no possible direct LCS ancestor-sequences.

An example of such morphological change contingent upon structural phonological change, leading to forms which preclude any direct LCS-stage reconstruction, is the replacement of i-stem endings with those of the corresponding jo- or jā-stems, in nouns whose stems end on labials or the subset of LCS dental consonants which lack palatal counterparts, viz. /d t s z/[20]. Evidence for such a change is furnished by the Old Russian masc gen./acc. form татѧ from the 1229 Treaty between Smolensk, Riga and Gotland (Version A). LCS *tatь is a masc. i-stem noun with genitive *tati, as it still appears in the Suprasliensis translation of John Chrysostom's Homily for Holy Thursday (...то кажетъ влад□к□ чловѣколюбьѥ҇· □҆ко прѣданника ра□боиника тати...), but in the dialect underlying the 1229 Treaty the rise of phonemically palatalised /t'/ after the Jer Shift means that the stem (and the nom. sg. тать /tat'/) of this noun now ends on the same class of "soft" consonants as original jo-stem nouns like *końь > /kon'/, where the original LCS palatal *ń has fallen together with secondarily-palatalised /n'/ from plain LCS *n before LCS front-vowels, in e.g. the original i-stem *bornь > /boron'/. This system thus no longer distinguishes between descendants of the original LCS palatals and the newly secondarily-palatalised consonants like /t'/: both are now together in the set of 'soft' consonants, opposed to their 'plain' or 'hard' counterparts, and so tend towards taking the same set of inflectional endings (in this case those of the original jo-stems)[21]. Consequently, a word like тать has begun to take jo-stem endings, including the Old Russian /a/ reflex of LCS *Ǣ in the genitive/accusative singular.

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[19] Conversely, sequences of *sk, *zg at prefix-boundaries which show PV1 reflexes, like Mar., Zogr. □□□□□□□□□□ {raštĭtetŭ} <*orš-čьtetъ (ECS *skĭt- > *ščĭt-), Psal. □□□□□□□□□□□ {raždiʒaetŭ} <*orž-ǯigajetъ (ECS *zgīg- > *žǯīg-) are kept as *šč, *žǯ. Such forms may well not go all the way back to the time of PV1, and instead be just the result of a synchronic rule prohibiting /zž/ and /sč/ (> /žǯ/ and /šč/) that remained active until much more recently, especially given prepositional-phrase forms like Psal. □□□□□□□ {ıštrěva} <*jьs ‿*červa, so this is arguably inconsistent with my treatment of *ss, *sš etc. My justification is firstly that *sk, *zg > *šč, *žǯ are *conspicuously* PV1-changes, which we *know* originated well before our target LCS point, whereas the precise timing of de-gemination or simplification of *sš is less clear-cut; and secondly that even in languages like Russian which *orthographically* have restored <сч> and <зж> spellings in compounds like *исчезнуть* and *разжечь*, the pronunciations are still arguably direct reflexes of LCS *šč and *žǯ, viz. [ɕː] and [ʑː] (or, in the conservative Moscow-dialect, the palatalised [ʐː] found also in *дождь* <*dъžǯь).

[20] In some dialects (notably East Slavic) the PV3 reflexes *ś and *ʒ became patalised counterparts to plain /s z/, i.e. /s' z'/, and merged with the /s' z'/ that developed from LCS *s,z before front-vowels, but in most OCS they seem to have just hardened to /s, z/: searching my database for the sequence *ьxǫ, for example, turns up exclusively <□□> spellings in Marianus, just one <□□> in Zogr., and exclusively <сѫ> in Suprasliensis, with only Assem. and Psal. containing a significant minority of <□□> spellings.

[21] Russian feminine i-stems like *весь* (<*vьsь, 'village') do not fall together with ja-stems in the way the masculines like *zvěrь fall together with jo-stems, but they do all still take the /am, ax, ami/ endings in the dat., loc. and instr. pl., e.g. *вéсям*, which contain the same LCS ja-stem *-Ǣ- vocalism which can only occur after LCS palatals, meaning they too end up totally unreconstructable due to an illegal **sǢ sequence.

LCS /Ǣ/, though, by definition can only occur after LCS palatal consonants (see above), so a reconstruction *tatǢ is just nonsensical. In the case of the dat. sg. /u/-desinence (which isn't attested in our Treaty but exists in modern Russian *татю*), we don't even have an LCS archiphoneme available to signal a preceding soft-consonant; there's simply no way of getting from LCS *tatu to Russian /tat'u/, because such a form was only made possible by the rise of phonemic /t'/, so our ability to index it with our LCS system is gone.

Were the same shift from i-stem to jo-stem to occur in a word like *zvěrь, then the structural change would not be so catastrophic, because our LCS system *does* contain a palatal *ŕ which any allophonically-softened LCS hard *r could easily be subsumed into. Indeed, interestingly Suprasliensis does in fact contain 3X gen. sg. звѣрѣ, with what looks like a jo-stem reflex of *ŕǢ (spelt with jat' as an overhang of the Glagolitic tradition, cf. 2X морѣ vs 1X мор□ spellings), suggesting that Russian-style secondary palatalisation of *r > /r'/ may have occured in the Bulgarian dialect underlying it[22]. You don't, though, get anything like татѧ[23], so the system-wide development of secondary-palatalisation does not seem to have advanced enough to have caused the sort of fundamental structural reorganising which shifted *tatь into the jo-stems in Old Russian.

Forms like татѧ, then, though they frustrate our goal of reconstructing entire texts, do provide us some objective measure of 'linguistic distance' between stages of a language, because their existence presupposes at least one intervening stage where the structure of the phonological system has changed enough from our LCS stage to have caused/allowed restructuring of the morphological system.

2 LCS Morphology and the Autoreconstructor

2.1 The existing TOROT data

The ten-place morphology-tags included as part of the word-level annotations in Eckhoff's TOROT corpus constitute a veritable goldmine of linguistic data, because, based as they are on the *form* of a word rather than the *function*, they bridge the gap between the higher (syntax, semantics etc.) and lower (phonology, orthography, morphology) levels of linguistics analysis. An example TOROT annotation for the word □□□□□□□□□ {vŭzvěštǫ} is given below:

```
<token id="3589172" form="възвѣштѫ" citation-part="70.17"
lemma="възвѣстити" part-of-speech="V-" morphology="1spia----i"
relation="pred" presentation-after=" "/>
```

Figure 1: TOROT annotation for Psal. Sin. Psalm 70 □□□□□□□□□ in XML format

TOROT token XML-tags include various attributes, but for the Autoreconstructor all that's needed are *form*, *lemma*, *part-of-speech*, and *morphology*. The *form* attribute is used to check for morphological variations/innovations, to ensure that what gets produced is the direct phonological ancestor of the actually-occurring word (see below for more about this deviance-detection). The *lemma* and *part-of-speech* attributes, when concatenated, serve as a unique key linking each word to

[22] Numerous spellings in Supr. like боура <*buŕǢ, ȣ̑карають <*ukaŕǢjetь, мороу <*moŕu etc., however, point to a hardening of LCS palatal *ŕ to plain /r/, so it's difficult to know whether звѣрѣ spellings stem from a genuine /zvěŕa/ form in the history of the language, or if they instead represent synchronic /zvěra/, i.e. with a hard o-stem ending, but with confusion by the scribe between <pa> and <p□>/<pѣ> spellings for what in his/her dialect would've all been /ra/.

[23] Except the numerous gen. sg. господа and dat. sg. господоу for the i-stem *gospodь, but this word seems to be an isolated special case, because it bafflingly turns up even in early Glagolitic OCS with endings like □□□ {g̃vi} (ju-stem dat. sg. *-evi, cf. Supr. господеви), □□ {g̃ü} (jo-stem dat. sg. *-u), and □□ {g̃ě} (jo-stem gen. sg. *-Ǣ). See Van Wijk (1929).

its lemma[24] and thus to its LCS reconstruction and inflection-class information[25]. Finally the *morphology* attribute consists of a 10-character string to hold values for the 10 morphological-features used by TOROT (and the wider PROIEL corpora). Not all features are relevant for all words, in which case a dash ‘-’ is used as a placeholder.

A detailed explanation of each feature can be found in Section 6 of Eckhoff et al. (2018: 41), but here it suffices to say that in this example the tag “1spia----i” is telling us that □□□□□□□□ {vŭzvěštǫ} is **1**st person, **s**ingular, **p**resent-tense, **i**ndicative-mood, **a**ctive-voice, has no gender, case, degree, or strength features, and is **i**nflectable rather than non-inflecting.

Of importance here is the *present* tense tagging, even though възвѣстити (even in OCS) can be taken as a perfective verb, opposed to its imperfective counterpart възвѣщати, and thus has future-tense meaning in its non-past indicative forms (and the Greek Septuagint here has ***ἀναγγελῶ τὰ*** *θαυμάσιά σου*, with a morphologically future-formed *ἀναγγελῶ* ‘I will proclaim’ from *ἀναγγέλλω*). The tagging thus follows the *inflectional*-morphology of □□□□□□□□ {vŭzvěštǫ}, rather than the future-meaning which is carried by the *derivational*-morphology. This is important because the Autoreconstructor works by *inflecting* LCS lemmas according to those morphology-tags; the annotation gives no information about its derivational-morphology or derivational relationship to its imperfective (and thus 1sg. *present* tense) counterpart □□□□□□□□□ {vŭzvěštaǫ̈}, since that is annotated with a separate lemma.

An even clearer example where this *form* over *function* annotation helps us is with the accusatives of animate nouns: it’s well known that a type of so-called differential-object-marking is beginning to take hold in Early Slavic, whereby syntactic accusatives use genitive endings to varying extents as a way of encoding the semantic ensouledness (and possibly definiteness) of the noun (the details aren’t important; for a recent thorough diachronic treatment of the topic see Eckhoff 2022), e.g. Supr. ра□бои́ника въ породѫ вьвͨеде ‘he brought the robber into paradise’. If these were all tagged as accusatives, the Autoreconstructor would produce the wrong ancestor to the text-form, i.e. *orzbojьnik-ъ, because the semantic information needed to decide whether this differential-object-marking is needed is not available. Happily though all such animate-accusatives are marked as genitive in TOROT: the morphology-tag for ра□бои́ника above is “-s---m**g**--i”, so the Autoreconstructor produces *orzbojьnik-a.

2.2 Computerised LCS inflectional-morphology and the Autoreconstructor

The Autoreconstructor reads the morphology-tag character-by-character and numerifies each field, and from that it computes a number which corresponds to the row of the inflection-table where the endings for that particular word’s inflection-class are stored. For verbs this will be a number between 1 and 44 (9 person/number combinations for each of present, aorist, imperfect, and imperative, plus 8 non-finite forms, 9*4+8), and for nominals between 1 and 63 (7 cases * 3 numbers * 3 genders). A version of the function which actually implements this tag-reading process can be seen in the OcsServer::numerifyMorphTag() function here: https://github.com/12401453/ocs_server/blob/main/OcsServer.cpp#L2714.[26]

Currently each inflection-class has up to three tables associated with it, the first of which is full (i.e. contains 44 or 63 entries) and holds the ‘basic’ or ‘correct’ (from the LCS perspective) endings. The second and third tables are ‘sparse’, in that they only contain entries for those parts of the paradigm where we expect to encounter alternative forms, with those I consider ‘deviant’ or ‘innovated’ held

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[24] Identical lemmas with the *same* part-of-speech tag, such as вести ‘to lead’ <*ved-ti and вести ‘to drive’ <*vez-ti, both of which have ‘V-’ for verb, are differentiated by appending #2 etc. to the extra homomorphs, i.e. вести vs. вести#2.

[25] The spreadsheet containing my reconstructions and inflection-class annotations for the TOROT OCS lemmas can be downloaded from https://github.com/12401453/torot_2023/blob/main/lemma_lists/chu_lemmas_master.xlsx

[26] Extra handling is required for forms of *byti, since that has a separate future-paradigm (and future-participle) which TOROT does actually specify separately with an ‘f’ value for the *tense* feature, as well as two variant imperfect sets (3sg. *bě vs. *běaše, which aren’t tagged, but which I detect), and a ‘conditional’ *bimь, *bi, *bǫ etc. Participles require an extra step to read their nominal-features and add their adjective-like endings.

in table 2, and ‘alternative’ but still ‘correct’ endings (i.e. LCS allomorphs) in table 3. An example of some of the endings in the three tables for basic class 1 verbs like *reħi is given below:

```
inner_map v_11_c0 = {
{1, "ǫ"},
{2, "eši"},
{3, "etQ"},
{4, "evě"},
{5, "eta"},
{6, "ete"},
{7, "emъ"},
{8, "ete"},
{9, "ǫtQ"},
{10, "ъ"},
{11, "e"},
{12, "e"},
{13, "ově"},
{14, "eta"},
{15, "ete"},
{16, "omъ"},
{17, "ete"},
{18, "ǫ"},
```

```
inner_map v_11_c1 = {
{6, "eta"},
{10, "oxъ"},
{13, "oxově"},
{14, "osta"},
{15, "oste"},
{16, "oxomъ"},
{17, "oste"},
{18, "ošę"},
{24, "ěašeta"},
{37, "ǫtj"},
};
```

```
inner_map v_11_c2 = {
{10, "sъ"},
{13, "sově"},
{14, "sta"},
{15, "ste"},
{16, "somъ"},
{17, "ste"},
{18, "sę"},
};
```

*Figure 2: Inflection-endings for class 1 verbs like *reħi, *greti stored using C++ std::unordered_map<int, std::string>*

Here the ‘deviant’ endings consist mostly of the ‘extended S-’, or *-ox- aorists (e.g. Assem. □□□□□□□□ {narekošɴ}), while the third ‘alternative’ table contains the primary S-aorist endings (e.g. Assem. □□□□□□□ {pogrěsɴ} <*pogrěb-sę). Clearly those primary S-aorist endings just being stuck onto a stem like *greb- would not produce the correct LCS form, so for certain inflection-classes there are post-processing functions which do things like replace all “ebsę” with “ěsę”, or (for stems which end on RUKI-consonants like *rek-) all “eks” with “ěx”. The full routines for verbs can be found in https://github.com/12401453/ocs_server/blob/main/autoreconstructor/verb_cleaning.h; to a large extent this is just enforcing the Slavic consonant-cluster restrictions discussed in Section 1.5.

The inflection-tables themselves are indexed by integer-keys associated with each inflection-class, such that the ‘correct’ table’s key always ends on ‘1’, meaning that shifting to the alternative tables is a matter of simply incrementing the key by either 1 or 2, and I can keep track of whether or not a form has required a deviant or alternative ending by inspecting the final value of this key[27]:

```
std::unordered_map<int, inner_map> verb_ = {

  {111, v_11_c0},
  {112, v_11_c1},
  {113, v_11_c2},
```

Figure 3: Integer-keys of the three inflection-tables in Figure 2

Multiple inflection-classes can share the same tables, for instance the *masc_o*, *nt_o*, and *fem_a* noun-classes all take endings from the *adj_hard* table, since the same set of 63 endings (21 for each gender) is used in all four paradigms[28].

---

[27] This is bad design and prevents me from easily handling more than one type of morphological innovation per class: for the *masc_jo* class nom. pl. I have the i-stem *-ьje ending in the ‘deviances’ table (as found in e.g. Supr. стражиѥ <*storž-ьje ‘guards’), which leaves no room for forms like Supr. зноѥве <*znoj-eve, Psal. □□□□□□ {zmıeve} <*zmьj-eve, which have the *-eve ending from the ju-stems.

[28] The long-form adjectives (and participles) are then formed by just sticking the inflected-form of the pronoun *jь onto the end, since this is the origin of such long-forms (Vaillant 1942). I agree with Townsend and Janda (1996: 178) that some simplification of these concatenated endings must have occurred by LCS, especially where the short-form endings contain *m or *x (e.g. the dat. and loc. pl. for all genders), because it’s unreasonable that e.g. Mar. Matt. 24 masc. loc. pl. □□□□□□□□□□□□□ {nebesŭskŭixŭ} could've developed directly from

Full paradigms for the 4,500-ish OCS lemmas I have so far reconstructed can be dynamically generated here https://ocstexts.co.uk/words; these are constructed from LCS lemmas in the same way the Autoreconstructor reconstructs individual forms, except it produces every form in the paradigm, rather than just the one specified by a text-word's morphology-tag. The LCS forms are then converted into 'normalised' OCS by the browser using the convertToOCS() function here: https://github.com/12401453/ocs_server/blob/main/HTML_DOCS/LCS_to_OCS.js#L197. Forms generated from the "deviant" and "alternative" tables are included, but the former are given less prominent styling to identify them as (probable) post-LCS innovations.

	Present	Aorist	Imperfect	Imperative
1st sg.	сътьрѫ	сътрьхъ	сътьрѣахъ	сътьрѣмь
2nd sg.	сътьреши	сътьре сътрь	сътьрѣаше	сътьри
3rd sg.	сътьретъ	сътьре сътрь	сътьрѣаше	сътьри
1st du.	сътьревѣ	сътрьховѣ	сътьрѣаховѣ	сътьрѣвѣ
2nd du.	сътьрета	сътрьста	сътьрѣашета	сътьрѣта
3rd du.	сътьрете сътьрета	сътрьсте сътрьста	сътьрѣашете сътьрѣашета	сътьрѣте
1st pl.	сътьремъ	сътрьхомъ	сътьрѣахомъ	сътьрѣмъ
2nd pl.	сътьрете	сътрьсте	сътьрѣашете	сътьрѣте
3rd pl.	сътьрѫтъ	сътьрѫ сътрьшѧ	сътьрѣахѫ	сътьрѫ

Participles	
PRAP1	сътьрꙑ
PRAP2	сътьрѫщь
PAP	сътьръ
L-Part.	сътрьлъ
PPP	сътрьтъ сътьренъ
PrPP	сътьромъ
Infinitive	сътрьти
Supine	сътрьтъ

Figure 4: Dynamically-generated paradigm for the OCS verb сътрьти, *based on the Autoreconstructor's computerised LCS inflectional-morphology*

The accuracy of the generated-forms can then be gauged by comparing them to tables populated only by forms which actually occur in Eckhoff's corpus-texts, using the 'Corpus-forms' switch:

---

*nebesьskěxъjixъ (here not least because of the lack of PV2-reflex). I am however relucant to adopt the (undiscussed) simplified LCS forms they present in the tables on p. 182-3 before I've done a more thorough investigation of early manuscript-forms, so for now I've left all such adjectivals wholly uncontracted (though the Autoreconstructor does actually mark long-adjectivals that contain such problematic concatenations and this information could be used to exclude them from searches).

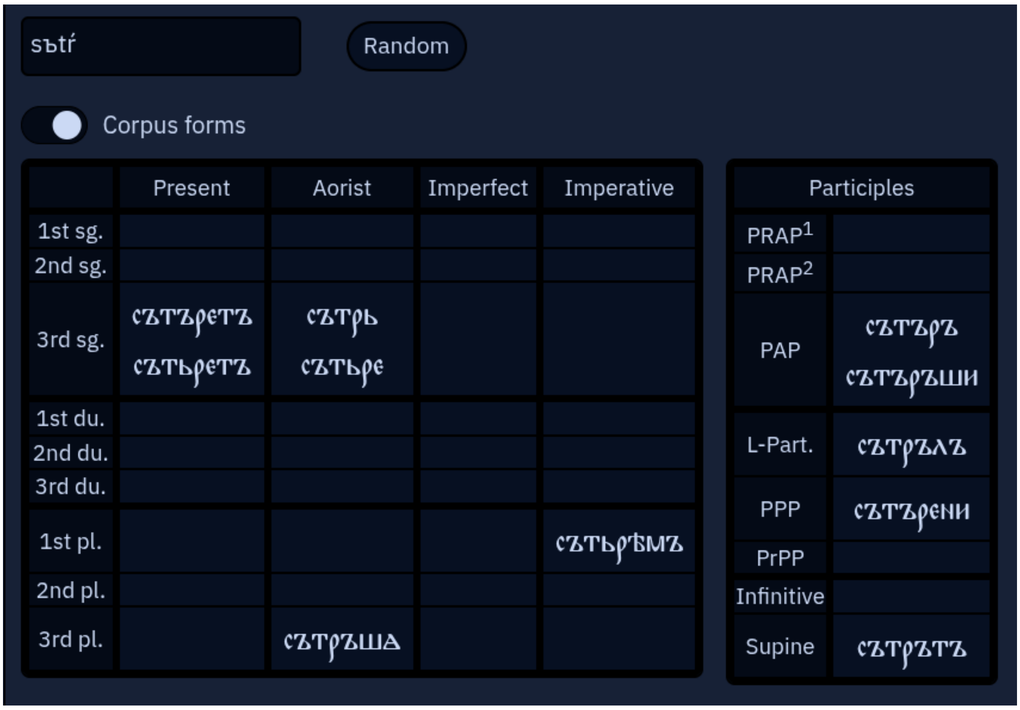


Figure 5: Forms of the same сътрьти *verb as they actually occur in Eckhoff's (Cyrillicised) TOROT corpus of Church Slavic texts*

An advantage of basing computer-generated paradigms on converted LCS (rather than directly on normalised OCS) is that it allows for the possibility of dialect or manuscript-specific conversions: e.g. Kiev Folia-flavoured paradigms that convert *ћ, *ђ to <ц, з> and use <ѣ> for *all* *Ǣ, or a Marianus-specific conversion that frequently shows vocalised strong-jers, or even an artificial pre-Jer Shift form of Old Russian that shows the denasalisation of *ę and its merger with *Ǣ via stochastic mixing up of <□, a> and <ѧ>:

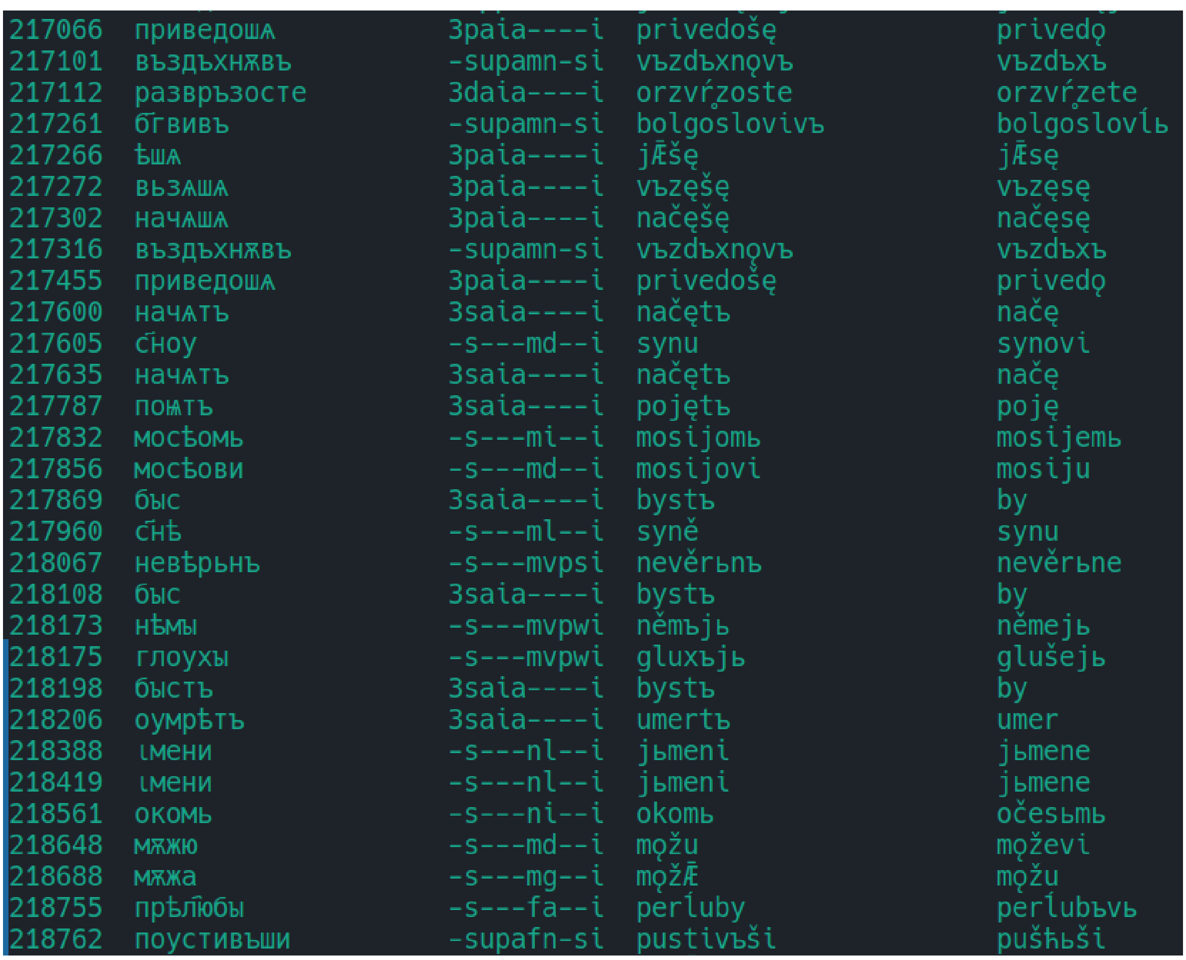
	Sing.	Dual	Plural
Nom.	кънѩзь кънѧгъ	кънѧзѩ	кънѧзи
Acc.	кънѧзь кънѧгъ	кънѧзѧ	кънѩзѣ кънѧгꙑ
Gen.	кънѧзѩ	кънѧзю	кънѩзь кънѩгъ
Dat.	кънѩзю	кънѩзема кънѩгома	кънѧземъ кънѧгомъ
Loc.	кънѩзи кънѩзѣ	кънѧзю	кънѩзихъ кънѧзѣхъ
Instr.	кънѧзьмь кънѩгъмь	кънѧзема кънѧгома	кънѩзи кънѧгꙑ
Voc.	кънѩже	кънѩзѧ	кънѩзи

*Figure 6: Dynamically-generated paradigm of *kъnęgь but with an Old Russian rather than OCS conversion*

	Sing.	Dual	Plural
Nom.	кънѧѕь кънѧгъ	кънѧѕа	кънѧѕи
Acc.	кънѧѕь кънѧгъ	кънѧѕа	кънѧѕѧ кънѧгꙑ
Gen.	кънѧѕа	кънѧѕоу	кънѧѕь кънѧгъ
Dat.	кънѧѕоу	кънѧѕема кънѧгома	кънѧѕемъ кънѧгомъ
Loc.	кънѧѕи кънѧѕѣ	кънѧѕоу	кънѧѕихъ кънѧѕѣхъ
Instr.	кънѧѕемь кънѧгомь	кънѧѕема кънѧгома	кънѧѕи кънѧгꙑ
Voc.	кънѧже	кънѧѕа	кънѧѕи

*Figure 7: *kъnęgь with the canonical OCS conversion for comparison*

2.3 Detecting and dealing with morphological innovations



217066	приведошѧ	3paia----i	privedošę	privedǫ
217101	въздъхнѫвъ	-supamn-si	vъzdъxnǫvъ	vъzdъxъ
217112	развръзосте	3daia----i	orzvŕ̥zoste	orzvŕ̥zete
217261	б҃гвивъ	-supamn-si	bolgoslovivъ	bolgoslovĺь
217266	ѣшѧ	3paia----i	jǢšę	jǢsę
217272	вьзѧшѧ	3paia----i	vъzęšę	vъzęsę
217302	начѧшѧ	3paia----i	načęšę	načęsę
217316	въздъхнѫвъ	-supamn-si	vъzdъxnǫvъ	vъzdъxъ
217455	приведошѧ	3paia----i	privedošę	privedǫ
217600	начѧтъ	3saia----i	načętъ	nače̜
217605	с҃ноу	-s---md--i	synu	synovi
217635	начѧтъ	3saia----i	načętъ	nače̜
217787	поѩтъ	3saia----i	pojętъ	poję
217832	мосѣомь	-s---mi--i	mosijomь	mosijemь
217856	мосѣови	-s---md--i	mosijovi	mosiju
217869	быс	3saia----i	bystъ	by
217960	с҃нѣ	-s---ml--i	syně	synu
218067	невѣрьнъ	-s---mvpsi	nevěrьnъ	nevěrьne
218108	быс	3saia----i	bystъ	by
218173	нѣмы	-s---mvpwi	němъjь	němejь
218175	глоухы	-s---mvpwi	gluxъjь	glušejь
218198	быстъ	3saia----i	bystъ	by
218206	оумрѣтъ	3saia----i	umertъ	umer
218388	ıмени	-s---nl--i	jьmeni	jьmene
218419	ıмени	-s---nl--i	jьmeni	jьmene
218561	окомь	-s---ni--i	okomь	očesьmь
218648	мѫжю	-s---md--i	mǫžu	mǫževi
218688	мѫжа	-s---mg--i	mǫžǢ	mǫžu
218755	прѣл҃юбы	-s---fa--i	perĺuby	perĺubъvь
218762	поустивъши	-supafn-si	pustivъši	pušħьši

Figure 8: Auto-detected and -reconstructed morphological deviances from a small part of the Book of Mark in Codex Zographensis

The screenshot above shows some raw data from my autoreconstructed SQLite database of the TOROT OCS texts; in this case it's forms from Zographensis (around Mark 7 to Mark 10) where the Autoreconstructor has detected morphological innovations. The fourth column shows what the Autoreconstructor thinks is the direct phonological ancestor to the text-form, but the ancestor of the 'original', 'correct', or 'default' morphological form is also generated and stored in the fifth column, so that such cases of innovation can be easily searched-for and counted (since non-innovated forms have NULL values in this column).

Types of innovation detected here include:

- extended S-aorists of class 1 verbs: 3rd pl. приведошѧ vs. приведѫ, 3rd dual развръзосте vs *разврьзете[29]
- unetymological extension of the RUKI-rule-produced *š in 3rd pl. primary sigmatic aorists: ѣшѧ vs. ѣсѧ <*jǢd-s-ę, вьзѧшѧ vs. възѧсѧ <*vъzьm-s-ę, начѧшѧ vs. начѧсѧ <*načьn-s-ę (neither *d, *m, nor *n have ever been RUKI sounds)
- extension of the *-nǫ- suffix to the past. act. part. of class 2 verbs like въздъхнѫти: въздъхнѫвъ (cf. Mar. □□□□□□□ {usŭxŭšǫ} from оусъхнѫти)
- addition of the *-tQ suffix from the 3rd sg. pres. (where Q stands for NSl. *ь vs SSl. *ъ, see fn. 5 above) to 3rd sg. aorist forms: начѧтъ, поѩтъ, оумрѣтъ
- original u/ju-stem nouns taking o/jo-stem endings: dat. sg. с҃ноу, мѫжю; loc. sg. с҃нѣ, gen. sg. мѫжа
- past act. part. of class 4 verbs using the suffix *-ivъ rather than *-jь: б҃гвивъ, поустивъши (cf. Mar. Mark 10 □□□□□□□ {puštŭši} <*pust-jьši)

Deciding upon the “correct” morphological endings for an unattested language inevitably entails some uncertainty and controversy: for instance, *-ox- aorists occur in both OCS and Old Russian, so why do I consider them LCS deviances? Basically because they **never**[30] occur in Marianus, which would be quite improbable if they were a discarded archaism (especially given their ubiquity in the closely related Zographensis). Additionally, West Slavic uses *-ex- here (e.g. Old Czech 1sg. *nesech* <*nes-ex-ъ), suggesting that the extended S-aorists were innovated independently in the post-LCS dialects (Meillet 1965: 256).
In other cases we are dealing with hodge-podge paradigms which are only ever attested with endings from multiple older classes, and sometimes dialectal or orthographic features of the manuscripts can make it difficult to distinguish potential LCS ancestors of certain endings. For instance, I have a *masc_tel* class used for agent-nouns like OCS дѣлател҄ь (i.e. Diels 1963: 166), which in the sg. and dual. behave exactly like masc jo-stems, but which in the gen. and instr. plural are attested also with basically consonant-stem endings on a hard *-tel- stem, e.g. Zogr. □□□□□□□□ {tɴžatelŭ} <*tężǢtelъ, Supr. св□тител□ <*svętitely. The nom. pl. appears also to take a consonant-stem *-e ending, but as Diels (op. cit.) points out, the spellings in Zogr. and Supr. (where use of the palatalisation-diacritic <˜> to denote LCS *ĺ is consistent enough to suggest a real phonemic /ĺ/ in the underlying dialects) like мѫчител҄е[31] mean we can’t follow Meillet (1965: 426) in setting up *-tele as the ‘correct’ ending, because spellings like □□□□□□ {žɴtele} in Mar. could just as easily descend from *žętele as from *žętele. Absent a manuscript which both consistently marks *ĺ *and* doesn’t use such a mark in this nom. pl. desinence, there’s no hard evidence of this *-tele ending ever actually existing. I thus use *-tele in the ‘correct’ table, and the jo-stem *-teĺi in the ‘deviances’ table[32].

2.4 Overcoming poor lemmatisation practice

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29 Koch (1990: 293) lists only sigmatic aorists as possibilities for the *-verz-/*-vŕz- stem verbs, and it seems that outside of the 3rd sg. (e.g. Psal. □□□□□□□□ {razvrŭze}, Zogr. □□□□□□□ {otvrĭze}) no root-aorists are attested in any Slavic text, so maybe I am wrong to set up asigmatic root-aorists like 3rd dual *-vŕzete as a possibility alongisde primary sigmatic *-verste (e.g. Mar. Mark 7 □□□□□□□□□ {razvrěste}). My justification is that the *-verg-/*-vŕg- stem verbs *do* attest such root-aorists, e.g. 3rd pl. □□□□□□□□ {otŭvrŭgǫ} in Psal. Psalm 77, and I don’t see what, apart from the nature of the final stem-consonant (obstruent vs. continuant), could be grounds for classifying these two verbs differently.

30 Having autoreconstructed all of Marianus I can verify this (admittedly already well-established) fact far more quickly and easily than was possible just with Eckhoff’s morphology and part-of-speech annotation-information: using TOROT the smallest net you could cast would be one that caught all aorist-tense verbs, whereas I can just search for reflexes of *oxъ, *oxov, *oxom, *ošę, *osta$, and *oste$ (the latter two using ‘regular-expression’ mode and $ to specify end-of-word), which for Mar. turns up nothing except the three occurrences of □□□□ {osta}.

31 Searching my database for *teĺe$ returns only a single властеле in Supr. without the diacritic; everything in Zogr. has it.

32 Diels mentions only Psal. Psalm 26 □□□□□□□□□□ {sŭvěděteli}, but the Autoreconstructor is intended for use with other early texts beyond canonical OCS which might also contain this type of assimilation to the jo-stems.

Sometimes Eckhoff's lemmatisation is too coarse-grained, in that forms which clearly descend from distinct doublets are subsumed under one lemma. To demonstrate just one example of how the Autoreconstructor deals with this sort of problem, take the numeral ѥдинъ <*jedinъ, which in the earliest OCS has straightforward hard pronominal endings and which therefore goes in my *pron_hard* class alongside demonstratives like *onъ. There is, however, what must be viewed as an LCS doublet *jedьnъ[33], which gives e.g. Serbian *jedan* and the modern Russian fem. nt. and oblique-case forms *одна, одного* etc., and which is used for the majority of non masc. nom./acc. sg. forms of the pronoun in Suprasliensis[34], e.g. ѥд’номоу. Notwithstanding Eckhoff's habit of lemmatising these with ѥдинъ instead of ѥдьнъ, it's trivial for the Autoreconstructor to check the form (which has in a previous step already been aggressively normalised to get rid of morphologically-irrelevant variation) for a <дін> sequence, which would never occur in the inflectional-ending and so always point to a *jedin-descended stem, and then replace our autoreconstruction with *jedьn- if such isn't found:

```
 // check for *jedьn-
if (lemma_ref.lemma_id == compileTimeHashString("Рхѥдинъ") || lemma_ref.lemma_id == compileTimeHashString("Маѥдинъ"))
{
  if (Sniff(cyr_id, "дін", 20) == false)
  {
    stem = "jedьn";
  }
}
```

*Figure 9: Part of the Autoreconstructor's code which checks for reflexes of *jedьn- that TOROT has mistakenly lemmatised under* ѥдинъ

In the long-term TOROT itself should fix such lemmatisation problems, but in the meantime checks like the above are computationally extremely cheap and prevent great swathes of the texts from being wrongly autoreconstructed.

3. Benefits and potential for future studies

We have already seen a few examples of minor questions where recourse to an LCS reconstructed layer makes exhaustive investigation of texts trivial; e.g. the reality of a difference between word-intial /je-/ and (foreign) /e-/ in the Codex Suprasliensis was shown by demonstrating statistically significant discrimination in the use of the <ѥ> letter (p.9-10). Another example, as hinted at in fn. 20, concerns the fate of the PV3 reflexes *ś and *ʒ, specifically whether they harden and merge with plain /s z/, or become palatalised counterparts to plain /s z/ (/s' z'/) as in East Slavic, which is a very important indicator of the extent to which a phonological system is developing Russian-style phonemic secondary palatalisation of consonants.

By far the largest benefit though of the type of annotation proposed here would be felt in questions of a more systematic and holistic nature, where cross-referencing of multiple orthographic (and potentially also morphological) features must be undertaken to corroborate hypotheses. The extremely important issue of the development of phonemic secondary consonant-palatalisation before various LCS front-vowels, for example, which is a critical isogloss in both Bulgaro-Macedonian and East Slavic, is just such a problem, but since there are no separate letters in either

33 This despite Meillet's (1965: 144) incoherent speculation about the /i/ in *jedinъ resulting from a phonetic development of strong *ь, analogous to what happens in the vicinity of *j, because "*or il s'agit d'un composé dont le second élément est *jĭnŭ*". While this interpretation of *jedinъ's derivation could be true (and according to Derksen's (2008: 212) etymology of the *jьnъ pronoun, its *j is prothetic, meaning it wouldn't develop if already attached to a stem ending in *d), the idea that a synchronic *-dьnъ by any purely phonological means could become /-din/, let alone early enough to completely displace by analogy the oblique forms in the earliest OCS where □□□□- {edin-} spellings are overwhelming, is ludicrous and contradicted by all the philological evidence.

34 According to my database there are 145 reflexes of *jedьn- in this category vs 46 from *jedin-, though all 107 reflexes of the masc. nom./acc. sg. are from *jedinъ. Interestingly the Uspenskij Sbornik (at least the parts included in TOROT), in contrast to later Russian, appears to completely lack *jedьn-, though all except the single одинои are spelt with the OCS <ѥ/е> reflex of intial *je-.

Glagolitic or Cyrillic for palatalised consonants, its existence can only be deduced indirectly by combining evidence from a range of other indicators. In addition to changes in nominal morphology mentioned in Section 1.6 (the adoption by *i-stems of *jo/*ja-stem endings), Townsend & Janda (1996: 107-8) bring our attention to a fascinating inverse-relationship in the Slavic dialects between the extent of phonemic tonality-distinctions (hard/soft opposition) in consonants on the one hand, and the longer maintenance of pitch-distinctions and distinctive vowel-length on the other. This is linked to earlier loss of jers in central vs. peripheral areas, with e.g. far southwestern Ukrainian being the most central part of East Slavic and thus most likely to have either lost or not developed secondary consonant palatalisation, and to have longer maintained distinctive vowel-length (on the East Slavic material specifically see also Trubetzkoy 1924).

An LCS index makes testing for such correlations much easier, not only because of the speed and ease with which things like the Jer Shift can be investigated, where there are simply so many data-points that exhaustive manual investigations would be a significant undertaking, but also because relevant changes often involve unpredictable or zero-reflexes, which it is simply not possible to reliably search for via surface-forms. The tolerance of bimoraic syllables and distinctive vowel-length, for instance, may be probeable by looking at things like loss of intervocalic /j/ and subsequent vowel-contractions (in e.g. the 3sg. present ending of class 5.4 verbs *-ajetQ, which in some OCS texts is reflected as <-аатъ>, or long-adjective endings like *o-je, *a-jǢ, which are the source of Ukrainian neuter *-e* and fem. *-a* endings, opposed to Russian uncontracted *-oe* and *-ая*), but one cannot search the surface-texts for the *absence* of something. One can however search the LCS index for sequences like *ajǢ or *ějǢ to see whether a certain text has more or fewer spellings indicative of contraction.
Similarly, one way that signs of the development of secondary consonant palatalisation before certain front-vowels *can* be sought out, at least in the case of /n’ l’ r’/, is by comparing spellings of the original LCS palatals *ĺ *ŕ *ń with those of plain *l *r *n. Shevelov (1956: 490) claims that the “ziemlich deutlich[e Unterscheidung]” of *ĺ,*ń from *l,*n before *e, but not *i, in the Archangel’sk Gospel of 1092 and the Dobrilo Gospel of 1166, is evidence that following *e never caused palatalisation of these consonants in the proto-Ukrainian dialects of the scribes, whereas *i did. Plain *ni,*li thus fell together with palatal *ńi,*ĺi as /n’i l’i/ and were no longer distinguishable. Whether or not this argument has validity, searching texts for relevant instances to compare is only possible with an LCS index that preserves *ŕ *ń *ĺ, because there is no way to search the surface-spellings for the *absence* of palatalisation-marking on, say, *ŕe *ńe *ĺe without just getting all the reflexes of *re *ne *le as well.
While detailed studies into individual manuscripts’ treatment of the jers were conducted more than a century ago by figures like Jagić (1876, Zogr.) and Leskien (1905, Zogr./Mar.), they contain little attempt to integrate their findings into something we would recognise as a coherent phonological description. Jagić’s study, which is the origin of the so-called ‘Jer Umlaut’ theory (the view that weak jers in Zographensis especially were fronted/backed by the vowel in the following syllable, p.17-18), in particular is full of incorrect etymologies[35], and although Leskien's is more systematic and free of etymological errors, the fact that he was writing before our concept of the phoneme shows itself everywhere in his reasoning and in the implausibility of his explanations. His ultimate conclusion is that the apparently ‘umlauted’ weak-jer spellings in fact reflect the hardness/softness of the consonant-cluster brought into being by the disappearance of the weak-jer (p.347-8), and therefore that front-vowels in following syllables softened in some way the preceding consonant, i.e. *zъlě > /zl'ě/ > /z'l'ě/ <□□□□> {zĭlě}. On p.325 though he explicitly denies "dass die Palatalisirung durch folgende weiche Vokale denselben Grad der Stärke besass wie die der altererbten palatalen Consonanten", and so must be envisaging a rather farfetched three-way phonological opposition between e.g. /n/, /n'/, and /ń/.[36] Lunt (1952: 322-5) then looked again at

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35 E.g. p.23 “кълати halte ich für richtigere als клати” (<*kolti), and on p.35 he assumes a jer-fronting in зърѣти based on his (also incorrect) assumption that the jer in бърати is original.

36 Leskien also missses a few spellings which contradict his argument, which I can trivially find using my LCS index, e.g. p.324-5 he claims that Zogr. *ь is prevented from being backed by a following *ń, despite

Zographensis, and adduces data he manually gathered from "about half the manuscript" to broadly support Jagić's 'Jer Umlaut' theory, but since he only gives rough percentages and we don't have the exact data he draws them from, we have no way of verifying it ourselves.
Perhaps the best indication of the need for a more holistic approach is the fact that Totomanova (2014: 14-28), drawing on Bulgarian dialect data, recently proposed yet *another* solution to this jer-question which is mutually incompatible with both Leskien's and Jagić's: a *merger* of the jers in the earliest Bulgarian after palatal and secondarily-palatalised consonants, and then only a later fronting of this merged /ʌ/ vowel to /e/ in strong-position if the preceding consonant remained soft, parallel to the better-known Middle Bulgarian merger of the nasals in the same environment. Under such a theory no conclusions whatsoever can be drawn about incorrect weak back-jer spellings before LCS hard consonants in OCS, because they could equally easily denote softening and then jer-merging as simple lack of softening and loss of the vowel.

Even very imperfect, wholly automatic autoreconstructions of manucripts not-yet manually morphology-tagged can provide a useful window into a text's features: included on ocstexts.co.uk is the Codex Assemanianus (Cyrillicised from Jouko Lindstedt's ASCII-encoded version[37]), which I tagged and lemmatised using a (slightly improved version of) the extremely outdated method detailed in Berdičeviskis, Eckhoff & Gavrilova (2016), and as shoddy as the autoreconstructions often are, one can still glean useful insights from searching around in it. For example, the affricate /ʒ/ reflex of the Second and Third Velar Palatalisations of *g, in e.g. OCS нозѣ 'leg' (fem. dat./loc. sg. of нога) and ѩѕа 'illness' (cf. the Polish cognates *nodze* and *jędza* 'witch'), which in all major manuscripts shows clear signs of deaffrication to /z/, looks to be pretty flawlessly preserved: in 100% of the forms with autoreconstructions implying such a PV2 or PV3 reflex of *g (i.e. those matching the regexes /gv?[ěęeiьr̄l̄]/ or /[ьięr̄l̄]g[auǫ]/), the spelling is <□> {ʒ}, never <□> {z}, and the only indications of its deaffrication to /z/ are the two spellings of the *-zęb- verb-root in □□□□□□□ {proʒɴbe} and □□□□□□□□□□ {proʒɴbnetŭ}[38].

LCS reconstructions are also useful for checking uncertain etymologies: if a text consistently spells a certain sound-group in securely-reconstructed words one way, but a different etymon which we less-assuredly reconstruct with that same sound-group is spelt differently, then that's a strong indication that our candidate-etymology is faulty. For example, the entry for Russian *брение* in Vasmer's etymological dictionary (Фасмер 1986) mentions Sobolevskiy's suggested *bŕ̥nьje reconstruction with syllabic *ŕ̥ instead of the more traditional *brьnьje with a TRЪT-group, based on East Slavic бьрние and берние spellings in a.o. the Izbornik 1073 and the Archangel'sk Gospel 1092. Sobolevskiy's view is that Russian *брение* is a written borrowing from Church Slavonic which reflects the OCS <liquid + jer> ordering of the letters for syllabic liquids, viz. брьнье.
If we search for this lexeme in Psalterium Sinaiticum, however, we find three spellings: □□□□□□□ {brenıě}, □□□□□□□ {brenĭe}, and □□□□□□□ {brŭnĭě}, two of which suggest a regular vocalised /e/ reflex of strong *ь in a TRЪT-group. Searching my autoreconstructed layer for *ŕ̥ turns up **zero** similar <□□> {re} spellings in the 410 reflexes of *ŕ̥ found: overwhelmingly this group is spelt <□□> {rŭ} in Psal. Conversely, if we use the regex [tŕrpsšdfgħђklĺzžxčvbnńmǯ]rь[tŕrpsšdfgħђklĺzžxčvbnńmǯ]+[ьъ] to search for the strong TrьT-group suggested by the traditional *brьnьje etymology, we find multiple <□□> {re} spellings: □□□□□□□□□□□□□ {poskrežĭćetŭ} <*poskrьžьħetQ, □□□□□□□□□ {vĭskresŭ} <*vъskrьsъ, □□□□□□□ {dьbrexŭ} <*dьbrьxъ, and □□□□□□□□□□□□□ {poskrežĭštetŭ} <*poskrьžьħetQ.
The evidence of Psal. is therefore unambiguously in favour of the *brьn- reconstruction: other reflexes of strong TrьT-groups show the same distribution of spellings as this etymon, while none of the *ŕ̥ reflexes do. Sobolevskiy is right to suspect a Church Slavonicism in Russian, but it's more

□□□□□□□□□□□□ {dĭnesŭnẽago} <*dьnьsьńǢjego and □□□□□□□□□ {sŭvɴzŭnẽ}< *sъvęzьńǢ (here perhaps he is led astray by the jat'-spelling of Zographensis /a/ <*Ǣ).

37 https://www.kielipankki.fi/download/ccmh-src/www/assemanianus.html

38 Though note that the modern Macedonian *ѕемне* <*zębne[tъ] 'shiver' also has an unetymological /dz/ in this root, as do many other Macedonian words with LCS *z such as ѕвер <*zvěrь.

likely to be in the early East Slavic бьрние etc. forms, where East Slavic <ьр> was introduced as a hypercorrection by scribes who mistakenly etymologised unfamiliar Bulgarian брьние as *bŕ̥nьje.

4. Conclusion

The foregoing article has laid out a method for leveraging existing morphosyntactic annotation-data to produce rigorous phonological annotations of Old Church Slavonic texts, and in so doing to make the results of large-scale corpus-building efforts like PROIEL and TOROT directly useful to historical phonologists, morphologists, and textologists, rather than just to the syntacticians, semanticists and pragmaticians these resources were primarily created for.

Both the resulting annotations and the computational apparatus used to produce them have value not only for researchers, but also in their potential for creating innovative pedagogical tools for learners. The work-in-progress ocstexts.co.uk website aims to demonstrate some of these things: for example 'normalised' spelling of a text's words, based on conversions from the regular autoreconstructed LCS forms, can help students learn about and cope with the significant dialectal and orthographic variation characteristic of all OCS texts (Figure 10), and the computerised LCS inflectional morphology which is used to generate the autoreconstructions of single forms can be repurposed to dynamically-generate the whole paradigm of every reconstructed lemma, resulting in comprehensive and high-quality inflection-tables (see p.16). We can even combine this paradigm-generating ability with each text-word's TOROT morphology-tag to visually locate a specific form in its generated table, which massively helps with the learning of paradigms (Figure 11).

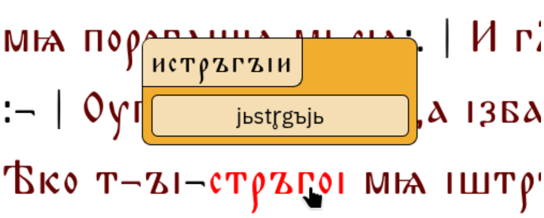

Figure 10: Tooltip showing the autoreconstructed LCS ancestor of the past active participle of the verb истрьгнѫти *in Psalterium Sinaiticum, along with its conversion into 'normalised' OCS*

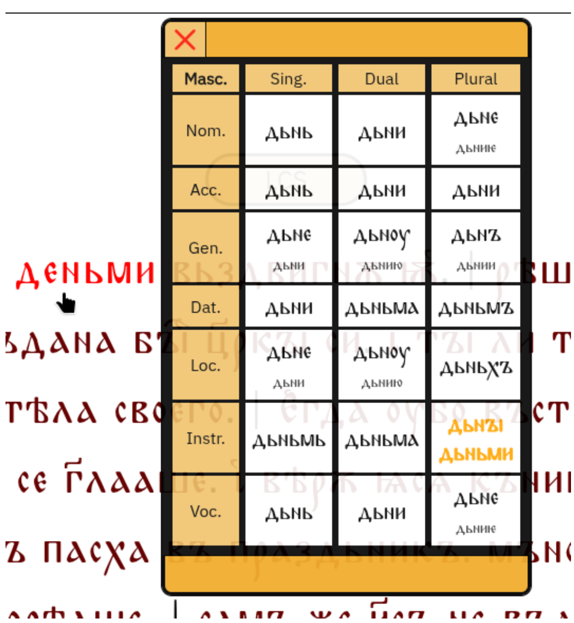

Figure 11: Paradigm of the noun дьнь *generated after Ctrl+clicking on an instrumental-plural occurrence in the Codex Marianus, with the corresponding table-entry animated*

In many ways the point of what I'm proposing with this article is simply to make maximally accessible and applicable for manuscript-investigators the knowledge already accumulated by the nearly two centuries of serious Slavonic scholarship, i.e. the sum of our etymological knowledge (as distilled in etymological dictionaries like Derksen (2008) and the ESSJa) combined with what we know about Common Slavonic phonology and morphology. Furnishing every word of a text with its LCS ancestor simply brings this accumulated knowledge directly 'on top' of the philological evidence at a granular, per-word level, so that deviations from this regular theoretical yardstick are visible, retrievable, and quantifiable.

Though important OCS and Old East Slavic texts are still lacking the sort of annotation-data needed for the method of autoreconstruction outlined here, it should be noted that recent advances in neural-network-based tagging and lemmatisation (e.g. Rabus & Besters-Dilger 2021) mean that modern taggers trained on the existing manually-annotated corpus will perform extremely well on the remaining canonical OCS texts, which massively cuts down the time and effort required to do for, say, the Savvina Kniga what has already been done for the Marianus. As the volume and quality of training-data increases, the performance of such automatic taggers can only improve, which will open up more and more of the early Slavic written record to the sort of automatic phonological annotation described here.

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Appendix 1: Glagolitic transliteration table

Glagolitic	Latin		Commonly-used Cyrillic
□	a		а
□	b		б
□	v		в
□	g		г
□	d		д
□	e		е
□	ž		ж
□	ʒ		ѕ, □
□	z		з, □
□	ï		ї, □
□	ı		і
□	i		и
□	ǵ		□, г̑, ђ
□	k		к
□	l		л
□	m		м
□	n		н
□	o		о
□	p		п
□	r		р
□	s		с
□	t		т
□	ϋ		ѵ, у
□	f		ф
□	x		х
□	ω		ѡ
□	θ		ѳ
□	c		ц
□	č		ч
□	š		ш
□	ć		щ
□	ŭ		ъ
□	ĭ		ь
□	ě		ѣ
□	u		оу, ѹ
□	ü		ю
□	ǫ		ѫ
□	ǫ̈		ѭ
□	ę		ѩ, ѧ
□	ɴ		ѧ
□	ɴ̨		□

Necromancing Diels: computerising the phonological analysis of early Slavonic texts using existing treebank data and a Late Common Slavonic computerised inflectional morphology

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ABSTRACT

This article demonstrates a new way of massively speeding up traditional historical phonological investigations into early Slavonic philological material by furnishing each word of a manuscript-text with its direct Late Common Slavonic (LCS) reconstructed ancestor. Such reconstructions act as a type of "phonological annotation", or index, since they make the reflexes of any LCS phoneme-sequence reliably and predictably searchable, regardless of a manuscript's surface-spellings. To automate the production of this LCS reconstructed layer, a comprehensive computerised LCS inflectional morphology has been developed, which is capable of using existing lemmatisation and morphology-tagging annotation-data from texts in the Tromsø Old Russian and Old Church Slavonic (TOROT) corpus in order to inflect LCS lemmas according to each text-word's morphological shape. A web-interface, ocstexts.co.uk, is provided to showcase some of the benefits of the resulting "phonological annotations", including searchable autoreconstructed versions of OCS texts from the TOROT corpus, with virtually 100% coverage of the Codex Marianus, Codex Zographensis, and Psalterium Sinaiticum.

0. Introduction

Much progress has been made in the last twenty years in early Slavonic corpus linguistics as a result of the Old Church Slavonic part of the PROIEL project (Haug & Jøhndal 2008) and its subsequent expansion as the TOROT treebank (Eckhoff & Berdičevskis 2015), such that currently just over 240,000 words of canonical OCS have been manually lemmatised, part-of-speech and morphologically-tagged, and syntactically parsed. The focus of these projects, however, has been exclusively on the higher-level linguistic domains of syntax, semantics, and pragmatics: surface-morphology has been of only incidental concern, for example in investigations into differential-object marking (Eckhoff 2015, 2022), but the sort of inflection-class-marking which would enable the retrieval of, say, masculine *o-stems vs *i-stems is lacking. The needs of historical phonologists especially are ill-served, since some of the texts contain quite severe typographical inconsistencies and errors that make them impossible to search reliably and dangerous to use without reference to the manuscripts.

That being said, enough information is included in TOROT's lemmatisation and morphology-tagging that, with a few exceptions (e.g. certain comparatives), the morphological shape of the inflected text-forms can be predicted from just the tag-information, provided that inflection-class annotations are added to the lemmas. This means that the immediate Late Common Slavonic ancestors of surface-text forms can be generated by reconstructing and inflection-class-marking the LCS lemmas, and then using a computerised LCS inflectional-morphology to inflect each text-word's lemma according to its morphology-tag[1]. Such LCS reconstructions are an extremely useful form of 'phonological annotation', because theoretically all the information required to give rise to an attested form must by definition be present in any correct reconstructed proto-form, and the complete regularity of the idealised LCS forms makes texts predictably searchable regardless of orthographic variability, abbreviations, or other irregularities in the surface-texts. When applied to whole texts, they make the exhaustive investigation of almost any phonological or orthographic question trivially easy compared to manually reading and extracting relevant forms, or using

1 Morphological innovations and variations are detected by inspecting the text-forms and then applying 'alternative' endings as specified in the inflectional-endings database; see Section 2.3.

TOROT's existing lemmatisation and morphology-tagging to try to gather morphological categories which might contain the sound-groups one is interested in.

The goal of this article is to describe just such a computerised LCS inflectional-morphology, and to show how it can be used to "autoreconstruct" different OCS texts from the TOROT corpus, while explaining how difficulties caused by things like morphological innovations, badly-integrated foreign loanwords, or insufficiently-precise tagging-data can be overcome. The resulting ‘phonologically annotated’ texts should allow investigators of the lower-level linguistic domains (phonology, orthography, and morphology) to benefit from the huge amount of manual annotation work that has gone into the OCS part of TOROT, and the computerised LCS inflectional-morphology by itself is valuable in that it allows linguistically-rigorous and comprehensive inflection and conjugation-tables to be generated[2].

A web-interface for autoreconstructed TOROT OCS texts is being developed at https://ocstexts.co.uk, which aims to demonstrate the benefits of this extra layer of annotation by offering things like LCS phoneme-search of texts, conversion of each autoreconstructed word into canonical "normalised" OCS, for easy comparison with the manuscript-form, and computer-generated inflection and conjugation-tables for every reconstructed lemma. It also includes TOROT's existing morphology, lemmatisation, and (for three of the texts) Greek source-alignment data, plus integration of the recently digitised OCS dictionaries hosted at gorazd.org.

The usefulness of computational methods for diachronic phonology has of course not escaped the notice of previous scholars, and Sims-Williams (2018) offers a useful survey of a field he calls "Mechanising Historical Phonology". Of the three main strands of work considered there, (“automatic cognate-recognition”, “computerised forwards-reconstruction”, and “computerised backwards-reconstruction”), only forwards-reconstruction is of any relevance to the task of holistically investigating the sound-systems of manuscripts, in that one could apply sound-changes and orthographic rules to the LCS reconstructed layer of individual texts to try to “reverse engineer” the manuscript spellings, and in so doing test hypotheses about the phonological features of the underlying dialect[s] that gave rise to them. The forwards-reconstruction work mentioned by Sims-Williams, though, is invariably aimed at producing sets of isolated, regularised forms, either from modern standard languages, or theoretical idealised forms from old languages, e.g. Borin’s (1988) OCSGen system that tried to produce “normalised” OCS from PIE. This type of work is useful for identifying problems with traditional diachronic accounts of ordered sound-rules[3], but it is of a fundamentally different nature to what I'm doing here, which is about elucidating the concrete philological facts of the manuscripts in order to better understand the synchronic linguistic systems responsible for them.
Most early Slavic texts are from dialect-areas (East Slavic and Bulgaro-Macedonian) whose modern phonological systems show evidence of sharp differentiation along deep structural lines, most importantly in the phonologisation of secondary consonant-palatalisation and the tolerance for bimoraic syllables (i.e. distinctive vowel-length). The seeds of these differences must have been

2 The lemma and inflection-class data included as a supplement to Polivanova’s (2013) comprehensive OCS grammar, https://integral.github.io/osd/, would in principle allow similar tables to be generated, but as far as I know Polivanova has not digitised the actual endings, nor written any scripts to generate paradigms. Additionally, because she bases her forms on a normalised ***OCS***, rather than ***LCS***, phonemic system, a massive amount of information has been needlessly destroyed. To give just one obvious example, the syllabic-liquids *ŕ̥ *ĺ̥ *r̥ *l̥ are represented by their OCS reflexes /рь ль ръ лъ/, meaning there’s no distinction between the syllable-rimes of roots like /-срьд-/ <*sŕ̥d and /-крьст-/ <*krьst. An inflectional-morphology based on her data would therefore only be good for generating the (in reality unattested and fictional) “normalised” form of OCS, whereas my LCS forms are trivially convertible into *any* dialect of OCS, as well as pre-Jer Shift Old East Slavic (see pg. 17 and Figure 6), because they represent the language at a stage before any such dialect-specific phonemic mergers.

3 For a recent example see Marr & Mortensen (2020), who found many problems with the traditional account of how Latin became French, and were able to improve upon it using their suite of software for performing forwards-reconstruction and "debugging" problematic sets of rules.

sown around the time of the radical restructuring occasioned by loss of the reduced-vowels *ъ,ь (the so-called "Jer Shift"), i.e. the very time at or shortly after which our manuscripts were written, but the limitations of the original writing-system (which was devised for a language which had *not* yet undergone the Jer Shift) mean that they do not find clear expression in the texts and can only be dimly groped at through indirect means. Much more thorough investigation of *all* aspects of the manuscript-spellings are thus required in order to uncover the nature of the linguistic systems they reflect.

Slavonicists are in the fortunate position of having not only a widely and diversely attested set of daughter-dialects from which to reconstruct the proto-language, but also a historical written record which is so close in time to the Common Slavonic period that the far-southeastern Bulgarian dialect of Thessaloniki was considered a suitable vehicle for proselytising the proto-Czech speakers of Moravia. The hypothetical LCS system which I employ for my reconstructions is therefore mostly uncontroversial and widely agreed-upon (except in trivial matters of notation); nevertheless, a corpus-focused project like mine where the aim is to reconstruct *all* words of a text requires an exhaustiveness and rigour that leaves no room for the dodging of difficult points, so Section 1 of this article is taken up with a comprehensive exposition and justification of my chosen LCS system. Section 2 then looks in more detail at my computerised LCS inflectional-morphology and the structure of the existing TOROT data, and shows that together they can be used to produce the direct phonemic ancestor-forms of text-words, despite various challenges, including the morphological variation and restructuring which is occurring in even the earliest texts. Section 3 highlights some of the benefits gained by having texts indexed with LCS reconstructions and explores some ways this could be used to improve and speed up investigations of manuscripts.

1. My Late Common Slavonic (LCS) phoneme-system

The premise of my chosen form of "phonological annotation" is that the earliest Slavic texts reflect languages which are **structurally** close enough to the broadly-agreed-upon system of Late Common Slavonic that the forms underlying the manuscript-spellings are more or less trivially derivable (by the application of sound-change rules) from their theoretical LCS ancestors.
By 'structurally' I am referring to structure at the phonological level; structural changes at higher levels of analysis (i.e. inflectional morphology, derivational morphology) are of no concern unless they are **made possible only by intervening phonological changes**.
My contention is that before about 1100 not enough of these structural changes are in evidence in any Slavic text, and thus texts can be relatively straightforwardly indexed using a well-chosen LCS system. Before giving examples of structural changes that are problematic for such an indexing-system, it's necessary to first lay out my LCS system in full:

In order to account for as much of the subsequently attested Slavic as possible, a point after the monophthongisation of diphthongs, but before the Second and Third Velar Palatalisations (PV2 and PV3) is chosen as the point of departure, because of the difference between the West Slavic /š/ and South/East /ś/ reflex of these two palatalisations of *x (Cz. loc. pl. *duších* vs Suprasliensis доусѣхъ <*duxěxъ; Polish *wszak* vs Supr. вьсакъ , Ru. *всяк[ий]* <*vьx-akъ), as well as the probable complete absence of PV2 in northern East Slavic[4] (Old Novgorodian, see Zaliznjak 2004: 42-45 for

4 The evidence regarding the possible absence of PV3 from Novgorodian is far less convincing: the Birchbark letters abound with examples of the PV3 reflex of *k (e.g. letter №439 from around 1200 has свинеце <*svinьkь and полотенеца <*poltьnьka), and those of *g are not unknown: Zaliznjak (2004: 47) admits that palatalised forms of the Germanic loan къназ- <*kъnęg- are the rule, but considers this to be a "supradialectal" word originating outside of the Novgorodian dialect-area; Галинская (2014: 10) is less convinced and adduces the form оусьразн (cf. Russian *серьга*, commonly assumed to be an Oghur, i.e. Bulgar, Turkic loan, cognate with e.g. Kazakh *сырға*) 'earrings' from letter №429 as a word of "вполне бытового характера" which thus supposedly shows a native Novgorodian reflex of PV3 of *g.

the evidence), and the blocking of PV2 by an intervening *v in West Slavic (Pol. *gwiazda*, Cz. *květ* <*gvězda, *květъ, etc.).
To be explicit, the native phonemes in my LCS system are given in the tables below:

Table 1: LCS Vowels after the monophthongisation of diphthongs

	Front		Back
High	i		y u
	ŕ̥ ь ĺ̥	y̨	r̥ ъ l̥
Mid	e ę		ǫ o
	ě ę̌		
Low	Ǣ		
		a	

Table 2: LCS consonants before PV2/PV3 (adapted from Winslow 2022: 304)

Labial	Dental	Palatal	Velar
m	n	ń	
b p	t d	ћ ђ	k g
	s z	š ž	x
		č	
	l	ĺ	
	r	ŕ	
v		j	

1.1 Foreign sounds

In addition, the following symbols are used for phonemes of wholly foreign origin in order to represent badly-integrated foreign borrowings, whose level of integration into the native system we deliberately do not take a position on: /ḱ ǵ x́ f ü/, e.g. in respectively кѵтъ <*ḱitъ, ıꙉемонъ <*iǵemonъ, χитонъ <*x́itonъ, иосифъ <*ijosifъ[5], and мѵро <*müro. Almost none of the words containing these symbols would actually have existed in the language during Common Slavonic times, but they need to be included in the indexing-system because they often contain native Slavic elements (f.ex. inflectional endings). Normally they represent specific sounds in the source-language (usually Greek), so including them is useful for investigating the process of these sounds' integration into the native systems. For instance, the extent to which Greek /ü/ is integrated into either native /i/ or /u/ can be seen in variations in the OCS spellings of the word for 'Egypt' (*eǵüpьtъ): ⰵⰼⱛⱂⱅⰰ {eǵüpta} vs ⰵⰳⰹⱂⱅⰰ {egıpta} vs ⰵⰳⱛⱂⱅⱏ {egüptŭ}vs є҆гоупьтъ vs ⰵ̄ⰼⱛⱂⱅ̄ⰰ {eǵüpta}[6], etc.. One might also ask whether a separate <ⰼ> letter for /ǵ/ (and the writing of <ⰽ̑>/<ⰳ̑

More importantly, as Галинская (op. cit.) points out, in all of the well-known Novgorodian forms of the pronoun *vьxъ 'all' which supposedly show a lack of PV3 by retaining both /x/ and back/hard desinences (e.g. fem. gen. sg. въχоѣ <*vьxojě̜ from letter №850), and which come from letters which otherwise correctly convey the jers (by writing <ь,є> for *ь and <о,ъ> for *ъ), the weak-jer is always written with <ъ,о>, unambiguously suggesting a /ъ/ pronunciation. These forms therefore more likely point to a LCS doublet-form *vъxъ which would never contain the conditioning environment for PV3 anyway, and thus you can't use them as evidence of a lack of PV3 in Novgorodian (on the plausibility of such a doublet see Галинская (2014: 14), though cf. Zaliznjak's (2004: 54) less convincing explanation of the /ъ/ in these words as an assimilation of original /ь/ to the back-vowels of the following syllable).

5 Of course the sequence /jo/ violates LCS phonotactics as well.

6 Forms are given as they appear in the manuscripts; modern fonts and Unicode symbols mean that the misleading and unhelpful practice of transcribing Glagolitic into Cyrillic is no longer defensible. Latin-based transliterations of Glagolitic forms are given in {curly braces} according to the table in Appendix 1, but it should be emphasised that no transliteration system can adequately convey the internal logic of the Glagolitic, and those who study OCS through transliterations are hamstringing themselves. Where specific forms from Eckhoff's corpus-texts are cited, they are hyperlinked to the corresponding part of the text as it appears on the ocstexts.co.uk website. The Glagolitic texts there are unfortunately still Cyrillicised and should be used with care, particularly ones like Psalterium Sinaiticum where certain ruesome decisions from the editors Severjanov (1922) and Mareš (1997) have been compounded by further information-destroying simplifications (for instance, Severjanov transliterates <ⰼ> with <г̑>, but Eckhoff then replaces Severjanov's roundy diacritic with a titlo, leading to nonsense transcriptions like Psalm 8 <ан҃г҃лъ> for ms. <ⰰⱀ̑ⰼⰾⱏ>). Cleaned up digitisations, using the Glagolitic originals as the base-text, and a

> with the palatalisation-diacritic) could be linked to the inadmissibility in the native systems of soft [k^j, g^j] sounds, and whether their replacement with regular <ⰳ,ⰽ> or <г, к> was more likely in systems with some level of native [k^j, g^j] (for instance, in Rus' after the so-called Fourth Velar Palatalisation *ky,*gy,*xy > [k^ji, g^ji, x^ji], or in Novgorod due to the retention of native velars before front-vowels because of the non-action of PV2, etc.). In any case, such questions are far easier to investigate if all relevant forms can be reliably retrieved by giving them even a consciously artificial LCS representation.

1.2 Vowels

I have deliberately not included accentual information in my reconstruction of vowels, even though such information is in fact required to explain certain differing manuscript-reflexes, e.g. Russkaja Pravda fem. acc. sg. робу < *orb-ǫ vs Uspenskij Sbornik nt. acc. sg. рало < *ordl-o, because for too large a proportion of the vocabulary this information is not sufficiently securely and uncontroversially reconstructed, and anyway the (often post-LCS) derivational processes which are responsible for most of the actual words in the attested texts (and the inevitable accentual levelling processes likely to have occurred in the course of these derivations) complicate things even further.

The two extra nasal-vowels /y̨/ and /ę̌/ are required to account for the split between North (East and West) and South Slavic forms of certain inflectional-endings:

- *y̨ is used for the nom. sg. masc./nt. pres. act. participle of certain verb-classes whose present-stem ends on a hard-consonant, which in South Slavic remains high and backed, e.g. Supr. ꙃовꙑ <*zovy̨, Psalterium Sinaiticum ⱄⱅⱃⱑⰳⱘⰹ {strěgǫï}<*stergy̨-jь, Codex Marianus ⱑⰴⱔⰹ {ědN̨i} <*jǼdy̨-jь (these forms lead Kortlandt (1979:260) to posit that some dialects of early OCS retained some kind of nasal character in this vowel and may even have developed the special "hooked" nasal letter <ⱔ> for it), but which in most of North Slavic lowered to /a/: Old Polish (Kazania Świętokrzyskie) has both *reca* and *recǫ* (with the special Old Polish letter for the merged reflex of *ę and *ǫ) <*reky̨, and in other texts also *biorǫ* <*bery̨; Russkaja Pravda река, Uspenskij Sbornik доида <*dojьdy̨, Ru.Ch.Sl. (Vita Methodii) вьсемогаи <*vьxemogy̨-jь. Kortlandt's positing of a CS *y̨ (which he writes as *aN) is far from universally accepted, and others consider these forms the result of various dialect-specific analogical process; see references and discussion in Olander (2015: 88-92). Whatever the truth of the matter, our *y̨ is a convenient placeholder which allows all the relevant evidence to be retrieved.
- *ę̌ is responsible for the NSl. /ě/ vs SSl. /ę/ shapes of jo-stem masc. acc. pl. and the ja-stem nom./acc. pl. and gen. sg. endings, which are reflected in respectively the post- and pre-revolutionary spellings of the Russian nom./acc. pl. long-adjective endings *-ые* < *yjě < *yję̌ vs *-ыя* < *yja < *yję < *yję̌.[7]

The need for the retention of the /Ǣ/ archiphoneme, which represents merged Early Common Slavonic *ē *ā in the position after palatal consonants, up to this point of LCS, is explored in detail in Winslow (2022), but the same archiphoneme (along with its short counterpart /Æ/) was explicitly posited by Kortlandt as far back as 1979 (p.266) as part of his ECS system. In short, a combination of:

1.) the lack of any device in the Glagolitic alphabet to render /ja ńa ŕa ĺa/ sequences (for which Glagolitic texts must use the jat' <ⱑ> letter whose base-value is /ě/);

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mechanism for displaying manuscript-images, are planned for the near future.

7 Other troublesome pre-LCS morphological isoglosses reflected in the texts include the masc./nt. instr. sg. *o- and *jo-stem endings *-ъmь/*-ьmь (N.Sl., e.g. KF ⱁⰱⱃⰰⰸⱏⰿⱐ {obraz<u>ŭmĭ</u>}, Uspensk. Sbor. кнѧзьмь) and *-omь/*-emь (S.Sl., e.g. Supr. ѻбразомъ, кнѧꙃемь), which are most commonly (e.g. Olander 2015: 168) thought to be analogical replacements of the original instr. sg. ending ECS *-ā which is preserved in the adverb *vьčera 'yesterday'; and the *-tь (N.Sl.) vs *-tъ (S.Sl.) verbal endings of 3rd sg. and pl. present (plus its extension to 2nd and 3rd sg. aorists like OCS начѧтъ, OR (Uspensk. Sbor.) бꙑстъ, начатъ). Here I have no choice but to index them with dummy-symbols in the database: *-Omь/*-Emь for the instr. sg. ending and *-tQ for the verb-endings.

2.) overwhelming spellings of palatal-letter (<ⱍⱎⰶⱋ>) + jat' in the Kiev Folia (the oldest and therefore least distant ms. from the 'original' OCS, as first codified by Cyrill and Methodius and for which the Glagolitic alphabet was devised) for the reflexes of LCS *č,*š,*ž,*ħ + *Ǣ (e.g. ⱁⰱⱑ<u>ⱌⱑ</u>ⰾⱏ {obě<u>cě</u>lŭ} <*ob-věħǢlъ, ⰴⱆ<u>ⱎⱑ</u>ⰿⰹ {du<u>šě</u>mı}<*dušǢmi), as well as occasional traces of such spellings in later Glagolitic OCS (e.g. Psal. <u>ⱍⱑ</u>ⱎⱔ {<u>čě</u>šę} <*<u>čǢ</u>šę̌); and

3.) the evidence of certain modern Bulgarian dialects, which have reflexes of LCS *ě in words like *<u>ж'е</u>ба* <*žǢba 'toad' (Stojkov 1954: 74–78),

all together point very strongly towards there having occurred a split on the Southeastern periphery of Slavic between LCS dialects which have /ě/ < *Ǣ and the majority of the rest which got /a/, and that original OCS ('Urkirchenslavisch') was an *Ǣ > /ě/ dialect. I posit that *Ǣ remained until the opposition /a/:/ě/ after palatal consonants was reintroduced when PV2 and PV3 brought new soft-consonants /c ś dź/ into the system, which could be followed by both /a/ and /ě/: fem. nom. sg. /stь<u>dźa</u>/ < *stьga (PV3) vs fem. loc. sg. /no<u>dźě</u>/ <*nogě (PV2) (Winslow 2022: 304-305). Thus since my LCS system is based on a point just *before* PV2 and PV3, I must also retain the *Ǣ archiphoneme.[8]

The syllabic liquids /ŕ̥ ĺ̥ r̥ l̥/ are included as unitary vocalic phonemes, following Schenker (1995: 94), rather than as combinations of /ь ъ/ + /ŕ ĺ r l/, because these groups descend from PIE syllabic liquids and many descendant South Slavic dialects which retain syllabic liquids in this position (including most of those underlying canonical OCS) do not show any evidence of an intervening oral-vowel + liquid stage (such a view is shared by Bethin (1998: 71-72); cf. also Bulgarian dialectal evidence in Stojkov (1954: 130-131), where hard consonants precede reflexes of the LCS /ĺ̥ ŕ̥/ even in dialects with secondarily-palatalised consonants before fallen weak LCS /ь/).
The need for both front and back *ŕ̥ *r̥ is unambiguously shown by the East Slavic reflexes /er/ and /or/ (Ru. *смерть*, *морковь*), but *ĺ̥ vs *l̥ is more complicated: PIE *pl̥nos, *wl̥kʷos > Lithuanian *pilnas* 'full', *wilkas* 'wolf' (LCS *pĺ̥nъ, *vĺ̥kъ) vs Lith. *stulpas* (LCS *stl̥pъ 'pillar') suggests that Balto-Slavic had differentiated front/back variants of the PIE syllabic *l̥ (Bethin 1998: 69), but the ancestor to East Slavic backed all vowels preceding tautosyllabic /l/ (Ru. *молоко* < Proto-ESl. *molko < LCS *melko > OCS млѣко), and thus only has /ol/ reflexes here: Ru. *волк*, *столб*, *полный*. It's true that Polish has *wilk* and *milczeć* (<*mĺ̥čǢti), but the Polish reflexes are complicated and likely have more to do with the surrounding consonants: *pĺ̥nъ by contrast gives *pełny* with hardened /l/ and the Polish non-palatalising-/e/ reflex of *ъ, and the differing reflexes in *wi<u>erz</u>ch* <*vŕ̥xъ, *śmi<u>er</u>ć* <*sъmŕ̥tь and *m<u>ar</u>twy* <*mŕ̥tvъjь rule out any explanation based on the nature of the LCS syllabic-liquid alone (for more discussion see Bethin op. cit.: 73-75).
While most OCS shows no sign at all of a front-back distinction in the syllabic-liquids and writes the reflexes of these groups overwhelmingly with <ръ> and <лъ>, the Kiev Folia, which is the only OCS text that reflects a pre-Jer Shift stage and is very nearly flawless in its etymologically correct rendering of the jers, also spells *ŕ̥ *r̥ and *ĺ̥ as one would expect: ⱄⰽⰲ<u>ⱃⱐ</u>ⱀ- {skv<u>rĭ</u>n-} ⱅⰲ<u>ⱃⱐ</u>ⰴ-

8 It's possible to argue that the short *Æ counterpart to *Ǣ persisted in East Slavic until after the Fall of the Jers, and that the ESl. so-called e > o shift before hard-consonants / back-vowelled syllables is actually just the resolution of this archiphoneme as /o/ (where palatalisation of the preceding consonant remained, in e.g. Ukr. *бджола* <*bьčÆla, or was newly phonemicised, in e.g. Ru. *вёсла* <*v'Æsla <*vesla), and that there was never a stage when these words had /e/ (based among other things on <o> spellings regardless of stress after palatal-letters in very early texts, and even after the letters for secondarily-soft LCS plain consonants in the Birchbark documents (Le Feuvre 1993, Nakonečnyj 1962), but there isn't space to elaborate on the issue here (see Winslow 2022: 304 fn.16). Unlike the situation with long *Ǣ, OCS shows no sign of anything but an /e/ reflex of short *Æ (and indeed the fact that the East Slavs inherited their writing system ultimately from the Urkirchenslavisch system designed for such a dialect, rather than one which had a clear way of writing /soft consonant/ + /o/, is likely the reason that /o/ reflexes are so rarely detectable in the early texts, since <e> had to be used for both /e/ and /’o/, cf. the spelling е̑вшанъ of the Kipchak word /jovşan/ ‘wormwood’ in the Hypatian Codex, whose modern cognates (Turkmen *ýowşan* /jowşan/, Kazakh *жусан* /žuwsan/, Azeri *yovşan*) unambiguously point to a Kipchak /o/), and the history of the East Slavic /o/ reflexes remains the subject of much disagreement, so it's simpler for everyone if I continue the traditional practice of writing LCS *e after palatals, even if that strictly speaking is inconsistent with my use of *Ǣ.

{tvrĭd-} ⱄⱃⱐⰴⱐⱌ- {srĭdĭc-} ⰴⱃⱐⰶ- {drĭž-} <*ŕ̥, ⱄⰽⱃⱏⰱⱐⱀⰹ {skrŭbĭnı} <*r̥, and ⱀⰰⱂⰾⱐⱀⰵⱀⰺ {naplĭnenï} <*ĺ̥ (Winslow 2022: 313), and even Zographensis spells all 5 occurences of *vĺ̥k- 'wolf' with ⰲⰾⱐⰽ-/ⰲⰾⱐⱌ- {vlĭk-/vlĭc-} and all 15 instances of its *-mĺ̥č- root with -ⰿⰾⱐⱍ- {-mlĭč-}(e.g. ⱆⰿⰾⱐⱍⰰⱎⱔ {umlĭčašɴ}). Therefore, taken as a whole the Slavic evidence pretty securely points to front and back variants of both syllabic liquids, and for searching purposes it's far preferable to denote them with separate symbols[9] rather than as the sequences /ьr ъr ьl ъl/[10].

1.3 Consonants - Dejotation

Reflexes of the so-called jot-palatalisation are all written either as unitary palatal phonemes, or in the case of jot-palatalised labials as /vĺ mĺ bĺ pĺ/, rather than as sequences of consonant + /j/, hence /ń ĺ ŕ/ for *nj *lj *rj. The ‘dejotated’ reflexes of *tj (and *kt+front-vowel) and *dj are denoted using the modern Serbian Cyrillic letters /ћ/ and /ђ/ respectively, because the commonly used alternatives, i.e. /ť ď/ (as used in e.g. Olander 2015) or /ḱ ǵ/ (as used in Winslow 2022), or variations thereof, are visually too close to symbols used elsewhere in the system. /ḱ, ǵ/ are anyway already used in my system for foreign /k, g/ before front-vowels, and /ť ď/ look too similar to the common denotations of secondarily-palatalised post-Jer Shift /t' d'/, as used in discussions of systems like Russian or Eastern Bulgarian where they arise.
The compelling hypothesis, first proposed by Durnovo (1929: 55-58) but most recently elaborated by Vermeer (2014: 209-214), and accepted by Mathiesen (2014: 197 fn. 22) and Winslow (2022: 310 fn.25), according to which the Urkirchenslavisch reflexes of *ђ,ћ were close enough to foreign /g k/ before front-vowels that the original Glagolitic system used <ⰼ ⱋ> for both sets of sounds (i.e. alongside attested ⰻⰼⰵⰿⱁⱀⱏ {iǵemonŭ}< ἡγεμών would have been **ⱁⱄⱘⰼⰵⱀⰻ {osǫǵeni}<*osǫђeni, and alongside attested ⰴⱏⱋⰵⱃⱐ {dŭćerĭ} <*dъћerъ would have been **ⱋⰻⱀⱏⱄⱏ {ćinŭsŭ} < κῆνσος[11]), does not prevent us from keeping the foreign sounds separate for our LCS stage, since clearly they differed enough in all the dialects underlying actually attested OCS to be written separately.

Pre-dejotation *stj and *zdj are differentiated from the PV1 reflexes of *sk and *zg by writing the former as *šћ and *žђ and the latter as *šč and *žǯ, even though their modern reflexes do not differ from each other anywhere and so must've fallen together in the CS period, because they often alternate with their respective un-palatalised counterparts morphologically and derivationally, e.g. očistiti:očišћenьje vs. jьskati:jьščǫ, jǢzditi:jǢžђǫ vs jьzgъnati:jьžǯenǫ.

There are convincing arguments for PV2/3 having preceded dejotation, at least in more central areas, most recently presented in e.g. Vermeer (2014: 197) and Wandl & Kavitskaya (2023: 244-

9 In the database I will have to use the single Unicode characters <ṛ ṝ ḷ ḹ>, rather than what's shown in Table 1, since the latter cannot actually be rendered without using the letters for /r ŕ l ĺ/ plus the 'combining ring below' U+0325 symbol, which means searches for the consonantal liquids on their own will also return results containing syllabic liquids. A similar problem affects /ę̌ y̨/, which I will have to replace with <ę̌ ỹ>.

10 To my mind the only evidence in support of a genuine jer + liquid stage comes from the paradigms of verbs like OCS сътрьти < *sъtŕ̥ti, where the syllabic /ŕ̥/ in the stem alternates with /ьr/ depending on the vocality of the following morpheme: the e.g. 3sg. pres. *sъtьretъ (Zogr., Supr. сътьретъ) or (one possibility of the) 3rd sg. aorist *sъtьre (Supr. сътьре) must have /ьre/, while the 3rd pl. aorist *sъtŕ̥šę (Supr. сътрьша) and the other possibility for the 3rd sg. aorist *sъtŕ̥ (Psal. ⱄⱏⱅⱃⱐ {sŭtrĭ}, or with a different prefix Mar. ⱁⱅⱃⱏ {otrŭ} <*otŕ̥), being word-final or pre-consonantal, must be syllabic /ŕ̥/. The same alternation occurs in the zero-grade forms of verbs like *umerti, as is clear from the Polish reflexes *umarł* <*umŕ̥lъ vs *umrę* <*umьrǫ. The argument could be that at some stage, before the LCS tendency towards Open Syllables became dominant, the stems in these paradigms were surely unitary /tĭr/, /mĭr/, i.e. 3sg. aor. /sŭ.tĭ.re/ vs 3rd. pl. aor. /sŭ.tĭr.šę/, and that the latter’s closed /tĭr/ syllable was only forced to open itself up by changing to /tŕ̥/ because of the Law of Open Syllables. Thus at least one source of the syllabic-liquids could be shown to have developed from a vowel + liquid stage, but that still doesn’t prove that they all did, or that the change of /ĭr/ to /ŕ̥/ in these verb-forms was not merely a move to an already-existing syllabic-liquid phoneme.

11 Interestingly, this aspect of the hypothesised Urksl. orthographic system has rearisen in the modern Macedonian standard due to Turkish loanwords: *ќемер* < Tk. *kemer* ‘belt’, *ќе* < *[хъ]ће[тъ]; *ѓон* < Tk. *gön* ‘leather’, *меѓу* <*meђu.

247), and therefore it could be objected that my system, which contains the dejotation reflexes /ħ ђ ń ĺ ŕ/ but not the PV2/3 reflexes /c ś dź/, is ahistorical. However, it should be reemphasised that the primary goal of my LCS reconstructions is to act as an index which allows reflexes in texts to be found, not to be a historically realistic description of some actually-existing LCS dialect. The absence of PV2 in Novgorodian shows that it can't have preceded dejotation everywhere in Slavic, and in any case the replacement of the sequences /tj dj nj lj rj/ by articulatorily distinct combined units, no longer associated by speakers with their /t/ and /j/ phonemes, is structurally completely irrelevant unless and until these new units merge with existing phonemes (or new sequences of dental + /j/ are introduced), as e.g. in the KF dialect where /tj/ merged with /c/ from PV2/3, or in ESl. where it merged with /č/ from PV1. A language which had distinct Czech-like palatal [c, ɟ] reflexes of *tj and *dj, and also no new sequences of [tj, dj], could not convicingly be argued to have undergone dejotation at the phonemic level, as these new units would just be phonetic realisations of /tj, dj/. Analysed like that, the symbols /ħ ђ ń ĺ ŕ/ in my system strictly speaking would really just be cover-symbols for the pre-jotation sequences, but such notation is preferable since it prevents searches for groups containing /j/ alone from returning results polluted by all the dejotation-groups. As explored in Winslow (2022), the status of /j/ as a phoneme in the earliest OCS texts is an intricate problem, so the ability to investigate the reflexes of *j in isolation from the dejotation-reflexes is important.

1.4 Issues involving the glide /j/

1.4.1 Word-initial *jǢ-/*a-

The tendency for ECS *ā- to have taken prothetic /j/ by LCS times (in accordance with the drive towards open syllables) can make it difficult to distinguish this group from *jǢ- in the absence of wider Indo-European evidence. Normally I've followed Derksen (2008), or the ESSJa (*Этимологический словарь славянских языков*, Trubačev 1974-), but for certain lexemes, e.g. *ama 'pit', which in OCS is spelt overwhelmingly with ⰵⰿ- {ěm-} or ѩм-, the single Greek cognate ἄμη adduced by ESSJa I p.70 in favour of jot-less *am- is not enough to categorically exclude the alternative *jǢma. In particular the 1sg. nom. pronoun *azъ/*jǢzъ is especially problematic: I follow ESSJa I p.100 which ultimately plumps for *azъ, but Derksen doesn't discuss it at all. (A lengthy discussion of the evidence can be found in Teneva's (2012) article on the subject.)
Forms with insecure etymologies can't under any methodology be used as good evidence in phonological investigations, so in difficult cases like the above I simply mark the lemma in the database and provide some short discussion, so that eventually the web-interface can flag such forms in some way and inform users of the specific difficulties.

Like Derksen, I assume that roots going back to PIE jot-less long *ē or dipthongal *oi-, e.g. the root for 'to eat', PIE *h_1ēd-, all took prothetic *j and merged with *jǢ- from other sources, unlike Durnovo (1929: 54), who seems to think that such a development was limited to Bulgarian and Macedonian dialects, including those underlying OCS (where in the Cyrillic mss. we get regular ѩсти etc.). Isolated nominal forms like Ru. *язва* (which Derksen (2008: 155) derives from a Balto-Slavic *oi- based on Lith. *aiža* and Old Prussian *eyswo*) suggest that *ě reflexes in the modern forms of verbs like Ru. *ехать*, Pol. *jeść* are later generalisations from prefixed forms like OR сънѣсти, where no jot-prothesis could take place (cf. Schenker 1995: 88, Winslow 2022: 302 fn.14).

1.4.2 Jers before *j

As explored more fully in Winslow (2022: 313-315), OCS spellings seem to suggest that free-variation between <ь,ъ> and <и,ъı> was a feature of the pre-Jer Shift Urkirchenslavisch orthographic system for conveying the reflexes of the sound-groups *ьj and *ъj (so-called 'tense jers'), regardless of whether they were in strong or weak position. The examples given were: Zogr.

ⰸⱀⰰⰿⰵⱀⱐⱑ {znamenĭě}vs ⰸⱀⰰⰿⰵⱀⰻⱑ {znamenie} <*znamenьjĀ̈, ⱆⰴⰰⱃⱐⰺ {udarĭï} <*udaŕь-jь vs ⱁⰿⱁⱍⰻⰺ {omočiı} <*omočь-jь; Mar. ⱁⱄⱘⰴⱀⱅⱏ ⰺ {osǫdɴtŭ ı} vs ⱁⱄⱘⰴⱀⱅⱏⰺ ⰻ {osǫdɴtŭı i} <*osǫdętъ ‿jь; KF ⰿⰻⰾⱁⱄⱅⱐⱙ {milostĭǫ̈} vs ⰿⰻⰾⱁⱄⱅⰻⱙ {milostiǫ̈} <*milostьjǫ, ⰲⱐⱄⰵⰿⱁⰳⱏⰺ {vĭsemogŭı} vs ⰲⱐⱄⰵⰿⱁⰳⱏⰻⰺ {vĭsemogŭiı} <*vьxomogy-jь). Of importance here is the fact that the same orthographic system characterises both pre- (i.e. KF) and post-Jer Shift texts; that even in strong position in a text like Zographensis, which shows pretty clear signs of having undergone the Jer Shift, spellings like ⱆⰴⰰⱃⱐⰺ {udarĭï}, ⰱⱁⰾⱐⰺ {boĺĭı}, ⰲⱀⱋⱐⰺ {vɴštĭı} for what in the live dialect underlying Zogr. must surely have been /udaŕij/, /boĺij/, /vęštij/, are not infrequent[12]. The fact that the same alternation occurs in the pre-Jer Shift KF (i-stem gen. plurals ⰸⰰⱂⱁⰲⱑⰴⱐⰺ {zapovědĭı} <*zapovědьjь vs ⰾⱓⰴⰻⰺ {lüdiï} <*ĺudьjь) suggests that it is a common inheritance from the Urkirchenslavisch spelling system, and thus that in pre-Jer Shift Slavic the difference between /ь ъ/ and /i y/ was neutralised before /j/, and we should perhaps posit archiphonemes (which I call /Î/ and /Ŷ/) in this position. These archiphonemes are, in slightly different terms, effectively posited by Trubetzkoy (1954: 70) in his analysis of the Urkirchenslavisch phoneme-system[13].

However, for simplicity and accessibility's sake it's better to avoid overburdening the indexing-system with unfamiliar and controversial archiphoneme-symbols, so I keep *ьj/*ъj as the denotations for these groups.

Difficulties arise though when deciding how to denote foreign sources of /ij/[14] which may or may not have been integrated into the native system as reflexes of /Îj/: words like мариꙗ < Μαρία, стадии < στάδιον, which are well-integrated into the morphological system as a fem. ja-stem and masc. jo-stem respectively, could either be reconstructed as consciously-foreign *marijĀ̈, *stadijь, or as nativised *marьjĀ̈, *stadьjь, but there are no occurrences of jer-spellings in these words in the OCS texts in TOROT. Other similarly-Greek words like диꙗволъ (< διάβολος), however, do show up in OCS with jer-spellings: Supr. дыꙗ́вола, Zogr. Luke 8 and Psal. Psalm 108 ⰴⱐⱑⰲⱁⰾⱏ {dĭěvolŭ}, which (alongside the modern Macedonian *ѓавол* with the reflex of *ђ produced by the Macedonian so-called 'new jotation' of /d/ after the fallen jer brought it into contact with /j/) clearly suggest an early adaptation of this foreign /ij/-group to native /Îj/. Old Russian texts even show spellings of мариꙗ suggestive of full nativisation: Laurentian Primary Chronicle м҃рьꙗ, м҃рьєю, Zadonshchina марꙗ, марьꙗ, as well as First Novgorod Chronicle gen. sg. васильꙗ (jo-stem василии < Βασίλειος).

Since we can't ever be sure of the precise timing or route by which these late borrowings entered the various Slavic dialects, or of the extent of their adoption by Slavs beyond a tiny and often Greek-knowing scribal-class, the best solution is to set all such foreign /ij/ groups apart from the native vocabulary by using an *ij reconstruction, even where we can be pretty sure that early nativisation to reflexes of *ьj occurred: *dijĀ̈volъ, *vasilijь, *marijĀ̈ etc.

<u>1.4.3 Word-initial *jь-/*ji-/*i-</u>

With native Slavic word-initial *ji-/*jь-, I follow Derksen's (2008: 16) practice of writing *jь-, even though Derksen himself (2003) has argued for a split between *ji- and *jь- conditioned partly by accentological factors (which, as stated above, I have chosen not to consider). Most of the modern languages reflect these groups as just /i-/, except for Czech and Ukrainian: forms like Cz. *jdou* and

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12 Marianus and Psalterium Sinaiticum, on the other hand, frequently show a Russian-style /ej/ reflex of strong tense *ьj: Psal. ⰲⱃⰰⰱⰵⰺ {vrabeı} <*vorbьjь, ⱂⰾⱏⱅⰵⰺ {plŭteı} <*plъtьjь, ⰿⱐⱀⰵⰺ {mĭneı} <*mьńьjь; Mar. ⰸⰰⱂⱁⰲⱑⰴⰵⰻ {zapovědei} <*zapovědьjь, ⱆⰴⰰⱃⰵⰻ {udarei} <*udaŕь-jь, ⰳⰲⱁⰸⰴⰵⰻⱀⱏⱔ {gvozdeinŭę} <*gvozdьjьny-ję̌. Psal. (Psalm 21) even has a past act. part. Nsg. masc. def. form ⱄⱅⱃⱏⰳⱁⰺ {strŭgoı} <*jьstrgъ-jь that suggests an /oj/ reflex of *ъjь.

13 Though Trubetzkoy, like me, believes Urkirchenslavisch to have been based on a /j/-less dialect, so in that particular system the archiphonemes would be conditioned by the position before *vowels*, rather than before /j/.

14 The sequence /ij/ is not totally banned from native words, since it appears to be preserved across morpheme-boundaries, such as in prefixed-verbs like приꙗти <*prijęti or long-form adjectives like masc. nom. pl. дроугии <*drugi-ji, but within roots it does seem restricted to these post-LCS loanwords.

Ukr. (after vowels) *йдуть* appear to have dropped the weak-jer in *jьdǫtь just like any other and retained the /j/, and Ukr. *ськати* <*jьskati (with the restricted meaning ‘look for nits/fleas in someone's hair’ after the base-meaning ‘seek’ was taken over by the Polonism *шукати*) shows the expected Ukr. softening of the /s/ after fallen weak-jer in *ьsk groups (cf. *польський*).
I make an exception for certain forms of the personal-pronoun *jь, however, and write *jimь, *jima, *jixъ *jimъ and *jimi for the masc/nt. instr. sg. and dat./instr. dual/pl., because Czech here has *jim jich jimi*.
In badly-integrated clearly post-LCS foreign words, such as Biblical names like иꙗковъ (borrowed via Gk. *Ἰακώβ*), or ıꙉемонъ (< *ἡγεμών*), I keep a bare initial *i-, though this is rather an arbitrary choice and done partly as a way of marking such words as non-native[15] (cf. my treatment of foreign initial *e- below). An exception is made for исоусъ < Gk. *Ἰησοῦς*, which I have as *jisusъ, because of the greater likelihood that Slavs will have heard of Jesus even before the first biblical translations, and because spellings like Zogr. ⰽⱏⰹ ⰺⱄⱛ {kŭı ĭsü} suggest that it causes the same /Ŷ/ archiphoneme reflex of *ъ before *j as you get in e.g. native Mar. ⰲⱏⰹ ⰻⱄⱅⰻⱀⱘ {vŭı istinǫ} < *vъ ‿ jьstinǫ (see above).
Prefixed forms like *do-jьti 'to come, arrive' for morphological reasons have to be distinguished from the class 4 verb *dojiti/dojiši/dojimъ etc. 'to breastfeed' (and its derived noun *dojidlika), a difference which is reflected in the modern Ukrainian *дійти* (<*dojьti with compensatorily-lengthened /o/ > /i/) vs *доїти*. Thus /i/ can follow /j/ when the former is part of a morpheme which just happens to be stuck onto a /j/-ending stem: I similarly allow words like *šujika (шюица) and *vojinъ ‘warrior’ (воинъ, as opposed to *vojьnъ, the gen. pl. of *vojьna), or the loc. sg./pl. desinences of any jo-stem noun whose stem ends on /j/, e.g. Psal. ⰶⱃⱑⰱⰹ {žrěbïı} <*žerbьji.

1.4.4 Word-initial *je-/*e-
No Glagolitic text makes any effort to distinguish /je/ (after vowels or word-initially) from post-consonantal /e/, writing both with <ⰵ>, unlike the situation with the reflexes of *ję vs *ę, where in Zogr. and Mar. and partially in Assem. (Velcheva 1981: 168) the full front-nasal digraph <ⰵⱔ> is reserved for *ję, while just the second 'nasalising component' <ⱔ> is used for post-consontal *ę, e.g. Mar. 3rd pl. aorist ⰵⱔⱄⱔ {ęsN} <*jęsę, as opposed to KF ⱂⱃⱐⰵⱔⱅⰻ {prıęti} <*prijęti vs ⰲⱏⰸⰵⱔⰾⱐ {vŭzęlı} <*vъzęli[16].
Glagolitic evidence alone therefore would suggest that foreign borrowings with word-initial /e-/ were simply adapted to whatever the reflex of native LCS *je was. Suprasliensis, though, which uses the jotated <ѥ> letter, does in fact make an extremely consistent spelling distinction between foreign borrowings and native Slavic words: of the 157 occurrences of the 13 foreign lemmas I have so far reconstructed with word-initial *e/*je- which appear in Supr. (*episkupъ, evanǵelьje, eǵüpьtъ, elisavetь, elinъ, evanǵelistъ, eǵüpьtьskъ, elinьskъ, episkupьstvo, evrejьskъ, elisejь, emъmausъ, etijopьskъ*), the only spellings with <ѥ> are ѥлисеи, Ѥ҃ппъ, ѥ҆лини, and ѥ҆лина, i.e. 4/157 or 2.5%. By contrast, of the 3172 native Slavic words in Suprasliensis which I Autoreconstruct as starting with *je- (not all of whose *lemmas* start with *je-, e.g. forms of **byti*), just 88 are written with initial <є>, vs 3070 with <ѥ>[17]. Thus 97.2% of native word-initial *je- in Suprasliensis is spelt with <ѥ>, while

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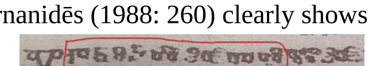
15 Spellings like Zogr. Mark 13:3 “ⱂⰵⱅⱃⱏ. ⰺ ⱑⰽⱁⰲⱏ. ⰺ ⱁⰰⱀⱀⱏ.” {petrŭ. ï ěkovŭ. ï oannŭ} “Peter and Jacob and John” would suggest that this initial *i- can get dropped after an /i/ of a preceding word, but whether this points to a dropping of the non-native *i-, simple deletion of a double /i i/ (haplology), or a native-like reflex of a weak-jer /*i *jьjǢkovъ/ > /i jakov/, is not really knowable, so indexing such words with a markedly foreign initial *ij- group is again the best way of allowing such difficult cases to be investigated.

16 Psalterium Sinaiticum contains at least five occurrences of non-digraph <є> for *ę: Eckhoff’s digitisation suggests ⰱⱃⱑⰿⱔ {brěmN}, ⰲⱏⰸⰾⰰⱍⱔ {vŭzlačN}, ⱄⱅⱐⰷⱔ {stĭʒN}, ⱅⱔ {tN}, ⱀⱔ {nN}, and ⱂⱃⱁⰽⰾⱔⱅⱏ {proklNtŭ}, but the last one ⱂⱃⱁⰽⰾⱔⱅⱏ (which is from Psalm 151 in the part discovered in 1975) comes from a mistake in Mareš’s (1997: 50) Cyrillicised edition: the manuscript-photograph in Tarnanidēs (1988: 260) clearly shows ⱂⱃⱁⰽⰾⰵⱔⱅⱏ {proklętŭ}:



17 The leftover 14 are things like 1st. pres. dual. ѩ҆мавѣ which Eckhoff’s corpus wrongly lemmatises as имати instead of имѣти, and which thus get reconstructed as *jemĺevě instead of *jьmavě. At the time of writing only 3227/6862 Suprasliensis lemmas have been reconstructed, but those 3227 cover 89713/99194, or 90.4%, of the words.

97.5% of the occurrences of the clearly post-LCS Greek-mediated foreign borrowings listed above instead use plain <є>, suggesting that *some* sort of difference was felt, at least by the scribes of Suprasliensis, and that we probably shouldn't index these with the same *je- as used for native forms. I therefore use non-jotated *e- for such foreign borrowings, and the extent to which they take prothetic *j- and fall together with the native vocabulary is left as something for investigators to determine based on the evidence of each manuscript.

1.5 Prefixes and consonant-clusters

The last particularity of my LCS indexing-system worth mentioning relates to the handling of consonant-clusters in prefixes: as exhaustively exemplified by Diels (1963: 121-125), Common Slavic permitted only a restricted set of consonant-combinations in the syllable onset, generally either combinations of the continuants *s/*z plus obstruent or sonorant (except *r, see below), or of obstruents plus sonorant (with some curiosities such as the seeming dialectal diversity in the tolerance of *bn but not *pn: OCS гъıбнѫти <*gyb-nǫti > Ukr. *гинути*, vs OCS оусънѫти <*usъp-nǫti (cf. 3sg. aor. оусъпє), Ru. *тонуть* <*top-nǫti, though see Meillet (1965: 142)). Geminate consonants were banned and either simplified (исѣшти <*jьs-sěkti) or dissimilated (процвисти <*prokvit-ti).

The ban on *sr/*zr is dealt with by insertion of *t and *d respectively, but the commonly-cited examples of *str <*sr (сєстра, строуıа, остръ) all concern root-internal *sr where insertion of *t is common also to the Germanic and sometimes Baltic cognates. The examples given by Meillet (1965: 136) include: (for строуıа) Lith. dial. *srauja* next to Latvian *strauja*, then Germanic *straum- (> Eng. *stream*, Old Norse *straumr* etc.); (for остръ) Lith. *aštrus*, Gk. *ἄκρος* (here the *s is from PIE *ḱ). As Meillet says, "*ce n'est pas un developpement germano-balto-slave; d'une part, le developpement d'un* -t- *dans le groupe* sr *est chose naturelle et se retrouve ailleurs (fr. pop.* castrole *de* casserole*) et, d'autre part, le developpement de* t *en ces conditions n'est pas general en baltique:* str *est regulier en lette, mais* sr *subsiste couramment en lituanien.*", so we can't really be sure when the Slavic change took place or whether it was still active during our LCS stage. The only indication of its activity in OCS is the single Psal. ⱄⱅⱃⰰⰿⱁⰿⱐ {stramomĭ} <*sorm-omь spelling cited by Diels (1963: 122); otherwise new /sr/ from metathesised *sErC groups is tolerated unchanged.

New occurences of *zr, on the other hand, are regularly generated in the language right up to OCS times, not only in the derivational-morphology because of the verb-prefixes *orz-, *vъz-, *jьz- (e.g. Supr. 3sg. aor. въздроу 'roared', from *vъz-ruti), but also because of the clitic prepositions *jьz and *bez, which form one phonological word with whatever follows them and thus cause OCS spellings like Mar. Luke 1 ⰻⰸⰴⱃⱘⰽⱏ {izdrǫkŭ} <*jь.z ‿rǫ.kъ. Meillet (p.136) also cites the Old Polish adverb *zdręki* <*jьz ‿rǫ.ky, which proves that the phenomenon is not limited to SSl. or OCS. Curiously, though, despite this overwhelming evidence of a synchronic /zr/ > /zdr/ rule in OCS, /zr/ from the metathesised *zork- root is never spelt <здрак> and so seems to be tolerated, even though Diels (p.122) cites prepositional forms like Supr. беꙁдраꙁоума <*bez ‿*orzuma, бєздрала[18] <*bez ‿*ordla, which come from metathesised *orT- groups but *do* show inserted /d/. Such inconsistency is hard to explain unless the addition of /d/ has been partly morphologised as a variant of specifically the prepositions before /r/.

With such a sound-change that appears most often at morpheme or straight-up word-boundaries, there is a strong drive to restore the underlying shape of the constituent parts, hence the modern languages have mostly restored /zr/ groups in e.g. Russian *разрешить*, and there are traces of this even in Psalterium Sinaiticum: Psalm 48 ⰻⰸⱃⱘⰽⱏⰺ {izrǫkŭı} (Diels 1963: 122). In Old Russian, the Uspenskij Sbornik is pretty consistent in keeping prefixed verb-forms like раздрүшить <*orzrušitь, but by the time of the Laurentian Codex we get forms like възрадүєм and нєизрєчєнноє.

Therefore even though *sr > *str and *zr > *zdr appear to be simply voiced and unvoiced variants of the same sound change, the practical effects are very different because the former is, from the

18 For some baffling reason this form is missing from Eckhoff's text, so I link instead to the relevant folio of David Birnbaum's online edition.

LCS perspective, totally 'opaque', since it only occurs in roots and thus is not analysable by speakers into constituent morphemes without the inserted stop, in the way that /bez ‿drǫky/ can be identified with separate /bez/ and /rǫky/.
For this reason I don't include /zdr/ <*zr at prefix or preposition-boundaries in my LCS system, so that investigators can see for themselves the extent of each text's adherence to the expected phonological development vs restoration of /zr/ under morphological pressure.

Following the same logic I also retain illegal *ss and *sš groups in prefixed-verbs like Psal. ⰺⱄⱔⱍⰵ {isęče} <*jьs-sęče, Mar. ⰻⱎⰵⰴⱏ {išedŭ} <*jьs-šьdъ, Assemanianus ⱃⰰⱎⰹⱃⱑⱘⱅ {rašırěǫ̈t} <*ors-širǼjǫtъ, because that same drive towards restoration of the underlying shapes of the prefixes *jьs-/*ors- etc. can be seen in modern Russian spellings *иссякнуть*, *расширять*, and Laurentian Codex расщедше.[19] This treatment is also more consistent with my handling of verbs like *jьs-kěliti (> OCS ицѣлити/исцѣлити) where simplification *must* have occurred posterior to our pre-PV2 LCS stage (since /sk/ is always totally permissible), and where manuscripts show great diversity, e.g. Zogr. and Assem. consistently have ⰺⱌⱑⰾ- {icěl-} while Mar. and Psal. keep ⰺⱄⱌⱑⰾ- {iscěl-} (more discussion of the wider Slavic reflexes of this group, including the OCS <ст> spellings, can be found in Meillet 1965: 133).

1.6 Morphological innovations that scupper LCS reconstruction

The units of the phoneme-system sketched above serve as the building-blocks for all higher-level linguistic systems, most immediately the inflectional-morphology and derivational-morphology systems, whose features are thus constrained by said phoneme-system and the distributional-restrictions of its units (i.e. *phonotactics*). Changes which occur in the phoneme-system between the time of our theoretical LCS and the time of our texts can therefore trigger (or allow) restructuring of these morphological systems, which in turn can produce forms containing phoneme-sequences with no possible direct LCS ancestor-sequences.

An example of such morphological change contingent upon structural phonological change, leading to forms which preclude any direct LCS-stage reconstruction, is the replacement of i-stem endings with those of the corresponding jo- or jā-stems, in nouns whose stems end on labials or the subset of LCS dental consonants which lack palatal counterparts, viz. /d t s z/[20]. Evidence for such a change is furnished by the Old Russian masc gen./acc. form татѧ from the 1229 Treaty between Smolensk, Riga and Gotland (Version A). LCS *tatь is a masc. i-stem noun with genitive *tati, as it still appears in the Suprasliensis translation of John Chrysostom's Homily for Holy Thursday (…то кажетъ владꙑкꙑ чловѣколюбьѥ· ꙗко прѣданика разбоиника тати…), but in the dialect underlying the 1229 Treaty the rise of phonemically palatalised /t'/ after the Jer Shift means that the stem (and the nom. sg. тать /tat'/) of this noun now ends on the same class of "soft" consonants as original

19 Conversely, sequences of *sk, *zg at prefix-boundaries which show PV1 reflexes, like Mar., Zogr. ⱃⰰⱎⱅⱏⱅⰵⱅⱏ {raštĭtetŭ} <*orš-čьtetъ (ECS *skĭt- > *ščĭt-), Psal. ⱃⰰⰶⰴⰹⰷⰰⰵⱅⱏ {raždiʒaetŭ} <*orž-ǯigajetъ (ECS *zgīg- > *žǯīg-) are kept as *šč, *žǯ. Such forms may well not go all the way back to the time of PV1, and instead be just the result of a synchronic rule prohibiting /zž/ and /sč/ (> /žǯ/ and /šč/) that remained active until much more recently, especially given prepositional-phrase forms like Psal. ⰻⱎⱅⱃⱑⰲⰰ {ıštrěva} <*jьs‿*červa, so this is arguably inconsistent with my treatment of *ss, *sš etc. My justification is firstly that *sk, *zg > *šč, *žǯ are *conspicuously* PV1-changes, which we *know* originated well before our target LCS point, whereas the precise timing of de-gemination or simplification of *sš is less clear-cut; and secondly that even in languages like Russian which *orthographically* have restored <сч> and <зж> spellings in compounds like *исчезнуть* and *разжечь*, the pronunciations are still arguably direct reflexes of LCS *šč and *žǯ, viz. [ɕː] and [ʑː] (or, in the conservative Moscow-dialect, the palatalised [ʑː] found also in *дождь* <*dъžǯь).

20 In some dialects (notably East Slavic) the PV3 reflexes *ś and *ʒ became patalised counterparts to plain /s z/, i.e. /s' z'/, and merged with the /s' z'/ that developed from LCS *s,z before front-vowels, but in most OCS they seem to have just hardened to /s, z/: searching my database for the sequence *ьхǫ, for example, turns up exclusively <ⱐⱄⱘ> spellings in Marianus, just one <ⱐⰷⱘ> in Zogr., and exclusively <ьсѫ> in Suprasliensis, with only Assem. and Psal. containing a significant minority of <ⱐⰷⱘ> spellings.

jo-stem nouns like *końь > /kon’/, where the original LCS palatal *ń has fallen together with secondarily-palatalised /n'/ from plain LCS *n before LCS front-vowels, in e.g. the original i-stem *bornь > /boron'/. This system thus no longer distinguishes between descendants of the original LCS palatals and the newly secondarily-palatalised consonants like /t'/: both are now together in the set of 'soft' consonants, opposed to their 'plain' or 'hard' counterparts, and so tend towards taking the same set of inflectional endings (in this case those of the original jo-stems)[21]. Consequently, a word like тать has begun to take jo-stem endings, including the Old Russian /a/ reflex of LCS *Ǣ in the genitive/accusative singular.
LCS /Ǣ/, though, by definition can only occur after LCS palatal consonants (see above), so a reconstruction *tatǢ is just nonsensical. In the case of the dat. sg. /u/-desinence (which isn't attested in our Treaty but exists in modern Russian *татю*), we don't even have an LCS archiphoneme available to signal a preceding soft-consonant; there's simply no way of getting from LCS *tatu to Russian /tat'u/, because such a form was only made possible by the rise of phonemic /t'/, so our ability to index it with our LCS system is gone.

Were the same shift from i-stem to jo-stem to occur in a word like *zvěrь, then the structural change would not be so catastrophic, because our LCS system *does* contain a palatal *ŕ which any allophonically-softened LCS hard *r could easily be subsumed into. Indeed, interestingly Suprasliensis does in fact contain 3X gen. sg. звѣрѣ, with what looks like a jo-stem reflex of *ŕǢ (spelt with jat’ as an overhang of the Glagolitic tradition, cf. 2X морѣ vs 1X морꙗ spellings), suggesting that Russian-style secondary palatalisation of *r > /r’/ may have occured in the Bulgarian dialect underlying it[22]. You don’t, though, get anything like тата[23], so the system-wide development of secondary-palatalisation does not seem to have advanced enough to have caused the sort of fundamental structural reorganising which shifted *tatь into the jo-stems in Old Russian.

Forms like тата, then, though they frustrate our goal of reconstructing entire texts, do provide us some objective measure of ‘linguistic distance’ between stages of a language, because their existence presupposes at least one intervening stage where the structure of the phonological system has changed enough from our LCS stage to have caused/allowed restructuring of the morphological system.

2 LCS Morphology and the Autoreconstructor

2.1 The existing TOROT data

The ten-place morphology-tags included as part of the word-level annotations in Eckhoff’s TOROT corpus constitute a veritable goldmine of linguistic data, because, based as they are on the *form* of a word rather than the *function*, they bridge the gap between the higher (syntax, semantics etc.) and lower (phonology, orthography, morphology) levels of linguistics analysis. An example TOROT annotation for the word ⰲⱏⰸⰲⱑⱋⱘ {vŭzvěštǫ} is given below:

---

21 Russian feminine i-stems like *весь* (<*vьsь, ‘village’) do not fall together with ja-stems in the way the masculines like *zvěrь fall together with jo-stems, but they do all still take the /am, ax, ami/ endings in the dat., loc. and instr. pl., e.g. *ве́сям*, which contain the same LCS ja-stem *-Ǣ- vocalism which can only occur after LCS palatals, meaning they too end up totally unreconstructable due to an illegal **sǢ sequence.

22 Numerous spellings in Supr. like боура <*buŕǢ, оукараѥтъ <*ukaŕǢjetъ, мороу <*moŕu etc., however, point to a hardening of LCS palatal *ŕ to plain /r/, so it’s difficult to know whether звѣрѣ spellings stem from a genuine /zvěŕa/ form in the history of the language, or if they instead represent synchronic /zvěra/, i.e. with a hard o-stem ending, but with confusion by the scribe between <ра> and <рꙗ>/<рѣ> spellings for what in his/her dialect would’ve all been /ra/.

23 Except the numerous gen. sg. господа and dat. sg. господоу for the i-stem *gospodь, but this word seems to be an isolated special case, because it bafflingly turns up even in early Glagolitic OCS with endings like ⰳ҃ⰲⰻ {g̃vi} (ju-stem dat. sg. *-evi, cf. Supr. господеви), ⰳ҃ⱓ {g̃ü} (jo-stem dat. sg. *-u), and ⰳ҃ⱑ {g̃ě} (jo-stem gen. sg. *-Ǣ). See Van Wijk (1929).

```
<token id="3589172" form="възвѣштѫ" citation-part="70.17"
lemma="възвѣстити" part-of-speech="V-" morphology="1spia----i"
relation="pred" presentation-after=" "/>
```

Figure 1: TOROT annotation for Psal. Sin. Psalm 70 ⰲⱏⰸⰲⱑⱎⱅⱘ *in XML format*

TOROT token XML-tags include various attributes, but for the Autoreconstructor all that's needed are *form*, *lemma*, *part-of-speech*, and *morphology*. The *form* attribute is used to check for morphological variations/innovations, to ensure that what gets produced is the direct phonological ancestor of the actually-occurring word (see below for more about this deviance-detection). The *lemma* and *part-of-speech* attributes, when concatenated, serve as a unique key linking each word to its lemma[24] and thus to its LCS reconstruction and inflection-class information[25]. Finally the *morphology* attribute consists of a 10-character string to hold values for the 10 morphological-features used by TOROT (and the wider PROIEL corpora). Not all features are relevant for all words, in which case a dash '-' is used as a placeholder.
A detailed explanation of each feature can be found in Section 6 of Eckhoff et al. (2018: 41), but here it suffices to say that in this example the tag "1spia----i" is telling us that ⰲⱏⰸⰲⱑⱎⱅⱘ {vŭzvěštǫ} is **1**st person, **s**ingular, **p**resent-tense, **i**ndicative-mood, **a**ctive-voice, has no gender, case, degree, or strength features, and is **i**nflectable rather than non-inflecting.

Of importance here is the *present* tense tagging, even though възвѣстити (even in OCS) can be taken as a perfective verb, opposed to its imperfective counterpart възвѣщати, and thus has future-tense meaning in its non-past indicative forms (and the Greek Septuagint here has ***ἀναγγελῶ*** *τὰ θαυμάσιά σου*, with a morphologically future-formed *ἀναγγελῶ* 'I will proclaim' from *ἀναγγέλλω*). The tagging thus follows the *inflectional*-morphology of ⰲⱏⰸⰲⱑⱎⱅⱘ {vŭzvěštǫ}, rather than the future-meaning which is carried by the *derivational*-morphology. This is important because the Autoreconstructor works by *inflecting* LCS lemmas according to those morphology-tags; the annotation gives no information about its derivational-morphology or derivational relationship to its imperfective (and thus 1sg. *present* tense) counterpart ⰲⱏⰸⰲⱑⱎⱅⰰⱙ {vŭzvěštaǫ̈}, since that is annotated with a separate lemma.
An even clearer example where this *form* over *function* annotation helps us is with the accusatives of animate nouns: it's well known that a type of so-called differential-object-marking is beginning to take hold in Early Slavic, whereby syntactic accusatives use genitive endings to varying extents as a way of encoding the semantic ensouledness (and possibly definiteness) of the noun (the details aren't important; for a recent thorough diachronic treatment of the topic see Eckhoff 2022), e.g. Supr. раꙁбойника въ породѫ вьведе 'he brought the robber into paradise'. If these were all tagged as accusatives, the Autoreconstructor would produce the wrong ancestor to the text-form, i.e. *orzbojьnik-<u>ъ</u>, because the semantic information needed to decide whether this differential-object-marking is needed is not available. Happily though all such animate-accusatives are marked as genitive in TOROT: the morphology-tag for раꙁбойника above is "-s---m**g**--i", so the Autoreconstructor produces *orzbojьnik-<u>a</u>.

<u>2.2 Computerised LCS inflectional-morphology and the Autoreconstructor</u>

The Autoreconstructor reads the morphology-tag character-by-character and numerifies each field, and from that it computes a number which corresponds to the row of the inflection-table where the endings for that particular word's inflection-class are stored. For verbs this will be a number

24 Identical lemmas with the *same* part-of-speech tag, such as вести 'to lead' <*ved-ti and вести 'to drive' <*vez-ti, both of which have 'V-' for verb, are differentiated by appending #2 etc. to the extra homomorphs, i.e. вести vs. вести#2.

25 The spreadsheet containing my reconstructions and inflection-class annotations for the TOROT OCS lemmas can be downloaded from https://github.com/12401453/torot_2023/blob/main/lemma_lists/chu_lemmas_master.xlsx

between 1 and 44 (9 person/number combinations for each of present, aorist, imperfect, and imperative, plus 8 non-finite forms, 9*4+8), and for nominals between 1 and 63 (7 cases * 3 numbers * 3 genders). A version of the function which actually implements this tag-reading process can be seen in the OcsServer::numerifyMorphTag() function here: https://github.com/12401453/ocs_server/blob/main/OcsServer.cpp#L2714.[26]

Currently each inflection-class has up to three tables associated with it, the first of which is full (i.e. contains 44 or 63 entries) and holds the 'basic' or 'correct' (from the LCS perspective) endings. The second and third tables are 'sparse', in that they only contain entries for those parts of the paradigm where we expect to encounter alternative forms, with those I consider 'deviant' or 'innovated' held in table 2, and 'alternative' but still 'correct' endings (i.e. LCS allomorphs) in table 3. An example of some of the endings in the three tables for basic class 1 verbs like *reħi is given below:

```
inner_map v_11_c0 = {
{1, "ǫ"},
{2, "eši"},
{3, "etQ"},
{4, "evě"},
{5, "eta"},
{6, "ete"},
{7, "emъ"},
{8, "ete"},
{9, "ǫtQ"},
{10, "ъ"},
{11, "e"},
{12, "e"},
{13, "ově"},
{14, "eta"},
{15, "ete"},
{16, "omъ"},
{17, "ete"},
{18, "ǫ"},
```

```
inner_map v_11_c1 = {
{6, "eta"},
{10, "oxъ"},
{13, "oxově"},
{14, "osta"},
{15, "oste"},
{16, "oxomъ"},
{17, "oste"},
{18, "ošę"},
{24, "ěašeta"},
{37, "ǫtj"},
};
```

```
inner_map v_11_c2 = {
{10, "sъ"},
{13, "sově"},
{14, "sta"},
{15, "ste"},
{16, "somъ"},
{17, "ste"},
{18, "sę"},
};
```

*Figure 2: Inflection-endings for class 1 verbs like *reħi, *greti stored using C++ std::unordered_map<int, std::string>*

Here the 'deviant' endings consist mostly of the 'extended S-', or *-ox- aorists (e.g. Assem. ⰐⰀⰓⰅⰍⰑⰞⰣ {narekošɴ}), while the third 'alternative' table contains the primary S-aorist endings (e.g. Assem. ⰒⰑⰃⰓⰡⰔⰣ {pogrěsɴ} <*pogrěb-sę). Clearly those primary S-aorist endings just being stuck onto a stem like *greb- would not produce the correct LCS form, so for certain inflection-classes there are post-processing functions which do things like replace all "ebsę" with "ěsę", or (for stems which end on RUKI-consonants like *rek-) all "eks" with "ěx". The full routines for verbs can be found in https://github.com/12401453/ocs_server/blob/main/autoreconstructor/verb_cleaning.h; to a large extent this is just enforcing the Slavic consonant-cluster restrictions discussed in Section 1.5.

The inflection-tables themselves are indexed by integer-keys associated with each inflection-class, such that the 'correct' table's key always ends on '1', meaning that shifting to the alternative tables is a matter of simply incrementing the key by either 1 or 2, and I can keep track of whether or not a form has required a deviant or alternative ending by inspecting the final value of this key[27]:

26 Extra handling is required for forms of *byti, since that has a separate future-paradigm (and future-participle) which TOROT does actually specify separately with an 'f' value for the *tense* feature, as well as two variant imperfect sets (3sg. *bě vs. *běaše, which aren't tagged, but which I detect), and a 'conditional' *bimъ, *bi, *bǫ etc. Participles require an extra step to read their nominal-features and add their adjective-like endings.

27 This is bad design and prevents me from easily handling more than one type of morphological innovation per class: for the *masc_jo* class nom. pl. I have the i-stem *-ьje ending in the 'deviances' table (as found in e.g. Supr. стражиѥ <*storž-ьje 'guards'), which leaves no room for forms like Supr. зноѥвє <*znoj-eve, Psal. ⰸⰿⰹⰵⰲⰵ {zmıeve} <*zmьj-eve, which have the *-eve ending from the ju-stems.

```
std::unordered_map<int, inner_map> verb_ = {

  {111, v_11_c0},
  {112, v_11_c1},
  {113, v_11_c2},
```

Figure 3: Integer-keys of the three inflection-tables in Figure 2

Multiple inflection-classes can share the same tables, for instance the *masc_o*, *nt_o*, and *fem_a* noun-classes all take endings from the *adj_hard* table, since the same set of 63 endings (21 for each gender) is used in all four paradigms[28].

Full paradigms for the 4,500-ish OCS lemmas I have so far reconstructed can be dynamically generated here https://ocstexts.co.uk/words; these are constructed from LCS lemmas in the same way the Autoreconstructor reconstructs individual forms, except it produces every form in the paradigm, rather than just the one specified by a text-word's morphology-tag. The LCS forms are then converted into 'normalised' OCS by the browser using the convertToOCS() function here: https://github.com/12401453/ocs_server/blob/main/HTML_DOCS/LCS_to_OCS.js#L197. Forms generated from the "deviant" and "alternative" tables are included, but the former are given less prominent styling to identify them as (probable) post-LCS innovations.

28 The long-form adjectives (and participles) are then formed by just sticking the inflected-form of the pronoun *jь onto the end, since this is the origin of such long-forms (Vaillant 1942). I agree with Townsend and Janda (1996: 178) that some simplification of these concatenated endings must have occurred by LCS, especially where the short-form endings contain *m or *x (e.g. the dat. and loc. pl. for all genders), because it's unreasonable that e.g. Mar. Matt. 24 masc. loc. pl. ⱀⰵⰱⰵⱄⱏⱄⰽⱏⰹⱈⱏ {nebesŭskŭıxŭ} could've developed directly from *nebesьskěxъjixъ (here not least because of the lack of PV2-reflex). I am however relucant to adopt the (undiscussed) simplified LCS forms they present in the tables on p. 182-3 before I've done a more thorough investigation of early manuscript-forms, so for now I've left all such adjectives wholly uncontracted (though the Autoreconstructor does actually mark long-adjectivals that contain such problematic concatenations and this information could be used to exclude them from searches).

sъtŕ Random

Corpus forms

	Present	Aorist	Imperfect	Imperative
1st sg.	сътьрѫ	сътрьхъ	сътьрѣахъ	сътьрѣмь
2nd sg.	сътьреши	сътьре сътрь	сътьрѣаше	сътьри
3rd sg.	сътьретъ	сътьре сътрь	сътьрѣаше	сътьри
1st du.	сътьревѣ	сътрьховѣ	сътьрѣаховѣ	сътьрѣвѣ
2nd du.	сътьрета	сътрьста	сътьрѣашета	сътьрѣта
3rd du.	сътьрете сътьрета	сътрьсте сътрьста	сътьрѣашете сътьрѣашета	сътьрѣте
1st pl.	сътьремъ	сътрьхомъ	сътьрѣахомъ	сътьрѣмъ
2nd pl.	сътьрете	сътрьсте	сътьрѣашете	сътьрѣте
3rd pl.	сътьрѫтъ	сътьрѫ сътрьшѧ	сътьрѣахѫ	сътьрѫ

Participles	
PRAP1	сътьрꙑ
PRAP2	сътьрѫщь
PAP	сътьръ
L-Part.	сътрьлъ
PPP	сътрьтъ сътьренъ
PrPP	сътьромъ
Infinitive	сътрьти
Supine	сътрьтъ

Figure 4: Dynamically-generated paradigm for the OCS verb сътрьти, *based on the Autoreconstructor's computerised LCS inflectional-morphology*

The accuracy of the generated-forms can then be gauged by comparing them to tables populated only by forms which actually occur in Eckhoff's corpus-texts, using the 'Corpus-forms' switch:

sъtŕ Random

Corpus forms

	Present	Aorist	Imperfect	Imperative
1st sg.				
2nd sg.				
3rd sg.	сътъретъ сътьретъ	сътрь сътьре		
1st du.				
2nd du.				
3rd du.				
1st pl.				сътьрѣмъ
2nd pl.				
3rd pl.		сътръшѧ		

Participles	
PRAP1	
PRAP2	
PAP	сътъръ сътъръши
L-Part.	сътрълъ
PPP	сътърени
PrPP	
Infinitive	
Supine	сътрътъ

Figure 5: Forms of the same сътрьти *verb as they actually occur in Eckhoff's (Cyrillicised) TOROT corpus of Church Slavic texts*

An advantage of basing computer-generated paradigms on converted LCS (rather than directly on normalised OCS) is that it allows for the possibility of dialect or manuscript-specific conversions: e.g. Kiev Folia-flavoured paradigms that convert *ћ, *ђ to <ц, ꙁ> and use <ѣ> for *all* *Ǣ, or a Marianus-specific conversion that frequently shows vocalised strong-jers, or even an artificial pre-Jer Shift form of Old Russian that shows the denasalisation of *ę and its merger with *Ǣ via stochastic mixing up of <ꙗ, а> and <ѧ>:

	Sing.	Dual	Plural
Nom.	кънꙗꙁь кънѧгъ	кънѧꙁꙗ	кънѧꙁи
Acc.	кънѧꙁь кънѧгъ	кънѧꙁѧ	кънꙗꙁѣ кънѧгꙑ
Gen.	кънѧꙁꙗ	кънѧꙁю	кънꙗꙁь кънꙗгъ
Dat.	кънꙗꙁю	кънꙗꙁема кънꙗгома	кънѧꙁемъ кънагомъ
Loc.	кънꙗꙁи кънꙗꙁѣ	кънѧꙁю	кънꙗꙁихъ кънѧꙁѣхъ
Instr.	кънѧꙁьмь кънꙗгъмь	кънѧꙁема кънагома	кънꙗꙁи кънѧгꙑ
Voc.	кънꙗже	кънꙗꙁѧ	кънꙗꙁи

*Figure 6: Dynamically-generated paradigm of *kъnęgь but with an Old Russian rather than OCS conversion*

	Sing.	Dual	Plural
Nom.	кънѧѕь кънѧгъ	кънѧѕа	кънѧѕи
Acc.	кънѧѕь кънѧгъ	кънѧѕа	кънѧѕѧ кънѧгꙑ
Gen.	кънѧѕа	кънѧѕоу	кънѧѕь кънѧгъ
Dat.	кънѧѕоу	кънѧѕема кънагома	кънѧѕемъ кънагомъ
Loc.	кънѧѕи кънѧѕѣ	кънѧѕоу	кънѧѕихъ кънѧѕѣхъ
Instr.	кънѧѕемь кънагомь	кънѧѕема кънѧгома	кънѧѕи кънѧгꙑ
Voc.	кънѧже	кънѧѕа	кънѧѕи

*Figure 7: *kъnęgь with the canonical OCS conversion for comparison*

2.3 Detecting and dealing with morphological innovations

217066	приведошѧ	3paia----i	privedošę	privedǫ
217101	въздъхнѫвъ	-supamn-si	vъzdъxnǫvъ	vъzdъxъ
217112	развръзосте	3daia----i	orzvŕ̥zoste	orzvŕ̥zete
217261	б͠гвивъ	-supamn-si	bolgoslovivъ	bolgoslovĺь
217266	ѣшѧ	3paia----i	jǣšę	jǣsę
217272	вьзѧшѧ	3paia----i	vъzęšę	vъzęsę
217302	начѧшѧ	3paia----i	načęšę	načęsę
217316	въздъхнѫвъ	-supamn-si	vъzdъxnǫvъ	vъzdъxъ
217455	приведошѧ	3paia----i	privedošę	privedǫ
217600	начѧтъ	3saia----i	načętъ	načę
217605	с͠ноу	-s---md--i	synu	synovi
217635	начѧтъ	3saia----i	načętъ	načę
217787	поѩтъ	3saia----i	pojętъ	poję
217832	мосѣомь	-s---mi--i	mosijomь	mosijemь
217856	мосѣови	-s---md--i	mosijovi	mosiju
217869	быс	3saia----i	bystъ	by
217960	с͠нѣ	-s---ml--i	syně	synu
218067	невѣрьнъ	-s---mvpsi	nevěrьnъ	nevěrьne
218108	быс	3saia----i	bystъ	by
218173	нѣмы	-s---mvpwi	němъjь	němejь
218175	глоухы	-s---mvpwi	gluxъjь	glušejь
218198	быстъ	3saia----i	bystъ	by
218206	оумрѣтъ	3saia----i	umertъ	umer
218388	ıмени	-s---nl--i	jьmeni	jьmene
218419	ıмени	-s---nl--i	jьmeni	jьmene
218561	окомь	-s---ni--i	okomь	očesьmь
218648	мѫжю	-s---md--i	mǫžu	mǫževi
218688	мѫжа	-s---mg--i	mǫžǣ	mǫžu
218755	прѣлю͠бы	-s---fa--i	perĺuby	perĺubъvь
218762	поустивъши	-supafn-si	pustivъši	pušħьši

Figure 8: Auto-detected and -reconstructed morphological deviances from a small part of the Book of Mark in Codex Zographensis

The screenshot above shows some raw data from my autoreconstructed SQLite database of the TOROT OCS texts; in this case it's forms from Zographensis (around Mark 7 to Mark 10) where the Autoreconstructor has detected morphological innovations. The fourth column shows what the Autoreconstructor thinks is the direct phonological ancestor to the text-form, but the ancestor of the 'original', 'correct', or 'default' morphological form is also generated and stored in the fifth column, so that such cases of innovation can be easily searched-for and counted (since non-innovated forms have NULL values in this column).

Types of innovation detected here include:

- extended S-aorists of class 1 verbs: 3rd pl. приведошѧ vs. приведѫ, 3rd dual развръзосте vs *развръзете[29]
- unetymological extension of the RUKI-rule-produced *š in 3rd pl. primary sigmatic aorists: ѣшѧ vs. ѣсѧ <*jǢd-s-ę, вьзѧшѧ vs. възѧсѧ <*vъzьm-s-ę, начѧшѧ vs. начѧсѧ <*načьn-s-ę (neither *d, *m, nor *n have ever been RUKI sounds)
- extension of the *-nǫ- suffix to the past. act. part. of class 2 verbs like въздъхнѫти: въздъхнѫвъ (cf. Mar. ⱆⱄⱏⱈⱏⱎⱘ {usŭxŭšǫ} from оусъхнѫти)

29 Koch (1990: 293) lists only sigmatic aorists as possibilities for the *-verz-/*-vŕ̥z- stem verbs, and it seems that outside of the 3rd sg. (e.g. Psal. ⱃⰰⰸⰲⱃⱏⰸⰵ {razvrŭze}, Zogr. ⱁⱅⰲⱃⱐⰸⰵ {otvrĭze}) no root-aorists are attested in any Slavic text, so maybe I am wrong to set up asigmatic root-aorists like 3rd dual *-vŕ̥zete as a possibility alongisde primary sigmatic *-verste (e.g. Mar. Mark 7 ⱃⰰⰸⰲⱃⱑⱄⱅⰵ {razvrěste}). My justification is that the *-verg-/*-vŕ̥g-stem verbs *do* attest such root-aorists, e.g. 3rd pl. ⱁⱅⱏⰲⱃⱏⰳⱘ {otŭvrŭgǫ} in Psal. Psalm 77, and I don't see what, apart from the nature of the final stem-consonant (obstruent vs. continuant), could be grounds for classifying these two verbs differently.

- addition of the *-tQ suffix from the 3rd sg. pres. (where Q stands for NSl. *ь vs SSl. *ъ, see fn. 5 above) to 3rd sg. aorist forms: начѧтъ, поѩтъ, оумрѣтъ
- original u/ju-stem nouns taking o/jo-stem endings: dat. sg. с҃ноу, мѫжю; loc. sg. с҃нѣ, gen. sg. мѫжа
- past act. part. of class 4 verbs using the suffix *-ivъ rather than *-jь: б҃гвивъ, поустивъши (cf. Mar. Mark 10 ⱂⱆⱎⱅⱏⱎⰹ {puštŭši} <*pust-jьši)

Deciding upon the "correct" morphological endings for an unattested language inevitably entails some uncertainty and controversy: for instance, *-ox- aorists occur in both OCS and Old Russian, so why do I consider them LCS deviances? Basically because they **never**[30] occur in Marianus, which would be quite improbable if they were a discarded archaism (especially given their ubiquity in the closely related Zographensis). Additionally, West Slavic uses *-ex- here (e.g. Old Czech 1sg. *nesech* <*nes-ex-ъ), suggesting that the extended S-aorists were innovated independently in the post-LCS dialects (Meillet 1965: 256).
In other cases we are dealing with hodge-podge paradigms which are only ever attested with endings from multiple older classes, and sometimes dialectal or orthographic features of the manuscripts can make it difficult to distinguish potential LCS ancestors of certain endings. For instance, I have a *masc_tel* class used for agent-nouns like OCS дѣлател҄ь (i.e. Diels 1963: 166), which in the sg. and dual. behave exactly like masc jo-stems, but which in the gen. and instr. plural are attested also with basically consonant-stem endings on a hard *-tel- stem, e.g. Zogr. ⱅⱔⰶⰰⱅⰵⰾⱏ {tɴžatelŭ} <*tęžĀ̈telъ, Supr. свѧтителꙑ <*svętitely. The nom. pl. appears also to take a consonant-stem *-e ending, but as Diels (op. cit.) points out, the spellings in Zogr. and Supr. (where use of the palatalisation-diacritic <ˆ> to denote LCS *ĺ is consistent enough to suggest a real phonemic /ĺ/ in the underlying dialects) like мѫчител҄е[31] mean we can't follow Meillet (1965: 426) in setting up *-tele as the 'correct' ending, because spellings like ⰶⱔⱅⰵⰾⰵ {žɴtele} in Mar. could just as easily descend from *žęteĺe as from *žętele. Absent a manuscript which both consistently marks *ĺ *and* doesn't use such a mark in this nom. pl. desinence, there's no hard evidence of this *-tele ending ever actually existing. I thus use *-teĺe in the 'correct' table, and the jo-stem *-teĺi in the 'deviances' table[32].

2.4 Overcoming poor lemmatisation practice

Sometimes Eckhoff's lemmatisation is too coarse-grained, in that forms which clearly descend from distinct doublets are subsumed under one lemma. To demonstrate just one example of how the Autoreconstructor deals with this sort of problem, take the numeral ѥдинъ <*jedinъ, which in the earliest OCS has straightforward hard pronominal endings and which therefore goes in my *pron_hard* class alongside demonstratives like *onъ. There is, however, what must be viewed as an LCS doublet *jedьnъ[33], which gives e.g. Serbian *jedan* and the modern Russian fem. nt. and

30 Having autoreconstructed all of Marianus I can verify this (admittedly already well-established) fact far more quickly and easily than was possible just with Eckhoff's morphology and part-of-speech annotation-information: using TOROT the smallest net you could cast would be one that caught all aorist-tense verbs, whereas I can just search for reflexes of *oxъ, *oxov, *oxom, *ošę, *osta$, and *oste$ (the latter two using 'regular-expression' mode and $ to specify end-of-word), which for Mar. turns up nothing except the three occurrences of ⱁⱄⱅⰰ {osta}.
31 Searching my database for *teĺe$ returns only a single властеле in Supr. without the diacritic; everything in Zogr. has it.
32 Diels mentions only Psal. Psalm 26 ⱄⱏⰲⰴⱑⰴⱑⱅⰵⰾⱐ {sŭvědětelĭ}, but the Autoreconstructor is intended for use with other early texts beyond canonical OCS which might also contain this type of assimilation to the jo-stems.
33 This despite Meillet's (1965: 144) incoherent speculation about the /i/ in *jedinъ resulting from a phonetic development of strong *ь, analogous to what happens in the vicinity of *j, because "*or il s'agit d'un composé dont le second élément est *jĭnŭ*". While this interpretation of *jedinъ's derivation could be true (and according to Derksen's (2008: 212) etymology of the *jьnъ pronoun, its *j is prothetic, meaning it wouldn't develop if already attached to a stem ending in *d), the idea that a synchronic *-dьnъ by any purely phonological means could become /-din/, let alone early enough to completely displace by analogy the oblique forms in the earliest OCS where ⰵⰴⰹⱀ- {edin-} spellings are overwhelming, is ludicrous and contradicted by all the philological evidence.

oblique-case forms *одна, одного* etc., and which is used for the majority of non masc. nom./acc. sg. forms of the pronoun in Suprasliensis[34], e.g. ѥд҆номѹ. Notwithstanding Eckhoff's habit of lemmatising these with ѥдинъ instead of ѥдьнъ, it's trivial for the Autoreconstructor to check the form (which has in a previous step already been aggressively normalised to get rid of morphologically-irrelevant variation) for a <дін> sequence, which would never occur in the inflectional-ending and so always point to a *jedin-descended stem, and then replace our autoreconstruction with *jedьn- if such isn't found:

```
 // check for *jedьn-
if (lemma_ref.lemma_id == compileTimeHashString("Рхѥдинъ") || lemma_ref.lemma_id == compileTimeHashString("Макдинъ"))
{
  if (Sniff(cyr_id, "дін", 20) == false)
  {
    stem = "jedьn";
  }
}
```

*Figure 9: Part of the Autoreconstructor's code which checks for reflexes of *jedьn- that TOROT has mistakenly lemmatised under* ѥдинъ

In the long-term TOROT itself should fix such lemmatisation problems, but in the meantime checks like the above are computationally extremely cheap and prevent great swathes of the texts from being wrongly autoreconstructed.

3. Benefits and potential for future studies

We have already seen a few examples of minor questions where recourse to an LCS reconstructed layer makes exhaustive investigation of texts trivial; e.g. the reality of a difference between word-intial /je-/ and (foreign) /e-/ in the Codex Suprasliensis was shown by demonstrating statistically significant discrimination in the use of the <ѥ> letter (p.9-10). Another example, as hinted at in fn. 20, concerns the fate of the PV3 reflexes *ś and *ʒ, specifically whether they harden and merge with plain /s z/, or become palatalised counterparts to plain /s z/ (/s' z'/) as in East Slavic, which is a very important indicator of the extent to which a phonological system is developing Russian-style phonemic secondary palatalisation of consonants.

By far the largest benefit though of the type of annotation proposed here would be felt in questions of a more systematic and holistic nature, where cross-referencing of multiple orthographic (and potentially also morphological) features must be undertaken to corroborate hypotheses. The extremely important issue of the development of phonemic secondary consonant-palatalisation before various LCS front-vowels, for example, which is a critical isogloss in both Bulgaro-Macedonian and East Slavic, is just such a problem, but since there are no separate letters in either Glagolitic or Cyrillic for palatalised consonants, its existence can only be deduced indirectly by combining evidence from a range of other indicators. In addition to changes in nominal morphology mentioned in Section 1.6 (the adoption by *i-stems of *jo/*ja-stem endings), Townsend & Janda (1996: 107-8) bring our attention to a fascinating inverse-relationship in the Slavic dialects between the extent of phonemic tonality-distinctions (hard/soft opposition) in consonants on the one hand, and the longer maintenance of pitch-distinctions and distinctive vowel-length on the other. This is linked to earlier loss of jers in central vs. peripheral areas, with e.g. far southwestern Ukrainian being the most central part of East Slavic and thus most likely to have either lost or not developed secondary consonant palatalisation, and to have longer maintained distinctive vowel-length (on the East Slavic material specifically see also Trubetzkoy 1924).

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34 According to my database there are 145 reflexes of *jedьn- in this category vs 46 from *jedin-, though all 107 reflexes of the masc. nom./acc. sg. are from *jedinъ. Interestingly the Uspenskij Sbornik (at least the parts included in TOROT), in contrast to later Russian, appears to completely lack *jedьn-, though all except the single ѻдиноі are spelt with the OCS <ѥ/є> reflex of intial *je-.

An LCS index makes testing for such correlations much easier, not only because of the speed and ease with which things like the Jer Shift can be investigated, where there are simply so many data-points that exhaustive manual investigations would be a significant undertaking, but also because relevant changes often involve unpredictable or zero-reflexes, which it is simply not possible to reliably search for via surface-forms. The tolerance of bimoraic syllables and distinctive vowel-length, for instance, may be probeable by looking at things like loss of intervocalic /j/ and subsequent vowel-contractions (in e.g. the 3sg. present ending of class 5.4 verbs *-ajetQ, which in some OCS texts is reflected as <-ѧѧтъ>, or long-adjective endings like *o-je, *a-jǞ, which are the source of Ukrainian neuter *-e* and fem. *-a* endings, opposed to Russian uncontracted *-oe* and *-ая*), but one cannot search the surface-texts for the *absence* of something. One can however search the LCS index for sequences like *ajǞ or *ějǞ to see whether a certain text has more or fewer spellings indicative of contraction.
Similarly, one way that signs of the development of secondary consonant palatalisation before certain front-vowels *can* be sought out, at least in the case of /n' l' r'/, is by comparing spellings of the original LCS palatals *ĺ *ŕ *ń with those of plain *l *r *n. Shevelov (1956: 490) claims that the "ziemlich deutlich[e Unterscheidung]" of *ĺ,*ń from *l,*n before *e, but not *i, in the Archangel'sk Gospel of 1092 and the Dobrilo Gospel of 1166, is evidence that following *e never caused palatalisation of these consonants in the proto-Ukrainian dialects of the scribes, whereas *i did. Plain *ni,*li thus fell together with palatal *ńi,*ĺi as /n'i l'i/ and were no longer distinguishable. Whether or not this argument has validity, searching texts for relevant instances to compare is only possible with an LCS index that preserves *ŕ *ń *ĺ, because there is no way to search the surface-spellings for the *absence* of palatalisation-marking on, say, *ŕe *ńe *ĺe without just getting all the reflexes of *re *ne *le as well.
While detailed studies into individual manuscripts' treatment of the jers were conducted more than a century ago by figures like Jagić (1876, Zogr.) and Leskien (1905, Zogr./Mar.), they contain little attempt to integrate their findings into something we would recognise as a coherent phonological description. Jagić's study, which is the origin of the so-called 'Jer Umlaut' theory (the view that weak jers in Zographensis especially were fronted/backed by the vowel in the following syllable, p.17-18), in particular is full of incorrect etymologies[35], and although Leskien's is more systematic and free of etymological errors, the fact that he was writing before our concept of the phoneme shows itself everywhere in his reasoning and in the implausibility of his explanations. His ultimate conclusion is that the apparently 'umlauted' weak-jer spellings in fact reflect the hardness/softness of the consonant-cluster brought into being by the disappearance of the weak-jer (p.347-8), and therefore that front-vowels in following syllables softened in some way the preceding consonant, i.e. *zьlě > /zl'ě/ > /z'l'ě/ <ⰸⰻⰾⱑ> {zĭlě}. On p.325 though he explicitly denies "dass die Palatalisirung durch folgende weiche Vokale denselben Grad der Stärke besass wie die der altererbten palatalen Consonanten", and so must be envisaging a rather farfetched three-way phonological opposition between e.g. /n/, /n'/, and /ń/.[36] Lunt (1952: 322-5) then looked again at Zographensis, and adduces data he manually gathered from "about half the manuscript" to broadly support Jagić's 'Jer Umlaut' theory, but since he only gives rough percentages and we don't have the exact data he draws them from, we have no way of verifying it ourselves.
Perhaps the best indication of the need for a more holistic approach is the fact that Totomanova (2014: 14-28), drawing on Bulgarian dialect data, recently proposed yet *another* solution to this jer-question which is mutually incompatible with both Leskien's and Jagić's: a *merger* of the jers in the earliest Bulgarian after palatal and secondarily-palatalised consonants, and then only a later fronting of this merged /ʌ/ vowel to /e/ in strong-position if the preceding consonant remained soft, parallel to the better-known Middle Bulgarian merger of the nasals in the same environment. Under such a

35 E.g. p.23 "къꙁлати halte ich für richtigere als клати" (<*kolti), and on p.35 he assumes a jer-fronting in зьрѣти based on his (also incorrect) assumption that the jer in въꙁрати is original.

36 Leskien also missses a few spellings which contradict his argument, which I can trivially find using my LCS index, e.g. p.324-5 he claims that Zogr. *ь is prevented from being backed by a following *ń, despite ⰴⰹⱀⰵⱄⱏⱀⱑⰰⰳⱁ {dĭnes<u>ŭńě</u>ago} <*dьnьsьńǞjego and ⱄⱏⰲⱀⰸⱏⱀⱑ {sŭvnz<u>ŭńě</u>}< *sъvęzьńǞ (here perhaps he is led astray by the jat'-spelling of Zographensis /a/ <*Ǟ).

theory no conclusions whatsoever can be drawn about incorrect weak back-jer spellings before LCS hard consonants in OCS, because they could equally easily denote softening and then jer-merging as simple lack of softening and loss of the vowel.

Even very imperfect, wholly automatic autoreconstructions of manucripts not-yet manually morphology-tagged can provide a useful window into a text's features: included on ocstexts.co.uk is the Codex Assemanianus (Cyrillicised from Jouko Lindstedt's ASCII-encoded version[37]), which I tagged and lemmatised using a (slightly improved version of) the extremely outdated method detailed in Berdičeviskis, Eckhoff & Gavrilova (2016), and as shoddy as the autoreconstructions often are, one can still glean useful insights from searching around in it. For example, the affricate /ʒ/ reflex of the Second and Third Velar Palatalisations of *g, in e.g. OCS но<u>ѕ</u>ѣ 'leg' (fem. dat./loc. sg. of нога) and ѩ<u>ѕ</u>а 'illness' (cf. the Polish cognates *no<u>dz</u>e* and *ję<u>dz</u>a* 'witch'), which in all major manuscripts shows clear signs of deaffrication to /z/, looks to be pretty flawlessly preserved: in 100% of the forms with autoreconstructions implying such a PV2 or PV3 reflex of *g (i.e. those matching the regexes /gv?[ěęeiьr̥̄l̥̄]/ or /[ьięr̥̄l̥̄]g[auǫ]/), the spelling is <ⰷ> {ʒ}, never <ⰸ> {z}, and the only indications of its deaffrication to /z/ are the two spellings of the *-zęb- verb-root in ⱂⱃⱁⰷⱀⰱⰵ {proʒɴbe} and ⱂⱃⱁⰷⱀⰱⱀⰵⱅⱏ {proʒɴbnetŭ}[38].

LCS reconstructions are also useful for checking uncertain etymologies: if a text consistently spells a certain sound-group in securely-reconstructed words one way, but a different etymon which we less-assuredly reconstruct with that same sound-group is spelt differently, then that's a strong indication that our candidate-etymology is faulty. For example, the entry for Russian *брение* in Vasmer's etymological dictionary (Фасмер 1986) mentions Sobolevskiy's suggested *bŕ̥nьje reconstruction with syllabic *ŕ̥ instead of the more traditional *brьnьje with a TRЪT-group, based on East Slavic **б<u>ьр</u>ние** and **б<u>ер</u>ние** spellings in a.o. the Izbornik 1073 and the Archangel'sk Gospel 1092. Sobolevskiy's view is that Russian *брение* is a written borrowing from Church Slavonic which reflects the OCS <liquid + jer> ordering of the letters for syllabic liquids, viz. брьнье.
If we search for this lexeme in Psalterium Sinaiticum, however, we find three spellings: ⰱⱃⰵⱀⰹⱑ {b<u>re</u>nıě}, ⰱⱃⰵⱀⱐⰵ {b<u>re</u>nĭe}, and ⰱⱃⱏⱀⱐⱑ {b<u>rŭ</u>nĭě}, two of which suggest a regular vocalised /e/ reflex of strong *ь in a TRЪT-group. Searching my autoreconstructed layer for *ŕ̥ turns up **zero** similar <ⱃⰵ> {re} spellings in the 410 reflexes of *ŕ̥ found: overwhelmingly this group is spelt <ⱃⱏ> {rŭ} in Psal. Conversely, if we use the regex [tŕrpsšdfgħђklĺzžxčvbnńmǯ]rь[tŕrpsšdfgħђklĺzžxčvbnńmǯ]+ [ьъ] to search for the strong TrьT-group suggested by the traditional *brьnьje etymology, we find multiple <ⱃⰵ> {re} spellings: ⱂⱁⱄⰽⱃⰵⰶⱐⱋⰵⱅⱏ {posk<u>re</u>žĭćetŭ} <*poskrьžьħetQ, ⰲⱐⱄⰽⱃⰵⱄⱏ {vĭsk<u>re</u>sŭ} <*vъskrьsъ, ⰴⱐⰱⱃⰵⱈⱏ {dьb<u>re</u>xŭ} <*dьbrьxъ, and ⱂⱁⱄⰽⱃⰵⰶⱐⱋⱅⰵⱅⱏ {posk<u>re</u>žĭštetŭ} <*poskrьžьħetQ. The evidence of Psal. is therefore unambiguously in favour of the *brьn- reconstruction: other reflexes of strong TrьT-groups show the same distribution of spellings as this etymon, while none of the *ŕ̥ reflexes do. Sobolevskiy is right to suspect a Church Slavonicism in Russian, but it's more likely to be in the early East Slavic бьрние etc. forms, where East Slavic <**ьр**> was introduced as a hypercorrection by scribes who mistakenly etymologised unfamiliar Bulgarian брьние as *bŕ̥nьje.

4. Conclusion

The foregoing article has laid out a method for leveraging existing morphosyntactic annotation-data to produce rigorous phonological annotations of Old Church Slavonic texts, and in so doing to make the results of large-scale corpus-building efforts like PROIEL and TOROT directly useful to historical phonologists, morphologists, and textologists, rather than just to the syntacticians, semanticists and pragmaticians these resources were primarily created for.

37 https://www.kielipankki.fi/download/ccmh-src/www/assemanianus.html
38 Though note that the modern Macedonian *семне* <*zębne[tъ] 'shiver' also has an unetymological /dz/ in this root, as do many other Macedonian words with LCS *z such as <u>ѕ</u>вер <*zvěrь.

Both the resulting annotations and the computational apparatus used to produce them have value not only for researchers, but also in their potential for creating innovative pedagogical tools for learners. The work-in-progress ocstexts.co.uk website aims to demonstrate some of these things: for example ‘normalised’ spelling of a text’s words, based on conversions from the regular autoreconstructed LCS forms, can help students learn about and cope with the significant dialectal and orthographic variation characteristic of all OCS texts (Figure 10), and the computerised LCS inflectional morphology which is used to generate the autoreconstructions of single forms can be repurposed to dynamically-generate the whole paradigm of every reconstructed lemma, resulting in comprehensive and high-quality inflection-tables (see p.16). We can even combine this paradigm-generating ability with each text-word’s TOROT morphology-tag to visually locate a specific form in its generated table, which massively helps with the learning of paradigms (Figure 11).

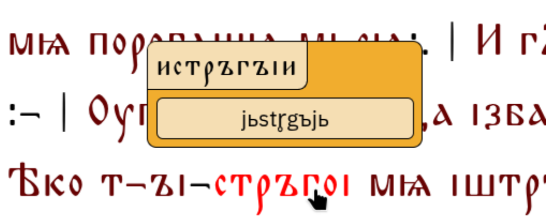

Figure 10: Tooltip showing the autoreconstructed LCS ancestor of the past active participle of the verb истрьгнѫти *in Psalterium Sinaiticum, along with its conversion into 'normalised' OCS*

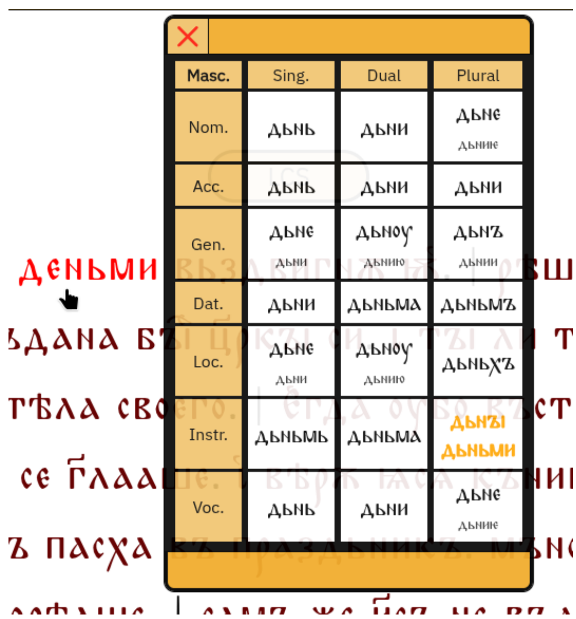

Figure 11: Paradigm of the noun дьнь *generated after Ctrl+clicking on an instrumental-plural occurrence in the Codex Marianus, with the corresponding table-entry animated*

In many ways the point of what I’m proposing with this article is simply to make maximally accessible and applicable for manuscript-investigators the knowledge already accumulated by the nearly two centuries of serious Slavonic scholarship, i.e. the sum of our etymological knowledge (as distilled in etymological dictionaries like Derksen (2008) and the ESSJa) combined with what we know about Common Slavonic phonology and morphology. Furnishing every word of a text with its LCS ancestor simply brings this accumulated knowledge directly ‘on top’ of the philological evidence at a granular, per-word level, so that deviations from this regular theoretical yardstick are visible, retrievable, and quantifiable.

Though important OCS and Old East Slavic texts are still lacking the sort of annotation-data needed for the method of autoreconstruction outlined here, it should be noted that recent advances in neural-network-based tagging and lemmatisation (e.g. Rabus & Besters-Dilger 2021) mean that modern taggers trained on the existing manually-annotated corpus will perform extremely well on the remaining canonical OCS texts, which massively cuts down the time and effort required to do for, say, the Savvina Kniga what has already been done for the Marianus. As the volume and quality of training-data increases, the performance of such automatic taggers can only improve, which will open up more and more of the early Slavic written record to the sort of automatic phonological annotation described here.

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Appendix 1: Glagolitic transliteration table

Glagolitic	Latin		Commonly-used Cyrillic
ⰰ	a		а
ⰱ	b		б
ⰲ	v		в
ⰳ	g		г
ⰴ	d		д
ⰵ	e		є
ⰶ	ž		ж
ⰷ	ʒ		ѕ, ꙃ
ⰸ	z		з, ꙁ
ⰹ	ï		ї, ꙇ
ⰺ	ı		і
ⰻ	i		и
ⰼ	ǵ		ꙉ, г̑, ђ
ⰽ	k		к
ⰾ	l		л
ⰿ	m		м
ⱀ	n		н
ⱁ	o		о
ⱂ	p		п
ⱃ	r		р
ⱄ	s		с
ⱅ	t		т
ⱛ	ü		ѵ, ү
ⱇ	f		ф
ⱈ	x		х
ⱉ	ω		ѡ
ⱚ	θ		ѳ
ⱌ	c		ц
ⱍ	č		ч
ⱎ	š		ш
ⱋ	ć		щ
ⱏ	ŭ		ъ
ⱐ	ĭ		ь
ⱑ	ě		ѣ
ⱆ	u		оү, оү
ⱓ	ü		ю
ⱘ	ǫ		ѫ
ⱙ	ǫ̈		ѭ
ⱗ	ę		ѩ, ѧ
ⱔ	ɴ		ѧ
ⱕ	ɴ̨		ѧ